Collision-oriented annihilation

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Abstract

Let $A = (a_{ij}) \in \mathbb{Q}^{n\times n}$. I attach to every cell (i, j) the binary address $w_{ij} = 2^{(i-1)n+j-1}$ and form the marked matrix $B_A(z)_{ij} = p_{ij} z^{w_{ij}}$ with $p_{ij} = 1 + a_{ij}^2 > 0$. I prove that the marked determinant $D_A(z) = \det B_A(z)$ annihilates exactly the column-collision assignments, and keeps each permutation σ at its own address $E(\sigma) = \sum_i w_{i,\sigma(i)}$, so that no two permutations ever share a monomial. The expanded polynomial has $n!$ monomials and degree $2^{\Theta(n^2)}$, yet it is computed exactly by an arithmetic circuit with fewer than $3n^4$ gates ($n \geq 2$), without ever listing its monomials. A rectangular extension is obtained by a Vandermonde complement. If every column carries an integer d_j, every surviving permutation has the same total frequency $D^* = \sum_j d_j$; a number-theoretic transform (NTT), a Hadamard product and an inverse transform therefore project the filtered output exactly onto the slot $D^* \bmod N$, and I show that this projection is itself computed by circuits of polynomial size. The signs carried by the monomials can be gathered into a single universal sign $(-1)^{n(n-1)/2}$ by doubling the orientation: each column carries a pair $q_j = x_j y_j$ of exterior generators. The resulting product is computed by an explicit circuit of $4n^2 + n - 2$ gates over the enlarged algebra; this is an algebra-valued count, and the scalar compilation I give has exponential size. One step of the resulting pipeline is not proved here: the correction of the signs carried by the coefficients of D_A. I formulate it as a precise conjecture about the signed polynomial D_A alone, prove that every other stage is polynomial, and prove that, conditionally on the conjecture, the whole pipeline is polynomial. Finally, I give an exact scalar encoding of the corrected polynomial as the central coefficient, at index $n^n - 1$, of an explicit polynomial in two variables computed by a circuit of $O(n^4)$ gates. This reduces the conjecture, exactly, to the computation of that central coefficient, which is the remaining step.

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1 Introduction

The determinant of an $n \times n$ matrix is a signed sum over the $n!$ permutations, and it is evaluated in polynomial time by elimination or by the characteristic-polynomial recurrences of Faddeev–LeVerrier, Csanky and Berkowitz [3, 5, 6, 4]. In that classical picture the individual terms of the sum are never visible: only their sum is. The purpose of this paper is to make the terms individually addressable while keeping the polynomial-time evaluation. To every cell (i, j) of the matrix I attach a power $z^{w_{ij}}$ of a formal variable z, where the exponent $w_{ij} = 2^{(i-1)n+j-1}$ is a *binary address* of the cell, and I study the polynomial

$$D_A(z) = \det\left(p_{ij} z^{w_{ij}}\right)_{i,j\in[n]}, \qquad p_{ij} = 1 + a_{ij}^2.$$

Because the addresses occupy pairwise disjoint bits, the exponent of the monomial attached to a permutation determines that permutation. Consequently the polynomial $D_A(z)$ is a *signed polynomial with $n!$ monomials in which each monomial records one permutation*, and the determinant is at the same time a filter: in the exterior algebra, every assignment of rows to columns that uses a column twice is annihilated.

The main results of the paper are the following. Throughout, $[m] = \{1, \dots, m\}$, S_n is the symmetric group, sgn is the signature, and for a map $s : [n] \to [n]$ I write

$$E(s) = \sum_{i=1}^{n} w_{i,s(i)}.$$

The total binary length of (the numerators and denominators of) the entries of A is denoted L.

Theorem 1.1 (Exact filtering and compact compilation)**.** *Let $n \geq 1$ and $A \in \mathbb{Q}^{n\times n}$. Put $p_{ij} = 1 + a_{ij}^2$, $w_{ij} = 2^{(i-1)n+j-1}$, $B_A(z)_{ij} = p_{ij} z^{w_{ij}}$ and $D_A(z) = \det B_A(z)$. Then the following hold.*

(i) $D_A(z) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\Big(\prod_{i=1}^{n} p_{i,\sigma(i)}\Big) z^{E(\sigma)}$.

(ii) *For all maps* $s,t:[n]\to[n]$, $E(s)=E(t)\Rightarrow s=t$.

(iii) $\operatorname{supp} D_A = \{E(\sigma):\sigma\in S_n\}$, *this set has exactly* $n!$ *elements, and* $[z^{E(s)}]D_A = 0$ *for every* $s:[n]\to[n]$ *which is not a permutation.*

(iv) $\deg_z D_A = E(\mathrm{id}) = \sum_{r=0}^{n-1} 2^{r(n+1)}$ *and* $\operatorname{ord}_z D_A = E(w_0) = \sum_{i=1}^{n} 2^{i(n-1)}$, *where* $w_0(i) = n+1-i$. *In particular* $2^{n^2-1} \le \deg_z D_A < 2^{n^2}$.

(v) *There is an arithmetic circuit* C_A *over* $\mathbb{Q}$ *computing* D_A *with*

$$G(n) \le 2n^4 - n^3 + 4n^2 - 1$$

gates; in particular $G(n) < 3n^4$ *for* $n \ge 2$.

(vi) *For* $n\ge 2$, C_A *has a binary encoding of length* $\operatorname{bits}(C_A) = O\big(L + n^4 \log n\big)$.

No list of the $n!$ *monomials of* (i) *is constructed.*

The second theorem explains why the surviving terms can be gathered by a spectral projection. Let K be a field, $N\ge 1$, and $\omega\in K^\times$ of exact multiplicative order N. Let Λ be the exterior algebra over $K[z]$ on generators $x_1,\dots,x_n$, and $\Omega = x_1\cdots x_n$.

Theorem 1.2 (Isochrony and spectral extraction)**.** *Let* $B=(B_{ij})\in K[z]^{n\times n}$ *be any matrix, let* $d_1,\dots,d_n\in\mathbb{Z}$ *and* $D^* = \sum_j d_j$. *Put* $M_{ij} = B_{ij}x_j$ *and*

$$P_i[k] = \sum_{j:\, d_j\equiv k \ (N)} M_{ij}, \quad A_i[f] = \sum_{k=0}^{N-1} P_i[k]\,\omega^{fk}, \quad \Psi_0[f]=1, \quad \Psi_i[f] = \Psi_{i-1}[f]\,A_i[f],$$

$$T[k] = \frac{1}{N}\sum_{f=0}^{N-1} \Psi_n[f]\,\omega^{-fk} \qquad (k\in\mathbb{Z}/N\mathbb{Z}).$$

Then $D_\sigma := \sum_i d_{\sigma(i)} = D^*$ *for every* $\sigma\in S_n$, $\Psi_n[f] = \omega^{fD^*}\det(B)\,\Omega$ *for every* f, *and*

$$T[k] = \mathbf{1}_{\{k\equiv D^* \ (N)\}}\,\det(B)\,\Omega \qquad \text{for all } k.$$

In particular $T[D^* \bmod N] = \det(B)\,\Omega$. *Applied to* $B = B_A$ (*with* $K=\mathbb{Q}(\zeta_N)$, *or with* $K=\mathbb{F}_Q$ *for a prime* Q *with* $N\mid Q-1$ *and* Q*-integral entries*), *the output is* $D_A\,\Omega$, *resp. its reduction modulo* Q.

The transform layer of Theorem 1.2 is written in the exterior algebra, whose intermediate elements $\Psi_i[f]$ have exponentially many coordinates (Proposition 11.1). The following result shows that this does not obstruct polynomial circuits, because the quantity I extract lies in the top exterior degree.

Theorem 1.3 (Polynomial circuits for the spectral layer)**.** *Assume that* $\operatorname{char} K = 0$ *or* $\operatorname{char} K > n$, *and let* $B = \bar{B}_A$ *be the matrix* B_A *with entries read in* K. *For every* $k\in\mathbb{Z}/N\mathbb{Z}$, $T[k] = c_k\Omega$

with $c_k \in K[z]$, where $c_k = 0$ if $k \not\equiv D^* \pmod N$, and $c_{D^* \bmod N} = \bar{D}_A$. The polynomial $c_{D^* \bmod N}$ is computed by an arithmetic circuit over K with at most

$$N\,(2n^4 - n^3 + 3n^2 + 2) + n^2$$

gates, obtained by evaluating the N frequency slots and applying the inverse transform; it is also computed, after collapsing the N determinants into one, by a circuit with at most $G(n)$ gates.

What remains is the following. By Theorem 1.1 (i), the coefficient of $z^{E(\sigma)}$ in D_A is $\operatorname{sgn}(\sigma)\prod_i p_{i\sigma(i)}$, hence carries the sign of the permutation. The sign is a function of the address alone (Proposition 12.3). For $e \in \mathbb{N}$ put

$$\chi_n(e) = \begin{cases} \operatorname{sgn}(\sigma) & \text{if } e = E(\sigma) \text{ for some } \sigma \in S_n, \\ 0 & \text{otherwise,} \end{cases}$$

and let $\mathcal{C}_n : \mathbb{Q}[z] \to \mathbb{Q}[z]$ be the $\mathbb{Q}$-linear operator defined on monomials by $\mathcal{C}_n(z^e) = \chi_n(e)\, z^e$. I call $\mathcal{C}_n$ the *sign corrector*. By Proposition 12.2, $\mathcal{C}_n(D_A) = \sum_e \big|[z^e]D_A\big|\, z^e$ is the coefficientwise absolute value of D_A.

The correction can be realized algebraically, and with a single universal sign, by doubling the orientation. Write

$$P_A = \mathcal{C}_n(D_A) = \sum_{\sigma \in S_n} \Big(\prod_{i=1}^{n} p_{i,\sigma(i)} \Big) z^{E(\sigma)}.$$

Let $\widetilde{\Lambda} = \Lambda_{\mathbb{Q}[z]}(x_1, \dots, x_n, y_1, \dots, y_n)$ be the exterior algebra on $2n$ generators, let $\widetilde{\Omega} = x_1 \cdots x_n y_1 \cdots y_n$, and attach to column j the pair $q_j = x_j y_j$ instead of x_j. Put $\gamma_n = (-1)^{n(n-1)/2}$ and

$$\mathscr{F}_A(z) = \prod_{i=1}^{n} \Big(\sum_{j=1}^{n} p_{ij} z^{w_{ij}} q_j \Big).$$

Theorem 1.4 (Uniform sign by double orientation)**.** (a) *q_j is central, $q_j^2 = 0$, $q_i q_j = q_j q_i$, and $q_{\sigma(1)} \cdots q_{\sigma(n)} = \gamma_n\, \widetilde{\Omega}$ for every $\sigma \in S_n$.*

(b) *$\mathscr{F}_A = \gamma_n\, P_A\, \widetilde{\Omega}$. In particular $[\widetilde{\Omega}]\mathscr{F}_A = \gamma_n P_A$ has $n!$ monomials, all carrying the same sign γ_n, and for every $e \in \mathbb{N}$, $[z^e]D_A = \chi_n(e)\,\gamma_n\,[z^e][\widetilde{\Omega}]\mathscr{F}_A$.*

(c) *Let K, N, ω, d_j, D^* be as in Theorem 1.2, and let $\widetilde{T}[k]$ be the spectral layer of Definition 13.10, built on the pairs q_j. Then $\widetilde{T}[k] = \mathbf{1}_{\{k \equiv D^* \ (N)\}}\, \gamma_n\, P_A\, \widetilde{\Omega}$.*

(d) *$\mathscr{F}_A$ is computed by an explicit circuit over $\widetilde{\Lambda}$ with exactly $G_{\mathrm{ext}}(n) = 4n^2 + n - 2$ gates, the p_{ij} being leaves (Section 13, Definition 13.6), and $P_A\widetilde{\Omega} = \gamma_n \mathscr{F}_A$ is obtained with at most one further gate, namely one multiplication by -1 if $n \equiv 2, 3 \pmod 4$ and none if $n \equiv 0, 1 \pmod 4$. These are gates of the enlarged algebra, not gates of a scalar circuit over $\mathbb{Q}[z]$.*

For $n = 2$, $\mathscr{F}_A = -\big(p_{11}p_{22}z^9 + p_{12}p_{21}z^6\big)\widetilde{\Omega}$. Theorem 1.4 gathers all the signs into the constant γ_n, but it does not make the resulting top coefficient cheap to compute: the count in (d) is a count of gates of the enlarged algebra, the partial products of $\mathscr{F}_A$ have exponentially many coordinates, the natural scalar compilation has at most $(n-1)2^n + 2n^2$ gates, and the determinantal identity of Theorem 1.1 does not apply to them (Propositions 13.13 and 13.14). The sign correction therefore remains the open step, and I formulate it as follows.

Conjecture 1.5 (Polynomial sign correction)**.** *There is a polynomial $\Pi \in \mathbb{N}[X]$ such that, for every $n \geq 1$ and every $A \in \mathbb{Q}^{n\times n}$, the polynomial $\mathcal{C}_n(D_A) \in \mathbb{Q}[z]$ is computed by an arithmetic circuit over $\mathbb{Q}$ with at most $\Pi(n)$ gates, all of whose constants are entries p_{ij} of B_A or rational numbers of binary length at most $\Pi(n)$. Moreover such a circuit can be constructed from (n, A) in time polynomial in n and L. Equivalently (Corollary 13.15), the top coefficient $[\widetilde{\Omega}]\mathscr{F}_A = \gamma_n P_A$ is computed by such a circuit, with $\Pi(n)$ replaced by $\Pi(n) + 1$. It is also equivalent to the polynomial-size computation of the central coefficient $[t^K]\widehat{H}_A$ of Theorem 1.7 (Corollary 14.10).*

The conjecture concerns the signed polynomial D_A and the diagonal operator $\mathcal{C}_n$ acting on its monomials, or equivalently the top coefficient of the doubly oriented product $\mathscr{F}_A$, and nothing else. I do not prove it. What I prove is that all other stages are polynomial, and that the conjecture is the only missing ingredient.

Theorem 1.6 (Polynomial pipeline under the sign-correction hypothesis)**.** *Assume Conjecture 1.5. Let $K \supseteq \mathbb{Q}$ be a field of characteristic 0 containing a primitive N-th root of unity ω, let $\mathcal{C}_n$ act K-linearly on $K[z]$, and let $T^+[k] = (\mathcal{C}_n \otimes \mathrm{id}_\Lambda)(T[k])$ be the corrected spectral layer. Then, for every n, every $A \in \mathbb{Q}^{n\times n}$ and every $k \in \mathbb{Z}/N\mathbb{Z}$:*

(a) *$P_A = \mathcal{C}_n(D_A) = \sum_{\sigma\in S_n} \big(\prod_i p_{i,\sigma(i)}\big) z^{E(\sigma)}$, a polynomial with $n!$ monomials and strictly positive coefficients, is computed by a circuit with at most $\Pi(n)$ gates;*

(b) *$T^+[k] = \mathbf{1}_{\{k\equiv D^*\ (N)\}} P_A\, \Omega$, and each coordinate is computed by a circuit with at most $\Pi(n)$ gates, so that all N slots together need at most $N\,\Pi(n)$ gates;*

(c) *every other stage of the pipeline (marking, filtering, spectral projection, coefficient extraction at a prescribed address) is polynomial without any hypothesis, by Theorems 1.1 and 1.3 and Proposition 8.1;*

(d) *the doubly oriented layer satisfies $\widetilde{T}[k] = \mathbf{1}_{\{k\equiv D^*\ (N)\}} \gamma_n P_A\, \widetilde{\Omega}$ (Theorem 1.4 (c)), and each coordinate is computed by a circuit with at most $\Pi(n) + 1$ gates.*

If moreover Conjecture 12.4 holds, then (a) *holds for the rectangular polynomials $D_A^{(r)}$ of Section 9, $1 \leq r \leq n$.*

The conjecture can be reduced, exactly, to the computation of a single coefficient. Let $n \geq 2$, $K = n^n - 1$, $u_i = t^{n^{i-1}}$ and

$$H_A(t, z) = \det\big(p_{ij} z^{w_{ij}} u_i^{j-1}\big)_{i,j=1}^{n} \prod_{1\leq i<\ell\leq n} (u_i - u_\ell), \qquad \widehat{H}_A(t, z) = 1 + t^{2K} + H_A(t, z).$$

Theorem 1.7 (Scalar encoding of P_A as a central coefficient)**.** *Let $n \geq 2$ and $A \in \mathbb{Q}^{n\times n}$.*

(a) *$[t^K]H_A = P_A$. Every exponent of t in H_A lies in $\{1, \dots, 2K - 1\}$, so that $\deg_t \widehat{H}_A = 2K$, $K = \frac{1}{2}\deg_t \widehat{H}_A$ and $[t^K]\widehat{H}_A = P_A$. Each surviving term carries its amplitude $\prod_i p_{i,\sigma(i)}$ and its address $z^{E(\sigma)}$, with the sign $+1$. For $n = 1$, $P_A = p_{11}z = [t^0]H_A$.*

(b) *$\widehat{H}_A$ is computed by a scalar circuit over $\mathbb{Q}$ in the variables z, t with at most $2n^4 - n^3 + 7n^2 + 4n\log_2 n \leq 4n^4$ gates, and a binary encoding of length $O(L + n^4 \log n)$.*

(c) *Any non-negative integer labels r_{ij} separating the permutations, used in place of $(j-1)n^{i-1}$ in the pairing of Proposition 14.2, force a central index $K \geq n! - 1$. Extraction of $[t^K]$ from a circuit of size s by truncation costs at most $s(K+1)^2$ gates, which for $\widehat{H}_A$ is not polynomial in n.*

(d) *Conjecture 1.5 holds if and only if $[t^K]\widehat{H}_A$ has a scalar circuit of polynomial size.*

Theorem 1.7 completes the construction of the encoding: $\widehat{H}_A$ is explicit, scalar, free of any exterior generator, polynomial in size, and contains no amplitude squared. What remains is the computation of its central coefficient, which by item (d) is exactly the remaining step to close Conjecture 1.5. Item (d) is an equivalence and not a proof of the conjecture.

The organization of the paper follows the expository template of Wang and Zahl [1]: the results are stated in this introduction, Section 2 explains the strategy on a complete toy example and lists which section proves which statement, and Sections 4–15 contain the proofs, each section opening with the precise statement it establishes. Section 16 states the scope and the limits of the results.

2 Overview of the argument

The proofs are organized in nine steps. Step 1 is combinatorial, Steps 2 to 5 are algebraic, Steps 6 and 7 concern the spectral layer, and Steps 8 and 9 concern the sign correction and the conjecture.

Step 1: binary addresses (Section 4). The bit positions $0, 1, \ldots, n^2 - 1$ are cut into n consecutive blocks $I_1, \ldots, I_n$ of length n, and the cell (i, j) is given the j-th bit of block I_i. For an assignment $s : [n] \to [n]$ the exponent $E(s)$ has exactly one bit in each block, so that s can be read off from the binary expansion of $E(s)$. This gives the injectivity $E(s) = E(t) \Rightarrow s = t$, the characterization of the set of addresses, and the comparison rule: E is ordered lexicographically by $(s_n, \ldots, s_1)$. The extremal permutations are the identity and the reversal.

Step 2: the exterior-algebra filter (Section 5). In the exterior algebra Λ on $x_1, \ldots, x_n$ one has $x_j^2 = 0$ and $x_j x_i = -x_i x_j$. The product $\prod_i \left(\sum_j B_{ij} x_j\right)$ expands over all n^n assignments. The assignments with a repeated column carry a factor x_j^2 after reordering and vanish; the $n!$ permutations carry $\operatorname{sgn}(\sigma)\Omega$. Hence the product equals $\det(B)\Omega$. With $B = B_A$ this yields items (i)–(iii) of Theorem 1.1.

Step 3: sizes of the expanded forms (Section 6). The degree and the valuation of D_A are obtained from the lexicographic rule of Step 1. The dense vector of coefficients has length $> 2^{n^2-1}$, the sparse list of exponents has at least $n!(n^2 - n + 1)$ bits, and the intermediate exterior layers have up to $\binom{n}{\lfloor n/2 \rfloor} \geq 2^n/(n+1)$ coordinates. This is why a circuit, and not an expansion, is needed.

Step 4: the determinantal circuit (Section 7). The markers $z^{w_{ij}}$ are all powers z^{2^r} and are produced by $n^2 - 1$ successive squarings. The determinant is then computed by the Faddeev–LeVerrier recurrence with an explicit gate count. A gate counts once regardless of the number of monomials of the polynomial it computes.

Step 5: extraction and the rectangular extension (Sections 8 and 9). The coefficient of D_A at a prescribed address e is obtained in polynomial time by decoding e. Replacing the last $n - r$ rows by Vandermonde rows yields polynomials $D_A^{(r)}$ on injections $[r] \hookrightarrow [n]$, with explicit nonzero coefficients.

Step 6: isochrony and spectral collapse (Section 10). If column j carries an integer d_j, then every permutation has the same total $D^* = \sum_j d_j$. The NTT groups assignments by their frequency $D_s = \sum_i d_{s_i}$ modulo N; after the filter of Step 2 only permutations remain, all in the class D^*, so the output sits in a single slot.

Step 7: polynomial circuits for the spectral layer (Section 11). The top-degree component $\Psi_n[f]$ equals $\omega^{fD^*}\det(B)\Omega$; hence the transform layer can be computed by N determinants, without materializing the exponentially large intermediate elements $\Psi_i[f]$.

Step 8: the sign corrector and the double orientation (Sections 12, 13 and 15). The signs of the coefficients of D_A are the signatures of the decoded permutations. I define the diagonal corrector $\mathcal{C}_n$, prove that it is computable monomial by monomial in polynomial time, that it maps D_A to $\sum_e |c_e| z^e$, and that it is not compatible with products. I then show that the correction is realized algebraically by doubling the orientation: replacing each x_j by the even element $q_j = x_j y_j$ keeps $q_j^2 = 0$, makes the generators commute, and gathers all the signs into the single constant $\gamma_n = (-1)^{n(n-1)/2}$; an explicit circuit of $4n^2 + n - 2$ gates over the enlarged algebra computes the product, with at most one further gate for the global sign. The price is that this count is algebra-valued: the determinantal identity of Step 4 is no longer available and the natural scalar compilation is exponential. The circuit complexity of the resulting top coefficient, equivalently of $\mathcal{C}_n(D_A)$, is Conjecture 1.5; under this conjecture I prove Theorem 1.6.

Step 9: the central-coefficient encoding (Section 14). I pair two determinants, with labels r_{ij} that separate the permutations; the two signatures then cancel on the coefficient of one degree in the new variable t. With the base-n labels $r_{ij} = (j-1)n^{i-1}$ the second determinant is a Vandermonde product, the central index is $K = n^n - 1$, and $P_A = [t^K]\widehat{H}_A$ for an explicit polynomial $\widehat{H}_A$ with a circuit of $O(n^4)$ gates. The conjecture is thereby reduced, exactly, to the polynomial-size computation of this central coefficient.

The following complete computation for $n = 2$ contains the two mechanisms of the paper, and I come back to it throughout. The addresses are $w_{11} = 1$, $w_{12} = 2$, $w_{21} = 4$, $w_{22} = 8$, hence

$$B_A(z) = \begin{pmatrix} p_{11}z & p_{12}z^2 \\ p_{21}z^4 & p_{22}z^8 \end{pmatrix}, \qquad D_A(z) = p_{11}p_{22}z^9 - p_{12}p_{21}z^6.$$

The four assignments (s_1, s_2) have exponents $E(1,1) = w_{11} + w_{21} = 5$, $E(1,2) = w_{11} + w_{22} = 9$, $E(2,1) = w_{12} + w_{21} = 6$, $E(2,2) = w_{12} + w_{22} = 10$. Thus $\operatorname{supp} D_A = \{6, 9\}$ and the collision addresses $5, 10$ are absent. In Λ, with $L_1 = p_{11}zx_1 + p_{12}z^2x_2$ and $L_2 = p_{21}z^4x_1 + p_{22}z^8x_2$,

$$L_1L_2 = p_{11}p_{21}z^5x_1^2 + p_{11}p_{22}z^9x_1x_2 + p_{12}p_{21}z^6x_2x_1 + p_{12}p_{22}z^{10}x_2^2 = D_A(z)\,x_1x_2,$$

since $x_1^2 = x_2^2 = 0$ and $x_2x_1 = -x_1x_2$. Now take $K = \mathbb{Q}$, $N = 2$, $\omega = -1$ (exact order 2), and $(d_1, d_2) = (0, 1)$, so $D^* = 1$. Write $M_{ij} = B_{ij}x_j$. Then $A_i[f] = M_{i1} + (-1)^f M_{i2}$, and

$$\Psi_2[0] = (M_{11} + M_{12})(M_{21} + M_{22}), \qquad \Psi_2[1] = (M_{11} - M_{12})(M_{21} - M_{22}),$$

$$T[0] = \tfrac{1}{2}\big(\Psi_2[0] + \Psi_2[1]\big) = M_{11}M_{21} + M_{12}M_{22} = p_{11}p_{21}z^5x_1^2 + p_{12}p_{22}z^{10}x_2^2 = 0,$$
$$T[1] = \tfrac{1}{2}\big(\Psi_2[0] - \Psi_2[1]\big) = M_{11}M_{22} + M_{12}M_{21} = D_A(z)\,x_1x_2.$$

The collisions $(1,1)$ and $(2,2)$ have frequencies $2d_1 = 0$ and $2d_2 = 2$, both in class 0; the permutations $(1,2)$ and $(2,1)$ have frequency $d_1 + d_2 = 1$. Without the exterior relations $T[0]$ would not vanish; with them, slot 0 is empty and slot 1 holds $D_A\Omega$. The remaining object of the

paper is the sign in $D_A = p_{11}p_{22}z^9 - p_{12}p_{21}z^6$: the corrector $\mathcal{C}_2$ turns it into $p_{11}p_{22}z^9 + p_{12}p_{21}z^6$, and the double orientation produces the same polynomial with the uniform sign $\gamma_2 = -1$.

The correspondence between statements and proofs is as follows. Theorem 1.1 (i)–(iii) follows from Corollary 4.2 and Proposition 5.5; (iv) from Proposition 4.4 and Proposition 6.1; (v) and (vi) from Propositions 7.3, 7.4 and 7.6. Theorem 1.2 is proved in Section 10, Theorem 1.3 in Section 11, Theorem 1.4 in Section 13, Theorem 1.7 in Section 14, and Theorem 1.6 in Section 15.

3 Conventions and notation

- $\mathbb{N} = \{0, 1, 2, \dots\}$ and $[m] = \{1, \dots, m\}$. An *assignment* is a map $s : [n] \to [n]$; it is a *collision assignment* if it is not injective. The set of assignments has n^n elements and S_n is the set of bijective ones.

- For $\sigma \in S_n$, $\mathrm{inv}(\sigma) = \#\{(a, b) : a < b,\ \sigma(a) > \sigma(b)\}$ and $\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)}$. I use the standard fact that if $\sigma(a) > \sigma(a+1)$ for some a, then $\tau = \sigma \circ (a\ a{+}1)$ satisfies $\mathrm{inv}(\tau) = \mathrm{inv}(\sigma) - 1$, and that $\sigma \neq \mathrm{id}$ has such an index a.

- $A = (a_{ij}) \in \mathbb{Q}^{n \times n}$, L is the total binary length of the numerators and denominators of the a_{ij}, $p_{ij} = 1 + a_{ij}^2$ and $w_{ij} = 2^{(i-1)n+j-1}$. If $a_{ij} = u/v$ in lowest terms then $p_{ij} = (u^2 + v^2)/v^2$, whose numerator and denominator have together at most $2(\mathrm{len}(u) + \mathrm{len}(v)) + 2$ bits; hence the total length of the p_{ij} is at most $2L + O(n^2)$.

- For a polynomial $f \in K[z]$, $\mathrm{supp}\, f$ is the set of exponents of its nonzero monomials, $\deg_z f$ its degree, $\mathrm{ord}_z f$ its valuation and $[z^e]f$ the coefficient of z^e.

- An *arithmetic circuit* over a field K is a finite directed acyclic graph whose sources (leaves) are labelled by the variable z or by elements of K, and whose other nodes (gates) have fan-in two and are labelled $+$ or $\times$. Each node computes a polynomial of $K[z]$ in the obvious way. The *size* is the number of gates. A multiplication by a constant, a subtraction and a negation each cost one gate. A gate is a pointer to two earlier nodes; it does not store the monomial expansion of the polynomial it computes (see [7, 8] for the standard framework).

- All rings are commutative with unit, except the exterior algebras of Section 5, which are graded-commutative.

4 Binary addresses

This section proves Theorem 1.1 (ii) and the structural facts about E that all later sections use. For $i \in [n]$ let

$$I_i = \{(i-1)n,\ (i-1)n+1,\ \dots,\ in-1\}, \qquad \beta_i(j) = (i-1)n + j - 1 \quad (j \in [n]),$$

so that $w_{ij} = 2^{\beta_i(j)}$.

Lemma 4.1 (Block decomposition)**.** (a) *The sets $I_1, \dots, I_n$ are pairwise disjoint and their union is $\{0, 1, \dots, n^2 - 1\}$.*

(b) *For each i, $j \mapsto \beta_i(j)$ is a bijection from $[n]$ onto I_i.*

(c) *For every assignment s, $E(s) = \sum_{i=1}^{n} 2^{\beta_i(s_i)}$, and the binary expansion of $E(s)$ has exactly one digit 1 in each block I_i, at position $\beta_i(s_i)$, and no other digit 1. In particular $0 < E(s) < 2^{n^2}$.*

Proof. (a) I_i consists of the n consecutive integers from $(i-1)n$ to $in-1$, so $I_1, \dots, I_n$ tile $\{0, \dots, n^2-1\}$ in order. (b) $j \mapsto (i-1)n + j - 1$ is strictly increasing from $[n]$ to I_i and $|I_i| = n = |[n]|$. (c) By definition $E(s) = \sum_i w_{i,s_i} = \sum_i 2^{\beta_i(s_i)}$. By (b), $\beta_i(s_i) \in I_i$, and by (a) the n exponents $\beta_1(s_1), \dots, \beta_n(s_n)$ are pairwise distinct. A sum of pairwise distinct powers of 2 has, by uniqueness of the binary expansion, digit 1 exactly at those exponents. All exponents are at most $n^2 - 1$, so $E(s) < 2^{n^2}$. □

Corollary 4.2 (Injectivity and image)**.** (a) *$E(s) = E(t) \Rightarrow s = t$ for all assignments s, t.*

(b) *An integer $e \in \mathbb{N}$ is of the form $E(s)$ if and only if $e < 2^{n^2}$ and its binary expansion has exactly one digit 1 in each block I_i; the assignment is then unique, $s = \mathrm{dec}(e)$, and $s_i = 1 + p_i - (i-1)n$ where p_i is the position of the digit 1 in I_i.*

(c) *$e \in E(S_n)$ if and only if e is in the image of E and $\mathrm{dec}(e)$ is bijective. In particular, if $s \notin S_n$ and $\sigma \in S_n$ then $E(s) \neq E(\sigma)$.*

Proof. (a) If $E(s) = E(t)$, the binary expansions coincide, hence for each i the unique digit 1 in I_i gives $\beta_i(s_i) = \beta_i(t_i)$, so $s_i = t_i$ since β_i is injective. (b) The necessity is Lemma 4.1 (c). Conversely, if $e < 2^{n^2}$ has exactly one digit 1 in each block, at positions $p_i \in I_i$, put $s_i = 1 + p_i - (i-1)n \in [n]$; then $\beta_i(s_i) = p_i$ and $E(s) = e$. Uniqueness is (a). (c) follows from (a) and (b). □

The next lemma compares two addresses.

Lemma 4.3 (Order of addresses)**.** *Let $s \neq t$ be assignments and let $i_0 = \max\{i : s_i \neq t_i\}$. Then $E(s) < E(t)$ if and only if $s_{i_0} < t_{i_0}$.*

Proof. Write $E(s) = H + 2^{\beta_{i_0}(s_{i_0})} + \Lambda_s$ and $E(t) = H + 2^{\beta_{i_0}(t_{i_0})} + \Lambda_t$, where $H = \sum_{i>i_0} 2^{\beta_i(s_i)}$ is common (the indices above i_0 agree), and $\Lambda_s = \sum_{i<i_0} 2^{\beta_i(s_i)}$, Λ_t likewise. All exponents in Λ_s, Λ_t are at most $(i_0-1)n - 1$ so $\Lambda_s, \Lambda_t < 2^{(i_0-1)n} \leq 2^{\beta_{i_0}(j)}$ for every $j \in [n]$. Suppose $s_{i_0} < t_{i_0}$, and put $b = \beta_{i_0}(s_{i_0})$, so that $\beta_{i_0}(t_{i_0}) \geq b + 1$. Then

$$E(s) = H + 2^b + \Lambda_s < H + 2^b + 2^b = H + 2^{b+1} \leq H + 2^{\beta_{i_0}(t_{i_0})} \leq E(t).$$

The case $s_{i_0} > t_{i_0}$ is symmetric, and since $E(s) \neq E(t)$ the equivalence follows. □

Let id be the identity permutation and $w_0(i) = n + 1 - i$ the reversal.

Proposition 4.4 (Extremal permutations)**.** *For every $\sigma \in S_n$, $E(\sigma) \leq E(\mathrm{id})$ and $E(\sigma) \geq E(w_0)$, with equality only for $\sigma = \mathrm{id}$, resp. $\sigma = w_0$. Moreover*

$$E(\mathrm{id}) = \sum_{r=0}^{n-1} 2^{r(n+1)} = \frac{2^{n(n+1)} - 1}{2^{n+1} - 1}, \qquad E(w_0) = \sum_{i=1}^{n} 2^{i(n-1)}.$$

Proof. Maximum. Let σ maximize E on S_n. I show by induction on $m \in [n]$ that $\sigma(n-l+1) = n-l+1$ for $l = 1, \dots, m$. Suppose this holds for $l < m$ (nothing is assumed for $m = 1$). Then the values $n, n-1, \dots, n-m+2$ sit at positions $n, \dots, n-m+2$, so the values available for positions $1, \dots, n-m+1$ are exactly $1, \dots, n-m+1$, and $\sigma(n-m+1) \le n-m+1$. If $\sigma(n-m+1) < n-m+1$, let τ be obtained from σ by exchanging the value $\sigma(n-m+1)$ with the value $n-m+1$ (the latter sits at some position $k < n-m+1$). Then τ and σ agree at all positions above $n-m+1$, and $\tau(n-m+1) = n-m+1 > \sigma(n-m+1)$; by Lemma 4.3, $E(\tau) > E(\sigma)$, contradicting maximality. Hence $\sigma(n-m+1) = n-m+1$. After $m = n$ steps, $\sigma = \mathrm{id}$, which is therefore the unique maximizer.

Minimum. The same argument with inequalities reversed forces $\sigma(n) = 1$, then $\sigma(n-1) = 2$, and so on, i.e. $\sigma(i) = n+1-i$; w_0 is the unique minimizer.

Values. $E(\mathrm{id}) = \sum_{i=1}^{n} 2^{(i-1)n+i-1} = \sum_{r=0}^{n-1} 2^{r(n+1)}$, a geometric sum of ratio 2^{n+1} with n terms. For w_0, $\beta_i(n+1-i) = (i-1)n + n - i = in - i = i(n-1)$. □

5 The exterior-algebra filter

Let R be a commutative ring with unit and let $\Lambda = \Lambda_R(x_1, \dots, x_n)$ be the R-algebra generated by $x_1, \dots, x_n$ subject to the relations

$$x_i^2 = 0 \ (i \in [n]), \qquad x_i x_j + x_j x_i = 0 \ (i \neq j). \tag{1}$$

I use the following standard facts [2]: Λ is a free R-module with basis the products $x_I = x_{i_1} \cdots x_{i_k}$ ($I = \{i_1 < \cdots < i_k\} \subseteq [n]$, with $x_\emptyset = 1$); it is graded by k; its top component $\Lambda^n = R\,\Omega$ is free of rank one on $\Omega = x_1 \cdots x_n$; and elements of R are central. In all applications $R = K[z]$ for a field K.

Lemma 5.1 (Anticommutation)**.** *For all $i, j \in [n]$, $x_i x_j = -x_j x_i$.*

Proof. For $i \neq j$ this is (1). For $i = j$ both sides are 0 since $x_i^2 = 0$. □

Lemma 5.2 (A collision annihilates)**.** *If $s : [n] \to [n]$ is not injective, then $x_{s_1} x_{s_2} \cdots x_{s_n} = 0$ in Λ.*

Proof. Since s is not injective, there are positions $a < b$ with $s_a = s_b =: j$. The letters between positions a and b may be arbitrary. Move the letter $x_{s_b} = x_j$ leftwards past the $b-a-1$ letters between the two occurrences, by as many adjacent transpositions; each multiplies the word by -1 by Lemma 5.1. The word becomes $\pm$ a word in which $x_{s_a} x_j = x_j^2 = 0$ appear adjacent, hence the word is 0. □

Lemma 5.3 (A permutation gives its signature)**.** *For every $\sigma \in S_n$, $x_{\sigma(1)} \cdots x_{\sigma(n)} = \mathrm{sgn}(\sigma)\,\Omega$.*

Proof. Induction on $\mathrm{inv}(\sigma)$. If $\mathrm{inv}(\sigma) = 0$ then $\sigma = \mathrm{id}$ and the word is Ω. If $\mathrm{inv}(\sigma) > 0$ choose a with $\sigma(a) > \sigma(a+1)$ and $\tau = \sigma \circ (a\ a{+}1)$, so $\mathrm{inv}(\tau) = \mathrm{inv}(\sigma) - 1$. The word of σ is obtained from that of τ by exchanging two adjacent letters, hence equals $-$ the word of τ by Lemma 5.1. By induction it equals $-\mathrm{sgn}(\tau)\Omega = \mathrm{sgn}(\sigma)\Omega$. □

Proposition 5.4 (Top-degree coefficient)**.** *For every commutative ring R and every $M \in R^{n\times n}$, in Λ_R one has, for the ordered product over $i = 1, \dots, n$,*

$$\prod_{i=1}^{n} \Big(\sum_{j=1}^{n} M_{ij} x_j \Big) = \det(M)\,\Omega.$$

Proof. Distributing the product, and using that the $M_{ij} \in R$ are central,

$$\prod_{i=1}^{n} \Big(\sum_{j} M_{ij} x_j \Big) = \sum_{s:[n]\to[n]} \Big(\prod_{i=1}^{n} M_{i,s_i} \Big) \, x_{s_1} \cdots x_{s_n}.$$

Split the sum according to whether s is a collision assignment or a permutation. By Lemma 5.2 the former terms vanish. By Lemma 5.3 the latter give $\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i M_{i,\sigma(i)} \, \Omega = \det(M)\Omega$ (Leibniz formula). □

Apply this to $R = K[z]$ ($K = \mathbb{Q}$ here) and $M = B_A(z)$. Put $L_i = \sum_j B_A(z)_{ij} x_j = \sum_j p_{ij} z^{w_{ij}} x_j$.

Proposition 5.5 (Filter identity)**.** $\displaystyle\prod_{i=1}^{n} L_i = D_A(z)\,\Omega$*, and before the relations* (1) *are imposed the* n^n *terms of the expansion sit at the* n^n *pairwise distinct addresses* $z^{E(s)}$*, of which* $n^n - n!$ *belong to collision assignments and are annihilated.*

Proof. The identity is Proposition 5.4 with $M = B_A(z)$. The expansion of the product is $\sum_s \big(\prod_i p_{i,s_i} \big) z^{E(s)} \, x_{s_1} \cdots x_{s_n}$, because $\prod_i z^{w_{i,s_i}} = z^{E(s)}$. By Corollary 4.2 (a) the exponents $E(s)$ are pairwise distinct over all n^n assignments. The number of collision assignments is $n^n - n!$. □

Proof of Theorem 1.1 (i)–(iii)*. (i).* Leibniz gives

$$D_A = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_i \big(p_{i,\sigma(i)} z^{w_{i,\sigma(i)}} \big) = \sum_{\sigma} \operatorname{sgn}(\sigma) \Big(\prod_i p_{i,\sigma(i)} \Big) z^{\sum_i w_{i,\sigma(i)}},$$

and $\sum_i w_{i,\sigma(i)} = E(\sigma)$.

(ii) is Corollary 4.2 (a).

(iii). By (ii), the map $\sigma \mapsto E(\sigma)$ is injective on S_n, so in the sum of (i) distinct permutations give distinct monomials and no two terms combine. The coefficient of $z^{E(\sigma)}$ is therefore $\operatorname{sgn}(\sigma) \prod_i p_{i,\sigma(i)}$, which is nonzero because each $p_{ij} = 1 + a_{ij}^2 \geq 1 > 0$. Hence $\operatorname{supp} D_A = \{E(\sigma) : \sigma \in S_n\}$, a set with $n!$ elements. If $s \notin S_n$, then $E(s) \neq E(\sigma)$ for all $\sigma \in S_n$ by Corollary 4.2 (c), so $z^{E(s)}$ does not occur in D_A. □

Remark 5.6 (Keeping the amplitudes). The positivity $p_{ij} > 0$ is used only to guarantee that no coefficient vanishes. For an arbitrary matrix $B \in K^{n\times n}$ with entries $b_{ij} z^{w_{ij}}$, the expansion of (i) holds unchanged, and $\operatorname{supp} \det = \{E(\sigma) : \prod_i b_{i,\sigma(i)} \neq 0\}$, because distinct permutations still have distinct exponents. Thus one may take $p_{ij} = a_{ij}$ and keep the amplitudes, and a permutation using a cell with $a_{ij} = 0$ is simply absent from the support. The same holds modulo a prime Q: the reduction of D_A has support $\{E(\sigma) : \prod_i \bar{p}_{i,\sigma(i)} \neq 0 \text{ in } \mathbb{F}_Q\}$, which may be strictly smaller than $\{E(\sigma)\}$ if some $\bar{p}_{ij} = 0$.

6 Degree, valuation and sizes of expanded forms

This section proves Theorem 1.1 (iv) and quantifies why an expanded representation of D_A is out of reach.

Proposition 6.1 (Degree and valuation)**.** *For every $A \in \mathbb{Q}^{n\times n}$, $D_A \neq 0$,*

$$\deg_z D_A = E(\mathrm{id}) = \sum_{r=0}^{n-1} 2^{r(n+1)}, \qquad \operatorname{ord}_z D_A = E(w_0) = \sum_{i=1}^{n} 2^{i(n-1)},$$

the leading coefficient is $\prod_i p_{ii} > 0$, the lowest coefficient is $(-1)^{n(n-1)/2}\prod_i p_{i,n+1-i}$, and $2^{n^2-1} \le \deg_z D_A < 2^{n^2}$.

Proof. By Theorem 1.1 (iii) the support is $E(S_n)$ with nonzero coefficients, so $D_A \neq 0$ and its degree and valuation are the maximum and minimum of E on S_n, given by Proposition 4.4. The corresponding coefficients are $\operatorname{sgn}(\mathrm{id})\prod_i p_{ii}$ and $\operatorname{sgn}(w_0)\prod_i p_{i,n+1-i}$, and $\operatorname{inv}(w_0) = n(n-1)/2$ since every pair $a < b$ is inverted. Finally $E(\mathrm{id}) \ge 2^{(n-1)(n+1)} = 2^{n^2-1}$ since the term $r = n-1$ is present, and $E(\mathrm{id}) < 2^{n^2}$ by Lemma 4.1 (c). □

This proves Theorem 1.1 (iv). It follows that

$$\lim_{n\to\infty} \frac{\log_2(\deg_z D_A + 1)}{n^2} = 1,$$

so a dense coefficient vector of D_A has length at least $2^{n^2-1} + 1$.

Proposition 6.2 (Sparse representation)**.** *Any list of the exponents of the $n!$ monomials of D_A, each written in binary without compression, needs*

$$S_{\mathrm{sparse}}(n) \;\ge\; n!\,(n^2 - n + 1) \text{ bits}, \qquad \log_2 S_{\mathrm{sparse}}(n) \;\ge\; n\log_2 n - (\log_2 e)\,n + \log_2 e,$$

hence $\liminf_{n\to\infty} \log_2 S_{\mathrm{sparse}}(n)/(n\log_2 n) \ge 1$.

Proof. Each $E(\sigma)$ has a digit 1 in the block I_n, at position $\ge (n-1)n = n^2 - n$, hence $E(\sigma) \ge 2^{n^2-n}$ and its binary expansion has at least $n^2 - n + 1$ digits; there are $n!$ exponents, all distinct. For the estimate on $n!$: $\ln$ is increasing, so $\ln k \ge \int_{k-1}^{k} \ln x\,dx$ for $k \ge 2$, and summing, $\ln n! \ge \int_1^n \ln x\,dx = n\ln n - n + 1$, i.e. $n! \ge e\,(n/e)^n$. Taking $\log_2$ and using $n^2 - n + 1 \ge 1$ gives the stated bound. □

Proposition 6.3 (Exterior layers)**.** *$\Lambda^k(V)$ has rank $\binom{n}{k}$, and $\binom{n}{\lfloor n/2\rfloor} \ge 2^n/(n+1)$.*

Proof. The rank is the number of increasing sequences $i_1 < \cdots < i_k$ in $[n]$. Since $2^n = \sum_{k=0}^{n}\binom{n}{k} \le (n+1)\max_k \binom{n}{k}$ and the maximum is attained at $k = \lfloor n/2\rfloor$, the bound follows. □

By Stirling's formula, $n! \sim \sqrt{2\pi n}\,(n/e)^n$ and $\binom{n}{\lfloor n/2\rfloor} \sim 2^n\sqrt{2/(\pi n)}$, which refines the two bounds above. The upshot is that neither the dense form, nor the sparse form, nor the exterior layers can be materialized in polynomial space; the circuit of the next section materializes none of them.

7 The determinantal circuit

This section proves Theorem 1.1 (v) and (vi). The Faddeev–LeVerrier, Csanky and Berkowitz methods are classical [3, 5, 6, 4]; I include complete proofs because the gate count is explicit and the identity $\det = (-1)^n c_n$ is used with a specific sign normalization.

Lemma 7.1 (Markers by successive squaring)**.** *Let $Z_0 = z$ and $Z_{r+1} = Z_r \times Z_r$. Then $Z_r = z^{2^r}$, and the $n^2 - 1$ gates $Z_1, \dots, Z_{n^2-1}$ provide every marker $z^{w_{ij}} = Z_{\beta_i(j)}$, $\beta_i(j) \in \{0, \dots, n^2 - 1\}$. The n^2 entries $B_A(z)_{ij} = p_{ij} \cdot Z_{\beta_i(j)}$ cost n^2 further gates.*

Proof. $Z_{r+1} = (z^{2^r})^2 = z^{2^{r+1}}$ by induction. By Lemma 4.1, $\beta_i(j)$ ranges over $\{0, \dots, n^2 - 1\}$ and $z^{w_{ij}} = z^{2^{\beta_i(j)}} = Z_{\beta_i(j)}$; $Z_0 = z$ is a leaf. No exponent of size 2^{n^2-1} is ever written in unary: the chain has length $n^2 - 1$. □

Let R be a commutative ring in which $1, 2, \dots, n$ are invertible (for example $K[z]$ with $\operatorname{char} K = 0$ or $> n$), and let $M \in R^{n \times n}$. Put $h(t) = \det(I - tM) = \sum_{k=0}^{n} h_k t^k$ with $h_0 = 1$.

Lemma 7.2 (Newton-type identity)**.** *For $1 \le k \le n$, $k\, h_k = -\sum_{j=0}^{k-1} h_j \operatorname{tr}(M^{k-j})$.*

Proof. In $R[[t]]$, $N(t) := I - tM$ is invertible with inverse $\sum_{m \ge 0} t^m M^m$, since

$$(I - tM) \sum_{m \ge 0} t^m M^m = I.$$

Jacobi's formula

$$\frac{d}{dt} \det N = \det N \cdot \operatorname{tr}(N^{-1} N')$$

holds over any commutative ring. Indeed, the product rule gives $\frac{d}{dt} \det N = \sum_i \det N^{[i]}$, where $N^{[i]}$ is N with its i-th row replaced by the derivative N_i'. Expanding along that row,

$$\det N^{[i]} = \sum_j N'_{ij} \operatorname{cof}_{ij}(N) = \det N \sum_j N'_{ij} (N^{-1})_{ji}.$$

Summing over i gives $\det N \operatorname{tr}(N' N^{-1}) = \det N \operatorname{tr}(N^{-1} N')$. Here $N' = -M$, so

$$h'(t) = -h(t) \operatorname{tr}\big((I - tM)^{-1} M\big) = -\Big(\sum_{j \ge 0} h_j t^j\Big)\Big(\sum_{m \ge 0} \operatorname{tr}(M^{m+1})\, t^m\Big).$$

The coefficient of t^{k-1} on the left is $k\, h_k$; on the right it is $-\sum_{j=0}^{k-1} h_j \operatorname{tr}(M^{k-j})$, where $m = k - 1 - j$ and $M^{m+1} = M^{k-j}$. □

Define the recurrence

$$H_0 = I, \qquad T_k = M H_{k-1}, \qquad c_k = -\frac{\operatorname{tr}(T_k)}{k}, \qquad H_k = T_k + c_k I \qquad (1 \le k \le n). \tag{2}$$

Proposition 7.3 (Faddeev–LeVerrier invariant)**.** *For $0 \le k \le n$, $H_k = \sum_{j=0}^{k} h_j M^{k-j}$, and for $1 \le k \le n$, $c_k = h_k$. Moreover $h_n = (-1)^n \det M$, hence*

$$\det M = (-1)^n c_n = \frac{(-1)^{n+1}}{n} \operatorname{tr}(T_n).$$

Proof. Induction on k. For $k = 0$, $H_0 = I = h_0 M^0$. Suppose $H_{k-1} = \sum_{j=0}^{k-1} h_j M^{k-1-j}$. Then $T_k = M H_{k-1} = \sum_{j=0}^{k-1} h_j M^{k-j}$ and, by linearity of the trace and Lemma 7.2, $\operatorname{tr}(T_k) = \sum_{j=0}^{k-1} h_j \operatorname{tr}(M^{k-j}) = -k\, h_k$. Hence $c_k = -\operatorname{tr}(T_k)/k = h_k$ and $H_k = T_k + h_k I = \sum_{j=0}^{k} h_j M^{k-j}$. For the last claim, expand $\det(I - tM) = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_i (\delta_{i\pi(i)} - t M_{i\pi(i)})$. Each factor has degree ≤ 1 in t, so the coefficient of t^n comes only from choosing $-tM_{i\pi(i)}$ in every factor: $h_n = \sum_\pi \operatorname{sgn}(\pi) \prod_i (-M_{i\pi(i)}) = \det(-M) = (-1)^n \det M$. □

Proposition 7.4 (Exact gate count)**.** *Let $M \in R^{n\times n}$ be given by n^2 nodes. The circuit (2), with the last scaling replaced by multiplication by the constant $(-1)^{n+1}/n$ so as to output* $\det M$ *directly, has at most $2n^4 - n^3 + 2n^2$ gates. Consequently the circuit C_A consisting of the markers of Lemma 7.1, the entries $p_{ij}Z_{\beta_i(j)}$ and this recurrence has*

$$G(n) \le (n^2 - 1) + n^2 + (2n^4 - n^3 + 2n^2) = 2n^4 - n^3 + 4n^2 - 1$$

gates, and $G(n) < 3n^4$ for $n \ge 2$.

Proof. A naive product of two $n \times n$ matrices computes n^2 entries, each with n multiplications and $n-1$ additions, i.e. $2n-1$ gates, for a total of $n^2(2n-1) = 2n^3 - n^2$ gates. At iteration k: $T_k = MH_{k-1}$ costs $2n^3 - n^2$; $\operatorname{tr}(T_k)$ costs $n-1$ additions; multiplication by the constant $-1/k$ costs 1; $H_k = T_k + c_k I$ changes only the n diagonal entries and costs n. Hence $\le 2n^3 - n^2 + 2n$ gates per iteration. The last iteration does not compute H_n, and uses the constant $(-1)^{n+1}/n$ instead of $-1/n$, which saves the factor $(-1)^n$ at no cost. Over n iterations the total is at most $n(2n^3 - n^2 + 2n) = 2n^4 - n^3 + 2n^2$. Adding the markers ($n^2 - 1$ gates, Lemma 7.1) and the entries (n^2 gates) gives the bound. Finally

$$3n^4 - (2n^4 - n^3 + 4n^2 - 1) = n^4 + n^3 - 4n^2 + 1 > 0 \quad (n \ge 2),$$

since it equals $16 + 8 - 16 + 1 = 9$ at $n = 2$ and is increasing for $n \ge 2$ (its derivative $4n^3 + 3n^2 - 8n$ is positive). □

By Proposition 7.3 with $R = \mathbb{Q}[z]$ and $M = B_A(z)$, C_A computes D_A. This proves Theorem 1.1 (v). Moreover the limit $G(n)/n^4 \to 2$ shows that the constant 2 is the one of the naive matrix product chosen here; a faster matrix product would lower the exponent but is not needed.

Proposition 7.5 (Syntactic compactness)**.** *Let C_r be the circuit consisting of the first r gates of C_A in topological order. Then $|V(C_{r+1})| = |V(C_r)| + 1$ and, for every $r \le G(n)$, $|V(C_r)| = O(n^4)$, independently of $|\operatorname{supp}\operatorname{poly}(v)|$ for any node v. The final node computes a polynomial with exactly $n!$ monomials, yet $|V(C_A)| < 3n^4 + n^2 + n + 2$.*

Proof. Each step adds exactly one gate. For a product gate $v = u \times w$, the polynomial is $\operatorname{poly}(u)\operatorname{poly}(w)$, which may have up to $|\operatorname{supp}\operatorname{poly}(u)| \cdot |\operatorname{supp}\operatorname{poly}(w)|$ monomials, but the gate is stored as the triple $(\times, \mathrm{id}(u), \mathrm{id}(w))$; no convolution of monomial lists is performed. The number of leaves is at most $1 + n^2 + n + 1$ (z, the p_{ij}, the constants $(-1)^{n+1}/n$ and $-1/k$, $1 \le k \le n-1$, and the constant 1). □

Proposition 7.6 (Binary encoding)**.** *For $n \ge 2$, the circuit C_A has a binary encoding of length* $\operatorname{bits}(C_A) = O(L + n^4 \log n)$.

Proof. The circuit has fewer than $3n^4$ gates and at most $n^2 + n + 2$ leaves, so at most $3n^4 + n^2 + n + 2 \le 4n^4$ nodes for $n \ge 2$ (as $n^2 + n + 2 \le n^4$ for $n \ge 2$). A node identifier needs at most $\lceil \log_2(4n^4) \rceil \le 2 + 4\lceil \log_2 n \rceil$ bits. Each gate is encoded by one bit for its type and two identifiers, for $O(n^4 \log n)$ bits in total. The constants p_{ij} have total length $\le 2L + O(n^2)$ (Section 3); the constants $1/k$ and $(-1)^{n+1}/n$ have length $O(\log k)$ each and total $\sum_{k \le n} O(\log k) = O(n \log n)$. Summing gives $O(L + n^2 + n \log n + n^4 \log n) = O(L + n^4 \log n)$. □

This proves Theorem 1.1 (vi) and completes the proof of Theorem 1.1. Comparing orders of magnitude,

$$\log_2 \operatorname{bits}(C_A) = O(\log L + \log n),$$
$$\log_2 S_{\mathrm{sparse}}(n) = \Theta(n \log n), \qquad \log_2(\deg_z D_A + 1) = \Theta(n^2),$$

so C_A is a syntactically compact representation of a polynomial whose natural expanded representations are exponentially larger.

8 Coefficient extraction at a prescribed address

Proposition 8.1 (Polynomial-time extraction)**.** *Given $A \in \mathbb{Q}^{n\times n}$ of total length L and an integer $e \geq 0$ in binary with $|e|$ digits, the coefficient $[z^e]D_A$ can be computed in $O\big(|e|+n^2 \log n + n(L+n^2)^2\big)$ bit operations, and is 0 unless $e \in E(S_n)$. In particular the extraction is polynomial in $L + n + |e|$, and in the same time one obtains $\big|[z^e]D_A\big| = \prod_i p_{i,\sigma(i)}$ when $e = E(\sigma)$.*

Proof. The algorithm has five steps.

1. If $|e| > n^2$, return 0: then $e \geq 2^{n^2} > E(s)$ for all s, so e is not an address (Lemma 4.1 (c)). Cost $O(|e|)$.

2. Otherwise pad e to n^2 bits and, for each block I_i, count the digits 1; if some block does not contain exactly one, return 0 (Corollary 4.2 (b)). Cost $O(n^2)$.

3. Decode $s_i = 1 + p_i - (i-1)n$ for each i (p_i the position of the unique digit 1 in I_i); then $E(s) = e$ (Corollary 4.2 (b)).

4. Test injectivity of s by comparing all pairs: $O(n^2 \log n)$ bit operations, the entries being $\leq n$. If s has a repeated value, return 0 (Corollary 4.2 (c)).

5. Otherwise $s = \sigma \in S_n$; compute $\operatorname{inv}(\sigma)$ by comparing all pairs in $O(n^2 \log n)$ operations, and $\prod_i p_{i,\sigma(i)}$ by $n-1$ schoolbook multiplications of rationals stored as numerator/denominator pairs, not necessarily reduced. The bit lengths add under multiplication, so the operands never exceed $\sum_{i,j} \operatorname{len}(p_{ij}) \leq 2L + O(n^2)$ bits, and the cost is $O\big(n(L+n^2)^2\big)$. Return $(-1)^{\operatorname{inv}(\sigma)} \prod_i p_{i,\sigma(i)}$.

Correctness: by Theorem 1.1 (iii) and Corollary 4.2, $[z^e]D_A = 0$ unless $e = E(\sigma)$, and then $[z^e]D_A = \operatorname{sgn}(\sigma) \prod_i p_{i,\sigma(i)}$, which is what steps 2–5 return. □

9 The rectangular extension

Fix $1 \leq r \leq n$. Consider the $n \times n$ matrix $M^{(r)}$ whose first r rows are $M^{(r)}_{ij} = p_{ij} z^{w_{ij}}$ $(1 \leq i \leq r)$ and whose last $n-r$ rows are the constant Vandermonde rows

$$M^{(r)}_{r+h,j} = j^{\,h-1} \qquad (1 \leq h \leq n-r).$$

Put $D^{(r)}_A(z) = \det M^{(r)}$. For $r = n$ there are no Vandermonde rows and $D^{(n)}_A = D_A$. Let $\operatorname{Inj}(n, r)$ be the set of injections $s : [r] \to [n]$ and, for $s \in \operatorname{Inj}(n, r)$,

$$E_r(s) = \sum_{i=1}^{r} w_{i,s(i)}, \quad S = s([r]), \quad T = [n] \setminus S = \{t_1 < \cdots < t_{n-r}\},$$

$$\pi_s = (s(1), \dots, s(r), t_1, \dots, t_{n-r}) \in S_n, \qquad \varepsilon(s) = \operatorname{sgn}(\pi_s).$$

Lemma 9.1 (Vandermonde)**.** *For $m \geq 0$ and elements $t_1, \dots, t_m$ of a commutative ring, $\det\left(t_\ell^{h-1}\right)_{1 \leq h, \ell \leq m} = \prod_{1 \leq u < v \leq m} (t_v - t_u)$, the empty determinant and the empty product being 1.*

Proof. For $m \leq 1$ both sides are 1. For $m \geq 2$ proceed by induction. For $h = m, m-1, \dots, 2$ in this order, replace row h by (row h) $- t_1 \cdot$(row $h-1$); this does not change the determinant. The new entry in row $h \geq 2$, column ℓ is $t_\ell^{h-1} - t_1 t_\ell^{h-2} = t_\ell^{h-2}(t_\ell - t_1)$, and column 1 becomes $(1, 0, \dots, 0)^{\mathsf{T}}$. Expanding along column 1 leaves the $(m-1) \times (m-1)$ determinant with entries $t_\ell^{h-2}(t_\ell - t_1)$, $2 \leq h, \ell \leq m$. Factor $(t_\ell - t_1)$ out of column ℓ and apply the induction hypothesis to $t_2, \dots, t_m$: the result is $\prod_{\ell \geq 2}(t_\ell - t_1) \prod_{2 \leq u < v}(t_v - t_u)$. □

Lemma 9.2 (Injectivity on partial assignments)**.** *For $s, s' \in \operatorname{Inj}(n, r)$, $E_r(s) = E_r(s') \Rightarrow s = s'$.*

Proof. $E_r(s) = \sum_{i \leq r} 2^{\beta_i(s_i)}$ with $\beta_i(s_i) \in I_i$ pairwise distinct; its binary expansion has its digits 1 exactly at these positions, one in each of the blocks $I_1, \dots, I_r$. Equality of the expansions gives $\beta_i(s_i) = \beta_i(s'_i)$ for $i \leq r$. □

Proposition 9.3 (Coefficients of the rectangular polynomial)**.**

$$D_A^{(r)}(z) = \sum_{s \in \operatorname{Inj}(n,r)} \varepsilon(s) \Big(\prod_{i=1}^{r} p_{i,s(i)} \Big) \Big(\prod_{\substack{u<v \\ u,v \in T(s)}} (v-u) \Big) \, z^{E_r(s)},$$

the exponents $E_r(s)$ are pairwise distinct, every coefficient is nonzero, and therefore

$$\operatorname{supp} D_A^{(r)} = \{E_r(s) : s \in \operatorname{Inj}(n,r)\}, \qquad \#\operatorname{supp} D_A^{(r)} = \frac{n!}{(n-r)!}.$$

The sign of the coefficient at $z^{E_r(s)}$ is $\varepsilon(s)$. For $r = n$ this is Theorem 1.1 (i)*.*

Proof. Leibniz expansion over $\pi \in S_n$ gives $D_A^{(r)} = \sum_\pi \operatorname{sgn}(\pi) \prod_{i \leq r} p_{i\pi(i)} z^{w_{i\pi(i)}} \prod_{h=1}^{n-r} \pi(r+h)^{h-1}$. Group the permutations by the injection $s = \pi|_{[r]}$. For fixed s, the permutations π with $\pi|_{[r]} = s$ are exactly those with $\pi(r+h) = t_{\rho(h)}$ for a unique $\rho \in S_{n-r}$; that is, $\pi = \pi_s \circ \tilde{\rho}$ where $\tilde{\rho} \in S_n$ fixes $1, \dots, r$ and maps $r + h \mapsto r + \rho(h)$. Indeed $\pi_s(\tilde{\rho}(r+h)) = \pi_s(r + \rho(h)) = t_{\rho(h)}$. Thus $\operatorname{sgn}(\pi) = \varepsilon(s) \operatorname{sgn}(\rho)$ and the contribution of s is

$$\varepsilon(s) \Big(\prod_{i \leq r} p_{i,s(i)} \Big) z^{E_r(s)} \sum_{\rho \in S_{n-r}} \operatorname{sgn}(\rho) \prod_{h=1}^{n-r} t_{\rho(h)}^{h-1}.$$

The inner sum is the Leibniz expansion of $\det\left(t_\ell^{h-1}\right)_{h,\ell}$ (the h-th row, $\ell = \rho(h)$), which equals $\prod_{u<v}(t_v - t_u)$ by Lemma 9.1. By Lemma 9.2, different injections give different exponents, so there is no cancellation between groups. Each coefficient is nonzero: $p_{ij} > 0$, $\varepsilon(s) = \pm 1$, and the t's are distinct integers, so every factor $v - u$ with $u < v$ in T is a positive integer; the empty product ($r \in \{n-1, n\}$) is 1. The number of injections $[r] \to [n]$ is $n!/(n-r)!$. □

Proposition 9.4 (Degree of $D_A^{(r)}$ and circuit)**.** $\deg_z D_A^{(r)} = \sum_{i=1}^{r} 2^{(i-1)n + (n-r+i) - 1}$*, attained at $s(i) = n - r + i$. The polynomial $D_A^{(r)}$ is computed by an arithmetic circuit over $\mathbb{Q}$ with at most $2n^4 - n^3 + 4n^2 - 1$ gates and of binary length $O\big(L + n^4 \log n\big)$, for every $n \geq 2$ and $1 \leq r \leq n$.*

Proof. By Lemma 4.3 applied to the blocks $I_r, \dots, I_1$ in turn, the maximum of E_r on injections is attained by $s(r) = n$, then $s(r-1) = n-1$, and so on, exactly as in Proposition 4.4; this gives $s(i) = n - r + i$ and $\beta_i(s(i)) = (i-1)n + (n-r+i) - 1$. For the circuit, take the circuit of Section 7 for the matrix $M^{(r)}$: the markers for rows $1, \dots, r$ are among $Z_1, \dots, Z_{n^2-1}$, the first r rows cost $rn \le n^2$ gates, the last $n-r$ rows are integer constants (leaves), and Proposition 7.3 applies with $R = \mathbb{Q}[z]$. The gate count is bounded as in Proposition 7.4. The constants $j^{h-1} \le n^{n-1}$ have $O(n \log n)$ bits each and there are at most n^2 of them, i.e. $O(n^3 \log n)$ bits, which is absorbed in $O(n^4 \log n)$. □

10 Isochrony and the spectral projection

This section proves Theorem 1.2. Let K, N, ω, B, d_j, D^*, Λ, Ω, M_{ij}, $P_i[k]$, $A_i[f]$, $\Psi_i[f]$, $T[k]$ be as in the statement. Since ω has exact order N, $\operatorname{char} K \nmid N$ (if a prime $\ell = \operatorname{char} K$ divided N, then $(\omega^{N/\ell} - 1)^\ell = \omega^N - 1 = 0$ would force $\omega^{N/\ell} = 1$, contradicting exact order), so $N^{-1} \in K$. The exponent of ω is only relevant modulo N, and ω^{-1} exists, so negative d_j and negative frequencies cause no difficulty.

Lemma 10.1 (Isochrony)**.** *For every $\sigma \in S_n$ and all integers $d_1, \dots, d_n$, $D_\sigma = \sum_{i=1}^n d_{\sigma(i)} = \sum_{j=1}^n d_j = D^*$.*

Proof. Since σ is a bijection of $[n]$, $\{\sigma(1), \dots, \sigma(n)\} = [n]$, and the sum of the d_j does not depend on the order of summation. □

Lemma 10.2 (Orthogonality)**.** *For every $m \in \mathbb{Z}$, $\dfrac{1}{N} \displaystyle\sum_{f=0}^{N-1} \omega^{fm} = \mathbf{1}_{\{m \equiv 0 \ (N)\}}$.*

Proof. If $N \mid m$ then $\omega^m = 1$ and the sum is N. If $N \nmid m$ then $\omega^m \neq 1$ because ω has exact order N, and $\sum_{f=0}^{N-1} (\omega^m)^f = \frac{(\omega^m)^N - 1}{\omega^m - 1} = 0$ since $(\omega^m)^N = (\omega^N)^m = 1$. (For $N = 1$ the statement is trivial.) □

Remark 10.3. For $K = \mathbb{F}_Q$ with Q prime and $N \mid Q - 1$, an element ω of exact order N exists because $\mathbb{F}_Q^\times$ is cyclic of order $Q - 1$. For $K = \mathbb{Q}(\zeta_N)$ or $\mathbb{C}$ one takes $\omega = \zeta_N$. In the first case, for $A \in \mathbb{Q}^{n \times n}$ whose entries have denominators prime to Q, the p_{ij} lie in $\mathbb{Z}_{(Q)}$, and reduction modulo Q is a ring homomorphism $\mathbb{Z}_{(Q)}[z] \to \mathbb{F}_Q[z]$ commuting with determinants; the output is then the reduction $\bar{D}_A$ of D_A, whose support may be smaller than that of D_A (Remark 5.6).

Proposition 10.4 (Frequency expansion before annihilation)**.** *For each $f \in \{0, \dots, N-1\}$, $A_i[f] = \sum_{j=1}^n M_{ij}\, \omega^{f d_j}$, and*

$$\Psi_n[f] = \sum_{s : [n] \to [n]} \omega^{f D_s}\, M_{1,s_1} M_{2,s_2} \cdots M_{n,s_n}, \qquad D_s = \sum_{i=1}^n d_{s_i}, \tag{3}$$

$$T[k] = \sum_{s :\, D_s \equiv k \ (N)} M_{1,s_1} \cdots M_{n,s_n} \qquad (k \in \mathbb{Z}/N\mathbb{Z}). \tag{4}$$

Proof. By definition, $A_i[f] = \sum_{k=0}^{N-1} \omega^{fk} \sum_{j :\, d_j \equiv k} M_{ij} = \sum_j M_{ij} \omega^{f d_j}$, since $\omega^{fk} = \omega^{f d_j}$ whenever $d_j \equiv k \pmod N$. Next, $\Psi_n[f] = \prod_{i=1}^n A_i[f]$ (ordered), and distributing the product, using

that the scalars $\omega^{fd_j} \in K$ are central, gives (3). Substituting (3) in the definition of $T[k]$,

$$T[k] = \sum_s M_{1,s_1} \cdots M_{n,s_n} \cdot \frac{1}{N} \sum_{f=0}^{N-1} \omega^{f(D_s - k)} = \sum_{s:\, D_s \equiv k} M_{1,s_1} \cdots M_{n,s_n}$$

by Lemma 10.2 with $m = D_s - k$. □

Proof of Theorem 1.2. Lemma 10.1 is the first claim. Since the $B_{ij} \in K[z]$ are central,

$$M_{1,s_1} \cdots M_{n,s_n} = \Big(\prod_i B_{i,s_i} \Big)\, x_{s_1} \cdots x_{s_n}.$$

In (3): the terms with s not injective vanish by Lemma 5.2, and for $s = \sigma \in S_n$, $D_\sigma = D^*$ by Lemma 10.1 and $x_{\sigma(1)} \cdots x_{\sigma(n)} = \operatorname{sgn}(\sigma)\Omega$ by Lemma 5.3. Therefore

$$\Psi_n[f] = \omega^{fD^*} \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i B_{i,\sigma(i)}\, \Omega = \omega^{fD^*} \det(B)\, \Omega.$$

Then, by the definition of $T[k]$ and Lemma 10.2 with $m = D^* - k$,

$$T[k] = \det(B)\, \Omega \cdot \frac{1}{N} \sum_{f=0}^{N-1} \omega^{f(D^* - k)} = \mathbf{1}_{\{k \equiv D^* \ (N)\}} \det(B)\, \Omega.$$

The last statements follow by taking $B = B_A$, resp. its reduction (Remark 10.3). □

Remark 10.5 (Two gradings). The proof shows that two independent mechanisms are at work. The grading by frequency ω^{fD_s} is blind to the addresses: it sorts assignments by D_s modulo N. The exterior relations act on the letters x_j and kill collisions *regardless* of their frequency class. Before annihilation, (4) shows that slot k contains every assignment whose frequency is $\equiv k$, including collisions; after annihilation only permutations remain, all in the class D^*. Theorem 1.2 holds for an arbitrary matrix B, not only for B_A: the identity $\Psi_n[f] = \omega^{fD^*} \det(B)\Omega$ is the factorization $\det\big(B \operatorname{diag}(\omega^{fd_1}, \ldots, \omega^{fd_n})\big) = \omega^{fD^*} \det B$ in disguise. The role of the addresses is to make the surviving polynomial $\det B_A$ a signed polynomial with $n!$ separately recoverable monomials.

11 Polynomial circuits for the spectral layer

The elements handled by the transform layer live in the exterior algebra, whose intermediate layers are exponentially large. I first make this precise.

Proposition 11.1 (Intermediate layers are exponentially large)**.** *Let K have characteristic 0, $B = B_A$ read in $K[z]$, $f \in \{0, \ldots, N-1\}$ and $0 \le i \le n$. Then*

$$\Psi_i[f] = \sum_{I \subseteq [n],\ |I| = i} \det\big(B^{(f)}_{[i],I}\big)\, x_I, \qquad B^{(f)}_{ab} = B_{ab}\, \omega^{fd_b},$$

where $B^{(f)}_{[i],I}$ is the submatrix with rows $1, \ldots, i$ and columns in I. All $\binom{n}{i}$ coefficients are nonzero polynomials when $B = B_A$. In particular the monomial-basis representation of $\Psi_{\lfloor n/2 \rfloor}[f]$ has at least $2^n/(n+1)$ nonzero coordinates.

Proof. By Proposition 10.4,

$$\Psi_i[f] = \prod_{a=1}^{i} \Big(\sum_b B^{(f)}_{ab} x_b \Big) = \sum_{t:[i]\to[n]} \Big(\prod_a B^{(f)}_{a,t(a)} \Big) x_{t(1)} \cdots x_{t(i)}.$$

Non-injective t contribute 0 (Lemma 5.2, whose proof applies verbatim to words of length i). An injective t has image $I = \{\iota(1) < \cdots < \iota(i)\}$ and $t = \iota \circ \rho$ for a unique $\rho \in S_i$, and $x_{t(1)} \cdots x_{t(i)} = \operatorname{sgn}(\rho) x_I$ by the argument of Lemma 5.3. The coefficient of x_I is thus $\sum_{\rho \in S_i} \operatorname{sgn}(\rho) \prod_a B^{(f)}_{a,\iota(\rho(a))} = \det(B^{(f)}_{[i],I})$. For $B = B_A$ the minor equals

$$\omega^{f \sum_{b\in I} d_b} \sum_{\rho \in S_i} \operatorname{sgn}(\rho) \prod_a p_{a,\iota(\rho(a))}\, z^{\sum_a w_{a,\iota(\rho(a))}};$$

by Lemma 9.2 distinct ρ give distinct exponents, and the coefficients $\prod p > 0$ are nonzero, so the minor is a nonzero polynomial (the scalar $\omega^{f \sum d_b}$ is a unit). Proposition 6.3 gives the last claim. □

Hence a circuit that evaluates the transform layer by computing the layers $\Psi_i[f]$ coordinate by coordinate cannot be polynomial. This is the reason for the determinantal route below.

Proof of Theorem 1.3. By Theorem 1.2, $T[k] = \mathbf{1}_{\{k \equiv D^*\}} \bar{D}_A\, \Omega$, so $c_k = 0$ for $k \not\equiv D^*$ and $c_{D^* \bmod N} = \bar{D}_A$. For the circuit, use $\Psi_n[f] = \omega^{fD^*} \det(B) \Omega$, that is $\Psi_n[f] = \det(B^{(f)})\, \Omega$ by Proposition 5.4 applied to $B^{(f)}$, and therefore

$$c_k = \frac{1}{N} \sum_{f=0}^{N-1} \omega^{-fk} \det\big(B^{(f)}\big), \qquad B^{(f)}_{ij} = \gamma^{(f)}_{ij}\, z^{w_{ij}}, \quad \gamma^{(f)}_{ij} = \bar{p}_{ij}\, \omega^{f d_j} \in K.$$

The circuit is built as follows. Compute once the $n^2 - 1$ markers Z_r (Lemma 7.1). For each f, form the n^2 entries $\gamma^{(f)}_{ij} \cdot Z_{\beta_i(j)}$ (n^2 gates) and apply the recurrence (2), which is legitimate since $1, \dots, n$ are invertible in K when $\operatorname{char} K = 0$ or $> n$; by Proposition 7.4 it computes $\det B^{(f)}$ with at most $2n^4 - n^3 + 2n^2$ gates. Multiply by the constant ω^{-fk} (1 gate), add the N results ($N - 1$ gates) and multiply by N^{-1} (1 gate). The total is

$$(n^2 - 1) + N\big(n^2 + 2n^4 - n^3 + 2n^2 + 1\big) + (N - 1) + 1 \;\le\; N\,(2n^4 - n^3 + 3n^2 + 2) + n^2.$$

For the collapsed circuit: since $\det B^{(f)} = \omega^{fD^*} \bar{D}_A$, for $k \equiv D^*$ one has

$$c_k = \bar{D}_A \cdot \frac{1}{N} \sum_{f=0}^{N-1} \omega^{f(D^* - k)} = \bar{D}_A,$$

and the circuit C_A of Section 7, read over K, has at most $G(n)$ gates. □

Remark 11.2 (Characteristic $\le n$). The hypothesis $\operatorname{char} K = 0$ or $> n$ is used only through the divisions by $k \le n$ in (2); Theorem 1.2 itself needs only $\operatorname{char} K \nmid N$. For small characteristic one replaces (2) by a division-free determinant circuit such as Berkowitz's [6], which also has $O(n^4)$ gates; I do not use this variant.

Remark 11.3 (Cost of the transform on its own). For $N = 2^m$ a radix-2 NTT of length N uses $O(N \log N)$ operations of the coefficient ring [10, 11, 9], hence $O(nN \log N)$ for n rows plus $O(nN)$ Hadamard products and $O(N \log N)$ for the inverse transform. The bound $O(nN \log N)$ is the count of *ring operations*; in the exterior algebra a single operation on elements of Λ^i costs up to $\binom{n}{i} \cdot n$ coefficient operations (Proposition 11.1), so this count is not a polynomial bound on circuit size. The size bound of Theorem 1.3 is obtained by restricting to the top component.

12 The sign corrector

Write $D_A^{(r)}$ for the polynomial of Section 9, $1 \le r \le n$, and recall $\mathrm{Inj}(n,r)$, E_r, ε. By Lemma 9.2, for each $e \in \mathbb{N}$ there is at most one $s \in \mathrm{Inj}(n,r)$ with $E_r(s) = e$.

Definition 12.1 (Sign character and sign corrector)**.** For $1 \le r \le n$ define $\chi_{n,r} : \mathbb{N} \to \{-1, 0, 1\}$ by

$$\chi_{n,r}(e) = \begin{cases} \varepsilon(s) & \text{if } e = E_r(s) \text{ for some } s \in \mathrm{Inj}(n,r), \\ 0 & \text{otherwise,} \end{cases}$$

and let $\mathcal{C}_{n,r} : K[z] \to K[z]$ be the K-linear map with $\mathcal{C}_{n,r}(z^e) = \chi_{n,r}(e)\, z^e$ for all $e \in \mathbb{N}$. For $r = n$ one has $\chi_{n,n}(E(\sigma)) = \mathrm{sgn}(\sigma)$, and I write $\chi_n = \chi_{n,n}$, $\mathcal{C}_n = \mathcal{C}_{n,n}$, as in the introduction.

Proposition 12.2 (Action on the signed polynomial)**.** *For every $A \in \mathbb{Q}^{n \times n}$ and $1 \le r \le n$,*

$$\mathcal{C}_{n,r}\big(D_A^{(r)}\big) = \sum_{s \in \mathrm{Inj}(n,r)} \Big(\prod_{i=1}^{r} p_{i,s(i)}\Big)\Big(\prod_{\substack{u<v \\ u,v \in T(s)}} (v-u)\Big) z^{E_r(s)} = \sum_{e} \Big|[z^e] D_A^{(r)}\Big|\, z^e,$$

a polynomial with $n!/(n-r)!$ monomials, all of whose coefficients are positive. For $r = n$, $\mathcal{C}_n(D_A) = \sum_\sigma \big(\prod_i p_{i,\sigma(i)}\big) z^{E(\sigma)} = \sum_e |[z^e] D_A| z^e$.

Proof. By Proposition 9.3, $D_A^{(r)} = \sum_s \varepsilon(s)\kappa(s) z^{E_r(s)}$ with $\kappa(s) = \prod_i p_{i,s(i)} \prod_{u<v \in T(s)} (v-u) > 0$, the exponents being pairwise distinct. By linearity and Definition 12.1, $\mathcal{C}_{n,r}(z^{E_r(s)}) = \varepsilon(s) z^{E_r(s)}$, and $\varepsilon(s)^2 = 1$, giving the first equality. Since $\kappa(s) > 0$, $|[z^{E_r(s)}] D_A^{(r)}| = \kappa(s)$, giving the second; monomials off the support do not occur. □

Proposition 12.3 (Pointwise computability and structure of $\mathcal{C}_{n,r}$)**.** (a) *For every integer $e \ge 0$, the value $\chi_{n,r}(e)$ can be computed from the binary expansion of e in $O(|e| + n^2 \log n)$ bit operations.*

(b) *$\mathcal{C}_{n,r}$ is diagonal in the monomial basis, and*

$$\mathcal{C}_{n,r}^2(z^e) = \mathbf{1}_{\{e \in E_r(\mathrm{Inj}(n,r))\}}\, z^e;$$

in particular $\mathcal{C}_{n,r}^2$ is the identity on $\operatorname{supp} D_A^{(r)}$.

(c) *$\mathcal{C}_{n,r}$ is not multiplicative: $\mathcal{C}_{2,2}(f^2) \ne \big(\mathcal{C}_{2,2} f\big)^2$ for $f = D_A$ with $n = 2$.*

Proof. (a) As in Proposition 8.1: if $|e| > n^2$ output 0; check that the first r blocks contain exactly one digit 1 each and the blocks $I_{r+1}, \dots, I_n$ none; decode s, check injectivity, and output $\varepsilon(s)$, computed from π_s by counting inversions in $O(n^2 \log n)$ operations. (b) is immediate from the definition since $\chi_{n,r} \in \{-1, 0, 1\}$ and $\chi_{n,r}^2 = \mathbf{1}_{\{\cdot \in E_r(\mathrm{Inj})\}}$. (c) For $n = 2$, $D_A = p_{11}p_{22}z^9 - p_{12}p_{21}z^6$, so $\mathcal{C}_2 D_A = p_{11}p_{22}z^9 + p_{12}p_{21}z^6$ and $(\mathcal{C}_2 D_A)^2$ contains the monomial z^{15} with coefficient $2p_{11}p_{22}p_{12}p_{21} \neq 0$. On the other hand D_A^2 is supported in $\{12, 15, 18\}$, none of which is an address (the addresses are $6, 9$), so $\mathcal{C}_2(D_A^2) = 0$. □

Proposition 12.3 (a) and Proposition 8.1 show that the *pointwise* sign correction is unconditional: for any prescribed address, the corrected coefficient $|[z^e]D_A|$ is obtained in polynomial time. What is open is whether the diagonal operator $\mathcal{C}_{n,r}$, acting on the *whole* polynomial $D_A^{(r)}$, can be realized by a circuit of polynomial size, without expanding the $n!/(n-r)!$ monomials. By Proposition 12.3 (c) the operator does not commute with products, so no argument based on a ring homomorphism (substitution, evaluation) applies to it.

Conjecture 12.4 (Rectangular form of Conjecture 1.5)**.** *There is a polynomial $\Pi \in \mathbb{N}[X]$ such that for every $n \geq 1$, every $1 \leq r \leq n$ and every $A \in \mathbb{Q}^{n \times n}$, the polynomial $\mathcal{C}_{n,r}\big(D_A^{(r)}\big) \in \mathbb{Q}[z]$ is computed by an arithmetic circuit over $\mathbb{Q}$ with at most $\Pi(n)$ gates, whose constants are entries p_{ij} or rationals of binary length at most $\Pi(n)$, constructible in time polynomial in n and L.*

Conjecture 1.5 is the case $r = n$ of Conjecture 12.4. Both statements concern exclusively the signed polynomials $D_A^{(r)}$ and the diagonal operators $\mathcal{C}_{n,r}$ defined above; no other object enters them. The restriction on the constants excludes the trivial realization in which the $n!/(n-r)!$ coefficients are hidden in a single huge constant.

13 Double orientation and a uniform sign

This section proves Theorem 1.4. The idea is to replace, for each column j, the odd generator x_j by the even element $q_j = x_j y_j$ built from two odd generators. Even elements commute, so the sign produced by reordering the columns disappears, while nilpotency $q_j^2 = 0$ is retained; the only sign left is a constant independent of the permutation. The entries $B_A(z)_{ij} = p_{ij} z^{w_{ij}}$ are not modified: neither a coefficient nor an address is squared or doubled, only the orientation factor attached to the column is.

Let R be a commutative ring with unit and let $\widetilde{\Lambda}_R = \Lambda_R(x_1, \dots, x_n, y_1, \dots, y_n)$ be the R-algebra generated by $x_1, \dots, x_n, y_1, \dots, y_n$ subject to

$$x_i^2 = y_i^2 = 0 \ (i \in [n]), \qquad x_i x_j = -x_j x_i, \quad y_i y_j = -y_j y_i \ \ (i \neq j), \qquad x_i y_j = -y_j x_i \ \ (i, j \in [n]). \tag{5}$$

Writing $u_a = x_a$ and $u_{n+a} = y_a$ for $a \in [n]$, these are exactly the relations (1) for the $2n$ generators $u_1, \dots, u_{2n}$. The standard facts recalled in Section 5 therefore apply: $\widetilde{\Lambda}_R$ is free over R with basis the increasing products u_I, $I \subseteq [2n]$; its top component is $R\,\widetilde{\Omega}$ with

$$\widetilde{\Omega} = u_1 \cdots u_{2n} = x_1 \cdots x_n \, y_1 \cdots y_n;$$

and Lemma 5.1 holds for every pair of generators, equal or not: $u_a u_b = -u_b u_a$. Elements of R are central. The relations (1) hold for $x_1, \dots, x_n$ in $\widetilde{\Lambda}_R$, and also for $y_1, \dots, y_n$; hence there are R-algebra homomorphisms

$$\iota_x, \iota_y : \Lambda_R(x_1, \dots, x_n) \to \widetilde{\Lambda}_R, \qquad \iota_x(x_j) = x_j, \quad \iota_y(x_j) = y_j.$$

Define

$$q_j = x_j y_j \quad (j \in [n]), \qquad \gamma_n = (-1)^{n(n-1)/2},$$
$$q_S = q_{j_1} \cdots q_{j_k} \quad \big(S = \{j_1 < \cdots < j_k\} \subseteq [n]\big), \qquad q_\emptyset = 1.$$

Lemma 13.1 (The constant γ_n)**.** *$\gamma_n = +1$ if $n \equiv 0$ or $1 \pmod 4$, and $\gamma_n = -1$ if $n \equiv 2$ or $3 \pmod 4$. Moreover $\gamma_n^2 = 1$, $\gamma_n^{-1} = \gamma_n$ and $\gamma_{n+1} = (-1)^n \gamma_n$. The first values are $\gamma_1 = 1$, $\gamma_2 = -1$, $\gamma_3 = -1$, $\gamma_4 = 1$.*

Proof. Write $n = 4m + r$ with $r \in \{0, 1, 2, 3\}$ and examine $\binom{n}{2} = n(n-1)/2$ in the four cases. If $r = 0$: $\binom{n}{2} = 2m(4m-1)$, even. If $r = 1$: $\binom{n}{2} = (4m+1)\,2m$, even. If $r = 2$: $\binom{n}{2} = (2m+1)(4m+1)$, a product of two odd numbers, odd. If $r = 3$: $\binom{n}{2} = (4m+3)(2m+1)$, odd. This gives the first claim; $\gamma_n = \pm 1$ gives $\gamma_n^2 = 1$. Finally $\binom{n+1}{2} - \binom{n}{2} = n$, whence $\gamma_{n+1} = (-1)^n \gamma_n$. The values for $n = 1, \ldots, 4$ are read off from $\binom{n}{2} = 0, 1, 3, 6$. □

Lemma 13.2 (Regrouping)**.** *For every $m \geq 0$ and every sequence $a_1, \ldots, a_m \in [n]$, repetitions allowed,*

$$\prod_{k=1}^{m} x_{a_k} y_{a_k} = (-1)^{\binom{m}{2}} \big(x_{a_1} \cdots x_{a_m}\big)\big(y_{a_1} \cdots y_{a_m}\big).$$

Proof. Induction on m. For $m = 0$ both sides are 1. Assume the identity for m, and multiply it on the right by $x_{a_{m+1}} y_{a_{m+1}}$. The right-hand side becomes $(-1)^{\binom{m}{2}} X_m Y_m \, x_{a_{m+1}} y_{a_{m+1}}$, with $X_m = x_{a_1} \cdots x_{a_m}$ and $Y_m = y_{a_1} \cdots y_{a_m}$. Move $x_{a_{m+1}}$ to the left across the m generators of Y_m. By (5), each exchange of $x_{a_{m+1}}$ with some y_{a_k} costs a factor -1, including when $a_k = a_{m+1}$. The cost is $(-1)^m$ and yields $X_m x_{a_{m+1}} \, Y_m y_{a_{m+1}} = X_{m+1} Y_{m+1}$. The total sign is $(-1)^{\binom{m}{2} + m} = (-1)^{\binom{m+1}{2}}$. □

Lemma 13.3 (Properties of the pairs)**.** *In $\widetilde{\Lambda}_R$, for all $i, j \in [n]$ and all $S \subseteq [n]$:*

(a) *q_j is central: $q_j \xi = \xi q_j$ for every $\xi \in \widetilde{\Lambda}_R$;*

(b) *$q_i q_j = q_j q_i$ (including $i = j$);*

(c) *$q_j^2 = 0$;*

(d) *q_S does not depend on the order in which the factors are written, $q_S = (-1)^{\binom{|S|}{2}} x_S y_S = (-1)^{\binom{|S|}{2}} u_{S \cup (n+S)}$ where $n + S = \{n + j : j \in S\}$, and the elements q_S, $S \subseteq [n]$, are R-linearly independent.*

Proof. (a) It suffices to show that q_j commutes with each generator u, since the generators and R generate $\widetilde{\Lambda}_R$ and R is central. Three cases. *Case 1:* $u \in \{x_k, y_k\}$ with $k \neq j$. Then u anticommutes with both x_j and y_j, so $u\, x_j y_j = -x_j\, u\, y_j = x_j y_j\, u$: two exchanges, sign $(-1)^2 = 1$. *Case 2:* $u = x_j$. Then $x_j\, x_j y_j = 0$ since $x_j^2 = 0$, and $x_j y_j\, x_j = -x_j x_j y_j = 0$; the two products are equal. *Case 3:* $u = y_j$. Then $y_j\, x_j y_j = -x_j y_j y_j = 0$ and $x_j y_j\, y_j = 0$, again equal.

(b) follows from (a). (c) $q_j^2 = x_j y_j x_j y_j = -x_j x_j y_j y_j = 0$, using $y_j x_j = -x_j y_j$ and $x_j^2 = 0$.

(d) By (b), q_S does not depend on the order of its factors. With $a_1 < \cdots < a_k$ the elements of S, Lemma 13.2 gives $q_S = (-1)^{\binom{k}{2}} x_{a_1} \cdots x_{a_k} y_{a_1} \cdots y_{a_k}$. The indices of the x's are $a_1 < \cdots < a_k \leq n$ and those of the y's are $n + a_1 < \cdots < n + a_k$ in the numbering $u_{n+a} = y_a$,

so this product is the basis element $u_{S\cup(n+S)}$ up to the displayed sign. Distinct S give distinct index sets $S\cup(n+S)$, since $S=(S\cup(n+S))\cap[n]$. Distinct basis elements of a free module are linearly independent. □

Lemma 13.4 (A collision annihilates the pairs)**.** *If $s:[n]\to[n]$ is not injective, then $q_{s_1}q_{s_2}\cdots q_{s_n}=0$.*

Proof. Since s is not injective, there are positions $a\neq b$ with $s_a=s_b=:j$. This covers every kind of collision: a column used twice, several times, or several columns each used more than once; one pair (a,b) is enough. The q's are central, so the n factors may be reordered arbitrarily and the product equals $q_j^2\prod_{c\neq a,b}q_{s_c}$, which is 0 by Lemma 13.3 (c). □

Lemma 13.5 (Product of the pairs along a permutation)**.** *For every $\sigma\in S_n$, $q_{\sigma(1)}\cdots q_{\sigma(n)}=\gamma_n\,\widetilde{\Omega}$. In particular $q_1\cdots q_n=\gamma_n\widetilde{\Omega}$.*

Proof. Step 1. By Lemma 13.2 with $m=n$ and $a_k=\sigma(k)$,

$$q_{\sigma(1)}\cdots q_{\sigma(n)}=(-1)^{\binom{n}{2}}X\,Y,\qquad X=x_{\sigma(1)}\cdots x_{\sigma(n)},\quad Y=y_{\sigma(1)}\cdots y_{\sigma(n)}.$$

Step 2. Applying the homomorphisms ι_x and ι_y to Lemma 5.3, $X=\iota_x(x_{\sigma(1)}\cdots x_{\sigma(n)})=\operatorname{sgn}(\sigma)\,x_1\cdots x_n$ and $Y=\iota_y(x_{\sigma(1)}\cdots x_{\sigma(n)})=\operatorname{sgn}(\sigma)\,y_1\cdots y_n$.

Step 3. Hence $q_{\sigma(1)}\cdots q_{\sigma(n)}=(-1)^{\binom{n}{2}}\operatorname{sgn}(\sigma)^2\,x_1\cdots x_ny_1\cdots y_n=\gamma_n\widetilde{\Omega}$, because $\operatorname{sgn}(\sigma)^2=1$.

Consistency check. By Lemma 13.3 (b), $q_{\sigma(1)}\cdots q_{\sigma(n)}$ is a rearrangement of the commuting factors $q_1,\dots,q_n$ (as σ is a bijection), hence equals $q_1\cdots q_n$, which is the case $\sigma=\mathrm{id}$ of Steps 1–3; this explains why the result cannot depend on σ. □

Proof of Theorem 1.4, items (a) *and* (b). Item (a) is Lemmas 13.3 and 13.5. For (b), take $R=\mathbb{Q}[z]$.

Step 1: expansion. Each factor $\sum_j p_{ij}z^{w_{ij}}q_j$ is a sum of n terms. The scalars $p_{ij}z^{w_{ij}}$ and the q_j are central, so distributing the product over the n factors gives a sum over the n^n assignments:

$$\mathscr{F}_A=\sum_{s:[n]\to[n]}\Big(\prod_{i=1}^{n}p_{i,s_i}\Big)z^{\sum_i w_{i,s_i}}\;q_{s_1}\cdots q_{s_n}=\sum_s\Big(\prod_i p_{i,s_i}\Big)z^{E(s)}\;q_{s_1}\cdots q_{s_n}.$$

Step 2: collisions. The $n^n-n!$ non-injective assignments contribute 0 by Lemma 13.4.

Step 3: permutations. Every injective $s:[n]\to[n]$ is a bijection, i.e. a permutation σ. By Lemma 13.5, its contribution is $\big(\prod_i p_{i,\sigma(i)}\big)z^{E(\sigma)}\gamma_n\widetilde{\Omega}$. Summing over S_n,

$$\mathscr{F}_A=\gamma_n\Big[\sum_{\sigma\in S_n}\Big(\prod_i p_{i,\sigma(i)}\Big)z^{E(\sigma)}\Big]\widetilde{\Omega}=\gamma_nP_A\,\widetilde{\Omega},$$

the bracket being $P_A=\mathcal{C}_n(D_A)$ by Proposition 12.2.

Step 4: top component. Each factor of $\mathscr{F}_A$ lies in the degree-2 component, so $\mathscr{F}_A$ lies in the degree-$2n$ component $\mathbb{Q}[z]\widetilde{\Omega}$, and $[\widetilde{\Omega}]\mathscr{F}_A=\gamma_nP_A$ is well defined since $\widetilde{\Omega}$ is a basis of that free module of rank one.

Step 5: monomials. By Theorem 1.1 (ii),(iii), the $n!$ exponents $E(\sigma)$ are pairwise distinct and each coefficient $\kappa_\sigma=\prod_i p_{i,\sigma(i)}$ is >0. Hence $[\widetilde{\Omega}]\mathscr{F}_A$ has exactly $n!$ monomials, each

with coefficient $\gamma_n\kappa_\sigma$, of sign γ_n. This holds for every $n \geq 1$; for $n = 1$, $\gamma_1 = 1$ and $\mathscr{F}_A = p_{11} z\, x_1 y_1 = P_A \widetilde{\Omega}$.

Step 6: coefficientwise relation. Let $e \in \mathbb{N}$. If $e = E(\sigma)$ with $\sigma \in S_n$, then $[z^e]D_A = \operatorname{sgn}(\sigma)\kappa_\sigma$, $\chi_n(e) = \operatorname{sgn}(\sigma)$ and $[z^e][\widetilde{\Omega}]\mathscr{F}_A = \gamma_n\kappa_\sigma$, so $\chi_n(e)\gamma_n[z^e][\widetilde{\Omega}]\mathscr{F}_A = \operatorname{sgn}(\sigma)\gamma_n^2\kappa_\sigma = [z^e]D_A$. If e is not of the form $E(\sigma)$, then $[z^e]D_A = 0$ by Theorem 1.1 (iii) and $\chi_n(e) = 0$. □

The case $n = 2$ is as follows. With $q_1^2 = q_2^2 = 0$, $q_2q_1 = q_1q_2$ and $q_1q_2 = x_1y_1x_2y_2 = -x_1x_2y_1y_2 = -\widetilde{\Omega}$,

$$\begin{aligned}\mathscr{F}_A &= \big(p_{11}zq_1 + p_{12}z^2q_2\big)\big(p_{21}z^4q_1 + p_{22}z^8q_2\big)\\ &= \big(p_{11}p_{22}z^9 + p_{12}p_{21}z^6\big)q_1q_2\\ &= -\big(p_{11}p_{22}z^9 + p_{12}p_{21}z^6\big)\widetilde{\Omega}.\end{aligned}$$

The terms in q_1^2 and q_2^2, at the collision addresses 5 and 10, have vanished; both surviving monomials have the same sign $\gamma_2 = -1$.

I now describe the circuit that computes $\mathscr{F}_A$ and count its gates. Its gates live in the enlarged algebra, and this must be distinguished from the scalar circuits of Section 7.

Definition 13.6 (Circuit over the enlarged algebra)**.** A *circuit over* $\widetilde{\Lambda}_{\mathbb{Q}[z]}$ is a finite directed acyclic graph whose leaves are labelled by z, by rational constants, or by generators x_j, y_j, and whose other nodes, the *gates*, have fan-in two and are labelled $+$ or $\times$; the two children of a $\times$ gate are ordered, since $\widetilde{\Lambda}_{\mathbb{Q}[z]}$ is not commutative. Each node computes an element of $\widetilde{\Lambda}_{\mathbb{Q}[z]}$, and the size is the number of gates. A circuit whose nodes all compute elements of $\mathbb{Q}[z]$ and whose leaves are z and constants is a scalar circuit in the sense of Section 3; for such circuits the two notions of size coincide. The count of gates of a circuit over $\widetilde{\Lambda}_{\mathbb{Q}[z]}$ does not bound the number of scalar gates needed to simulate it.

The circuit $\widetilde{C}_A$ takes z, x_j, y_j and the p_{ij} (as constants) as leaves, and has four blocks.

- *Block A, markers.* $Z_0 = z$ and $Z_{r+1} = Z_r \times Z_r$ for $0 \leq r < n^2 - 1$.
- *Block B, oriented entries.* $q_j = x_j \times y_j$ $(j \in [n])$, $B_{ij} = p_{ij} \times Z_{\beta_i(j)}$ and $M_{ij} = B_{ij} \times q_j$.
- *Block C, row sums.* For each i: $U_{i,1} = M_{i1}$, $U_{i,j} = U_{i,j-1} + M_{ij}$ for $2 \leq j \leq n$, and $L_i = U_{i,n}$.
- *Block D, product of the rows.* $\Psi_1 = L_1$ and $\Psi_i = \Psi_{i-1} \times L_i$ for $2 \leq i \leq n$. The output is Ψ_n.

No coefficient extraction occurs in $\widetilde{C}_A$.

Proposition 13.7 (Correctness of the circuit)**.** *For every $n \geq 1$ and $A \in \mathbb{Q}^{n\times n}$, the nodes of $\widetilde{C}_A$ compute: $Z_r = z^{2^r}$ and $Z_{\beta_i(j)} = z^{w_{ij}}$; $M_{ij} = p_{ij}z^{w_{ij}}x_jy_j = p_{ij}z^{w_{ij}}q_j$; $L_i = \sum_j p_{ij}z^{w_{ij}}q_j$; $\Psi_i = L_1 \cdots L_i$; and the output is $\Psi_n = \mathscr{F}_A$.*

Proof. Block A. $Z_0 = z = z^{2^0}$, and if $Z_r = z^{2^r}$ then $Z_{r+1} = (z^{2^r})^2 = z^{2^{r+1}}$. The recursion runs for $0 \leq r < n^2 - 1$ and so defines $Z_1, \ldots, Z_{n^2-1}$. By Lemma 4.1, $\beta_i(j) = (i-1)n + j - 1$ ranges over $\{0, \ldots, n^2 - 1\}$, so every index $\beta_i(j)$ is available, and $Z_{\beta_i(j)} = z^{2^{\beta_i(j)}} = z^{w_{ij}}$. For $n = 1$ the recursion is empty and $Z_0 = z = z^{w_{11}}$.

Block B. $B_{ij} = p_{ij} z^{w_{ij}}$ and $M_{ij} = B_{ij} q_j$, where the scalar B_{ij} is central, so $M_{ij} = p_{ij} z^{w_{ij}} x_j y_j$. The scalar p_{ij} appears once and the marker $z^{w_{ij}}$ once.

Block C. By induction on j, $U_{i,j} = \sum_{j' \le j} M_{ij'}$; for $j = n$ this is L_i. For $n = 1$ the block is empty and $L_1 = M_{11}$.

Block D. By induction on i, $\Psi_i = L_1 \cdots L_i$ (the left child is Ψ_{i-1} and the right child L_i, matching the order of the product). For $n = 1$, $\Psi_1 = L_1$. For $i = n$, $\Psi_n = \prod_{i=1}^{n} L_i = \mathscr{F}_A$ by definition of $\mathscr{F}_A$. □

Proposition 13.8 (Exact gate count over the enlarged algebra)**.** *The circuit $\widetilde{C}_A$ has exactly*

$$G_{\mathrm{ext}}(n) = 4n^2 + n - 2$$

gates, independently of A, with the entries p_{ij} counted as leaves.

Proof. Count block by block:

Gates	Number
markers $Z_1, \dots, Z_{n^2-1}$	$n^2 - 1$
pairs $q_j = x_j \times y_j$	n
entries $B_{ij} = p_{ij} \times Z_{\beta_i(j)}$	n^2
oriented entries $M_{ij} = B_{ij} \times q_j$	n^2
row sums ($n-1$ additions in each of the n rows)	$n(n-1)$
product of the rows	$n - 1$

The total is $(n^2 - 1) + n + n^2 + n^2 + n(n-1) + (n-1) = 4n^2 + n - 2$. For $n = 1, 2, 3$ this gives $3, 16, 37$ gates respectively; for $n = 1$ the three gates are q_1, B_{11} and M_{11}. □

Remark 13.9 (Computing the amplitudes). If the $p_{ij} = 1 + a_{ij}^2$ are not available as leaves but must be computed from the a_{ij}, each costs two further gates (a square and an addition of the constant 1), and the total becomes $4n^2 + n - 2 + 2n^2 = 6n^2 + n - 2$.

Proof of Theorem 1.4 (d). By Propositions 13.7 and 13.8, $\widetilde{C}_A$ computes $\mathscr{F}_A$ with $4n^2 + n - 2$ gates. Let $\mathcal{R}_n(F) = \gamma_n F$. Since $\gamma_n^2 = 1$ (Lemma 13.1) and $\mathscr{F}_A = \gamma_n P_A \widetilde{\Omega}$ (item (b)),

$$\mathcal{R}_n(\mathscr{F}_A) = \gamma_n \mathscr{F}_A = \gamma_n^2 P_A \widetilde{\Omega} = P_A\, \widetilde{\Omega}.$$

By Lemma 13.1, there are two cases. If $n \equiv 0, 1 \pmod 4$ then $\gamma_n = 1$ and $\mathcal{R}_n$ is the identity: no gate is added. If $n \equiv 2, 3 \pmod 4$ then $\gamma_n = -1$ and $\mathcal{R}_n$ is one multiplication by the constant -1: one gate is added. In both cases no exponentiation, expansion into monomials or coefficient extraction is performed. □

I now treat the spectral layer built on the pairs. Let K, N, ω, $d_1, \dots, d_n$, D^* be as in Theorem 1.2.

Definition 13.10 (Doubled spectral layer)**.** For $B \in K[z]^{n \times n}$ put $\widetilde{M}_{ij} = B_{ij}\, q_j \in \widetilde{\Lambda}_{K[z]}$ and

$$\widetilde{A}_i[f] = \sum_{k=0}^{N-1} \omega^{fk} \sum_{j:\, d_j \equiv k \ (N)} \widetilde{M}_{ij},$$

$$\widetilde{\Psi}_0[f] = 1, \qquad \widetilde{\Psi}_i[f] = \widetilde{\Psi}_{i-1}[f]\, \widetilde{A}_i[f],$$

$$\widetilde{T}[k] = \frac{1}{N} \sum_{f=0}^{N-1} \widetilde{\Psi}_n[f]\, \omega^{-fk}.$$

Proposition 13.11 (Doubled spectral extraction)**.** *Let $B = B_A$ read in $K[z]$, with K of characteristic 0 containing $\mathbb{Q}$, or $K = \mathbb{F}_Q$ as in Remark 10.3, in which case P_A is replaced by its reduction modulo Q. Then for every $N \geq 1$, every $d_1, \dots, d_n \in \mathbb{Z}$ (possibly negative), every f and every k,*

$$\widetilde{\Psi}_n[f] = \omega^{fD^*} \gamma_n \, P_A \, \widetilde{\Omega}, \qquad \widetilde{T}[k] = \mathbf{1}_{\{k \equiv D^* \ (N)\}} \, \gamma_n \, P_A \, \widetilde{\Omega}.$$

Proof. Step 1. Since $\omega^{fk} = \omega^{fd_j}$ whenever $d_j \equiv k \pmod N$, $\widetilde{A}_i[f] = \sum_j \widetilde{M}_{ij} \, \omega^{fd_j}$.

Step 2. The scalars ω^{fd_j} and B_{ij} and the elements q_j are central, so distributing the product gives

$$\widetilde{\Psi}_n[f] = \sum_{s:[n]\to[n]} \omega^{fD_s} \Big(\prod_i B_{i,s_i} \Big) \, q_{s_1} \cdots q_{s_n}, \qquad D_s = \sum_i d_{s_i}.$$

Step 3. Non-injective s contribute 0 by Lemma 13.4.

Step 4. For $s = \sigma \in S_n$, $D_\sigma = D^*$ by Lemma 10.1 and $q_{\sigma(1)} \cdots q_{\sigma(n)} = \gamma_n \widetilde{\Omega}$ by Lemma 13.5. Hence $\widetilde{\Psi}_n[f] = \omega^{fD^*} \gamma_n \big(\sum_\sigma \prod_i B_{i,\sigma(i)} \big) \widetilde{\Omega}$, and for $B = B_A$ the bracket is P_A.

Step 5. By Lemma 10.2 with $m = D^* - k$, $\widetilde{T}[k] = \gamma_n P_A \widetilde{\Omega} \cdot \frac{1}{N} \sum_f \omega^{f(D^*-k)} = \mathbf{1}_{\{k \equiv D^*\}} \gamma_n P_A \widetilde{\Omega}$. If $N = 1$ then $\omega = 1$ and the single slot contains everything; the formula holds with $k \equiv D^*$ always true. □

Remark 13.12 (The two filters). Theorem 1.2 and Proposition 13.11 treat the same assignments, the same addresses $E(\sigma)$ and the same spectral slot $D^* \bmod N$. The exterior algebra on the x_j produces $D_A\Omega$, where the monomial at the address e carries the sign $\chi_n(e)$; the doubled algebra produces $\gamma_n P_A \widetilde{\Omega}$, where every monomial carries the same sign γ_n. By Theorem 1.4 (b), the *only* difference between the two outputs, monomial by monomial, is the factor $\chi_n(e)\gamma_n$.

The uniform sign has a price, which the following propositions quantify. The determinantal identity of Section 7 is no longer available, and the natural scalar compilation of $\widetilde{C}_A$ is exponential.

Proposition 13.13 (The cost moves)**.** *Let $1 \leq k \leq n$ and $\mathscr{F}_{A,k} = \prod_{a=1}^{k} \big(\sum_j p_{aj} z^{w_{aj}} q_j \big)$, the partial product over the first k rows; this is the node Ψ_k of $\widetilde{C}_A$.*

(a) *$\mathscr{F}_{A,k} = \sum_{S \subseteq [n], |S|=k} F_{k,S}(z) \, q_S$ with*

$$F_{k,S}(z) = \sum_{\tau:[k]\to S \ \text{bijective}} \Big(\prod_{a=1}^{k} p_{a,\tau(a)} \Big) z^{E_k(\tau)} \neq 0,$$

where $E_k(\tau) = \sum_{a \leq k} w_{a,\tau(a)}$. Hence $\mathscr{F}_{A,k}$ has exactly $\binom{n}{k}$ nonzero coordinates in the basis (q_S), and $\binom{n}{\lfloor n/2 \rfloor} \geq 2^n/(n+1)$ for $k = \lfloor n/2 \rfloor$.

(b) *If $R \neq 0$ is a ring in which 2 is not a zero divisor and $n \geq 2$, there is no R-algebra homomorphism $\Lambda_R(x_1, \dots, x_n) \to \widetilde{\Lambda}_R$ with $x_j \mapsto q_j$ for all j.*

(c) *For $n = 2$ and A arbitrary, $P_A - D_A = 2p_{12}p_{21}z^6 \neq 0$ and $P_A + D_A = 2p_{11}p_{22}z^9 \neq 0$, so $D_A \neq \pm P_A$. In particular the top coefficient of $\mathscr{F}_A$ is not the determinant $\det B_A$ (Proposition 5.4), and the circuit of Section 7 does not compute it.*

Proof. (a) Distributing, $\mathscr{F}_{A,k} = \sum_{t:[k]\to[n]} \big(\prod_a p_{a,t(a)}\big) z^{E_k(t)} q_{t(1)} \cdots q_{t(k)}$. If t is not injective the term vanishes, by the argument of Lemma 13.4 (applied to a word of length k). If t is injective with image S, the q's commute, so $q_{t(1)} \cdots q_{t(k)} = q_S$ with no sign; grouping by S gives the formula, $\tau = t$ being a bijection $[k] \to S$. Each $F_{k,S}$ is a sum of $k!$ monomials with strictly positive coefficients and, by Lemma 9.2, pairwise distinct exponents; hence $F_{k,S} \neq 0$. The q_S are linearly independent by Lemma 13.3 (d), so the coordinates are exactly the $F_{k,S}$. The count is the number of k-subsets of $[n]$; the bound is Proposition 6.3.

(b) Such a homomorphism φ would satisfy $0 = \varphi(x_1x_2 + x_2x_1) = q_1q_2 + q_2q_1 = 2q_1q_2$. Now $q_1q_2 = -u_{\{1,2,n+1,n+2\}}$ (Lemma 13.3 (d) with $S = \{1,2\}$) is, up to sign, a basis element of a free module, so $2q_1q_2 = 0$ forces $2 \cdot 1 = 0$ in R. Since 2 is not a zero divisor, $2 \cdot 1 = 0$ gives $1 = 0$, contradicting $R \neq 0$.

(c) For $n = 2$, $D_A = p_{11}p_{22}z^9 - p_{12}p_{21}z^6$ and $P_A = p_{11}p_{22}z^9 + p_{12}p_{21}z^6$ with all $p_{ij} > 0$; the difference and the sum are as stated. Proposition 5.4 identifies the top coefficient of $\prod_i(\sum_j B_{ij}x_j)$ with $\det B$, whereas Theorem 1.4 (b) identifies that of $\mathscr{F}_A$ with $\gamma_n P_A$. □

Proposition 13.14 (Scalar compilation of the doubled circuit)**.** *Let $F_{0,\emptyset} = 1$. For $1 \le i \le n$ and $U \subseteq [n]$ with $|U| = i$,*

$$F_{i,U} = \sum_{j \in U} p_{ij} z^{w_{ij}} F_{i-1,U\setminus\{j\}},$$

and $P_A = F_{n,[n]}$. Consequently P_A is computed by a scalar arithmetic circuit over $\mathbb{Q}$ with at most

$$(n-1)\,2^n + 2n^2$$

gates, whose constants are the p_{ij} and the constant 1.

Proof. By Proposition 13.13 (a), $F_{i,U} = \sum_{\tau:[i]\to U \text{ bij.}} \prod_{a\le i} p_{a,\tau(a)} z^{E_i(\tau)}$. Split the bijections according to $j = \tau(i) \in U$: the remaining part is a bijection $[i-1] \to U \setminus \{j\}$ and contributes $F_{i-1,U\setminus\{j\}}$, while the factor for row i is $p_{ij}z^{w_{ij}}$. This is the recurrence. For $i = n$, Proposition 13.13 (a) gives $\mathscr{F}_A = F_{n,[n]}\, q_{[n]} = F_{n,[n]}\, \gamma_n \widetilde{\Omega}$ (Lemma 13.5), while Theorem 1.4 (b) gives $\mathscr{F}_A = \gamma_n P_A \widetilde{\Omega}$. Since $\widetilde{\Omega}$ is a basis of a free module of rank one and $\gamma_n = \pm 1$ is invertible, $F_{n,[n]} = P_A$.

For the circuit, compute the markers ($n^2 - 1$ gates, Lemma 7.1) and the n^2 entries $m_{ij} = p_{ij}z^{w_{ij}}$ (n^2 gates). For each pair (i, U) with $|U| = i$, the recurrence needs i multiplications and $i - 1$ additions, that is $2i - 1$ gates. There are $\binom{n}{i}$ such sets U. The total over $i = 1, \dots, n$ is

$$\sum_{i=1}^{n} (2i-1)\binom{n}{i} = 2 \cdot n2^{n-1} - (2^n - 1) = (n-1)2^n + 1,$$

using $\sum_i i\binom{n}{i} = n2^{n-1}$ and $\sum_{i\ge 1}\binom{n}{i} = 2^n - 1$. Adding $(n^2 - 1) + n^2$ gives at most $(n-1)2^n + 1 + 2n^2 - 1 = (n-1)2^n + 2n^2$ gates. □

By Proposition 13.13 (a), any evaluation of $\widetilde{C}_A$ that materializes the coordinates of its nodes is exponential, exactly as for the layers $\Psi_i[f]$ of Proposition 11.1; Proposition 13.14 gives the corresponding upper bound $(n-1)2^n + 2n^2$ for the scalar compilation. By Proposition 13.13 (b),(c), the determinantal route does not apply. This does not prove that no compact scalar circuit for P_A exists: it shows that the algebra-valued count $4n^2 + n - 2$ of Proposition 13.8 and the scalar size asked for in Conjecture 1.5 are different quantities.

Corollary 13.15 (Equivalent form of the sign-correction conjecture)**.** *Let $\Pi \in \mathbb{N}[X]$. For $n \geq 1$ and $A \in \mathbb{Q}^{n\times n}$, the polynomial $P_A = \mathcal{C}_n(D_A)$ is computed by a scalar arithmetic circuit over $\mathbb{Q}$ with at most $\Pi(n)$ gates, constants being entries p_{ij} or rationals of length at most $\Pi(n)$, if and only if the polynomial $\widetilde{c}_A = [\widetilde{\Omega}]\mathscr{F}_A$ is computed by such a circuit with at most $\Pi(n)+1$ gates, the constants being the same together with ± 1. Consequently Conjecture 1.5 holds if and only if the same statement holds with $\mathcal{C}_n(D_A)$ replaced by $[\widetilde{\Omega}]\mathscr{F}_A$ (for a possibly different polynomial Π).*

Proof. By Theorem 1.4 (b), $\widetilde{c}_A = \gamma_n P_A$ and $P_A = \gamma_n \widetilde{c}_A$, with $\gamma_n \in \{\pm 1\}$. Multiplying the output of a circuit by the constant γ_n costs at most one gate (none if $\gamma_n = 1$), and the constant ± 1 has length 1. Uniform constructibility transfers in the same way. □

Remark 13.16 (Status). The double orientation shows that the correction of the signs of D_A can be realized algebraically by a single universal sign γ_n, with an explicit circuit of $4n^2+n-2$ gates over the enlarged algebra and at most one further gate for the global correction, at no cost in addresses, amplitudes or spectral slot. It does not resolve the sign-correction problem: the gate count is algebra-valued, the natural scalar compilation has exponential size, and the question of a compact scalar circuit for the top coefficient of $\mathscr{F}_A$ is Conjecture 1.5.

14 A scalar encoding of P_A as a central coefficient

This section proves Theorem 1.7. The double orientation of Section 13 lives in an enlarged algebra. Here I give an encoding that is entirely scalar: P_A is obtained as a single coefficient of an explicit polynomial in two variables z and t, and that polynomial has a scalar circuit of $O(n^4)$ gates. The mechanism is to pair two determinants whose signs cancel exactly on the coefficient of one degree in t, after attaching to each cell an auxiliary label that separates the permutations. The labels are carried by the new variable t; they do not replace the addresses $z^{E(\sigma)}$.

Circuit model. In this section a scalar circuit is a circuit over $\mathbb{Q}$ in the two variables z and t, as in Section 3: the leaves are z, t and rational constants, the gates are $+$ and $\times$ of fan-in two, a subtraction or a multiplication by a constant costs one gate, and the size is the number of gates. A coefficient of a polynomial in z, t with respect to t is a polynomial in z.

Lemma 14.1 (Coefficient of the central label in a product of two determinants)**.** *Let $n \geq 1$, let R be a commutative ring, and let $b_{ij} \in R$ and $r_{ij} \in \mathbb{N}$ for $i, j \in [n]$. Put*

$$c_i = \max_{j\in[n]} r_{ij}, \qquad K = \sum_{i=1}^{n} c_i, \qquad \rho(\sigma) = \sum_{i=1}^{n} r_{i,\sigma(i)} \quad (\sigma \in S_n),$$

and

$$H = \det\left(b_{ij} t^{r_{ij}}\right)_{i,j\in[n]} \cdot \det\left(t^{c_i - r_{ij}}\right)_{i,j\in[n]} \in R[t].$$

Then:

(a) *all exponents $c_i - r_{ij}$ are non-negative integers, and*

$$H = \sum_{\sigma,\tau\in S_n} \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)\Big(\prod_{i=1}^{n} b_{i,\sigma(i)}\Big)\, t^{\,K+\rho(\sigma)-\rho(\tau)};$$

(b) $0 \le \rho(\sigma) \le K$ *for every* $\sigma \in S_n$*, so every exponent of* t *in* H *lies in* $\{0, 1, \dots, 2K\}$*;*

(c) $[t^K]H = \displaystyle\sum_{\substack{\sigma,\tau \in S_n \\ \rho(\sigma)=\rho(\tau)}} \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau) \prod_{i=1}^{n} b_{i,\sigma(i)}$*;*

(d) *if* ρ *is injective on* S_n*, then* $[t^K]H = \displaystyle\sum_{\sigma \in S_n} \prod_{i=1}^{n} b_{i,\sigma(i)}$*.*

Proof. (a) By definition of c_i, $c_i - r_{ij} \ge 0$. The Leibniz expansion of the first determinant is $\sum_\sigma \operatorname{sgn}(\sigma) \prod_i b_{i,\sigma(i)}\, t^{\rho(\sigma)}$, since $\prod_i t^{r_{i,\sigma(i)}} = t^{\rho(\sigma)}$. That of the second is $\sum_\tau \operatorname{sgn}(\tau)\, t^{\sum_i (c_i - r_{i,\tau(i)})} = \sum_\tau \operatorname{sgn}(\tau)\, t^{K-\rho(\tau)}$. Multiplying the two expansions gives the formula.

(b) Each $r_{i,\sigma(i)} \in [0, c_i]$, so $0 \le \rho(\sigma) \le \sum_i c_i = K$. Hence $K + \rho(\sigma) - \rho(\tau) \in [0, 2K]$.

(c) By (a), the term indexed by (σ, τ) has t-exponent K if and only if $\rho(\sigma) = \rho(\tau)$.

(d) If ρ is injective on S_n, the condition $\rho(\sigma) = \rho(\tau)$ means $\sigma = \tau$, and then $\operatorname{sgn}(\sigma)\operatorname{sgn}(\tau) = \operatorname{sgn}(\sigma)^2 = 1$. Only the diagonal pairs remain in (c). □

Lemma 14.1 (c) also shows that the injectivity of ρ cannot be dropped: without it, pairs (σ, τ) with $\sigma \ne \tau$ and $\rho(\sigma) = \rho(\tau)$ contribute with the sign $\operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$, which may be -1.

Specialize to $R = \mathbb{Q}[z]$ and $b_{ij} = p_{ij} z^{w_{ij}}$. Then $\prod_i b_{i,\sigma(i)} = \big(\prod_i p_{i,\sigma(i)}\big) z^{E(\sigma)}$ and $\sum_\sigma \prod_i b_{i,\sigma(i)} = P_A$.

Proposition 14.2 (Pairing for arbitrary separating labels)**.** *Let* $n \ge 1$*,* $A \in \mathbb{Q}^{n \times n}$ *and let* $r_{ij} \in \mathbb{N}$ *be such that*

$$\rho(\sigma) = \rho(\tau) \implies \sigma = \tau \qquad (\sigma, \tau \in S_n). \tag{6}$$

Let c_i*,* K *be as in Lemma 14.1 and*

$$H_A(t, z) = \det\left(p_{ij} z^{w_{ij}} t^{r_{ij}}\right) \det\left(t^{c_i - r_{ij}}\right) \in \mathbb{Q}[t, z].$$

Then $[t^K]H_A(t, z) = P_A(z)$*: every surviving term carries its amplitude* $\prod_i p_{i,\sigma(i)}$ *and its address* $z^{E(\sigma)}$*, with the sign* $+1$*.*

Proof. Lemma 14.1 (d) with $b_{ij} = p_{ij} z^{w_{ij}}$. □

Remark 14.3 (Using the addresses as labels)*.* Take $r_{ij} = w_{ij}$. Then $c_i = w_{i,n} = 2^{in-1}$, and (6) is the injectivity $E(\sigma) = E(\tau) \Rightarrow \sigma = \tau$ of Corollary 4.2. Since $p_{ij} z^{w_{ij}} t^{w_{ij}} = p_{ij} (zt)^{w_{ij}}$, the first determinant is $D_A(zt)$, and

$$H_A(t, z) = D_A(zt)\, \det\left(t^{\,2^{in-1} - w_{ij}}\right), \qquad K = \sum_{i=1}^{n} 2^{in-1} \ge 2^{n^2-1}.$$

This works, but the central index is of order 2^{n^2}. The base-n labels below bring it down to $n^n - 1$.

Lemma 14.4 (Base-n labels)**.** *Let* $n \ge 2$ *and* $r_{ij} = (j-1)n^{i-1}$*. Then:*

(a) $\rho(\sigma) = \sum_{i=1}^{n} (\sigma(i) - 1) n^{i-1}$*, and* ρ *is injective on all assignments* $s : [n] \to [n]$*, hence on* S_n*;*

(b) $c_i = (n-1)n^{i-1}$ *and* $K = n^n - 1$;

(c) $\rho(\sigma) \le K - 1$ *for every* $\sigma \in S_n$.

For $n = 1$, $r_{11} = 0$, $c_1 = 0$ *and* $K = 0 = n^n - 1$.

Proof. (a) The integer $\sum_i (s_i - 1)n^{i-1}$ has base-n digits $s_1 - 1, \dots, s_n - 1$, each in $\{0, \dots, n-1\}$. By uniqueness of the base-n expansion, the integer determines every digit, hence s. (b) $\max_j (j-1)n^{i-1} = (n-1)n^{i-1}$, and $K = (n-1)\sum_{i=1}^{n} n^{i-1} = (n-1)\frac{n^n-1}{n-1} = n^n - 1$. (c) Since $n \ge 2$ and σ is a bijection, σ is not the constant map n; pick i_0 with $\sigma(i_0) \le n-1$. Then $\rho(\sigma) \le K - n^{i_0-1} \le K - 1$. □

Put $u_i = t^{n^{i-1}}$ for $i \in [n]$. Then $t^{r_{ij}} = u_i^{j-1}$ and $t^{c_i - r_{ij}} = u_i^{n-j}$.

Lemma 14.5 (The second determinant is a Vandermonde product)**.** *For elements* $u_1, \dots, u_n$ *of a commutative ring,* $\det\left(u_i^{n-j}\right)_{i,j\in[n]} = \prod_{1\le i<\ell\le n}(u_i - u_\ell)$.

Proof. Let $V = (u_i^{n-j})_{i,j}$, with rows indexed by the variable i and columns by the exponent $n-j$. Its transpose V^{T} has rows indexed by j, with exponent $n-j$. Reversing the order of these n rows, $j \mapsto h = n+1-j$, gives the matrix $W = (u_i^{h-1})_{h,i}$, whose determinant is $\prod_{i<\ell}(u_\ell - u_i)$ by Lemma 9.1. Reversing n rows is a permutation with $\binom{n}{2}$ inversions, hence $\det W = (-1)^{\binom{n}{2}} \det V^{\mathsf{T}} = (-1)^{\binom{n}{2}} \det V$. Therefore $\det V = (-1)^{\binom{n}{2}} \prod_{i<\ell}(u_\ell - u_i) = \prod_{i<\ell}(u_i - u_\ell)$, since the product has $\binom{n}{2}$ factors. □

Now define, for $n \ge 2$ and $A \in \mathbb{Q}^{n\times n}$,

$$H_A(t,z) = \det\left(p_{ij} z^{w_{ij}} u_i^{j-1}\right) \prod_{1\le i<\ell\le n} (u_i - u_\ell), \qquad u_i = t^{n^{i-1}}, \tag{7}$$

$$\widehat{H}_A(t,z) = 1 + t^{2K} + H_A(t,z), \qquad K = n^n - 1. \tag{8}$$

Proof of Theorem 1.7(a). *Case* $n \ge 2$. By Lemma 14.4 the labels $r_{ij} = (j-1)n^{i-1}$ satisfy (6), with $c_i = (n-1)n^{i-1}$ and $K = n^n - 1$. By Lemma 14.5, $\det(t^{c_i - r_{ij}}) = \det(u_i^{n-j}) = \prod_{i<\ell}(u_i - u_\ell)$, and $t^{r_{ij}} = u_i^{j-1}$. So the polynomial H_A of (7) is the polynomial of Proposition 14.2, and $[t^K]H_A = P_A$.

Exponents. By Lemma 14.1 (a), the exponents of t in H_A are $K + \rho(\sigma) - \rho(\tau)$. By Lemma 14.4 (c), $\rho(\sigma) \le K - 1$, and $\rho(\tau) \ge 0$, hence the exponents are $\le 2K - 1$. Also $\rho(\sigma) \ge 0$ and $\rho(\tau) \le K - 1$, hence the exponents are ≥ 1. So every exponent of t in H_A lies in $\{1, \dots, 2K-1\}$.

Centering. Since $K = n^n - 1 \ge 3 > 0$, the exponents 0 and $2K$ differ from K, and H_A has no monomial at t^0 or t^{2K}. Hence $\widehat{H}_A$ has t-degree exactly $2K$ with leading coefficient 1 and constant term 1, and $[t^K]\widehat{H}_A = [t^K]H_A = P_A$, with $K = \frac{1}{2}\deg_t \widehat{H}_A$.

Case $n = 1$. Here $H_A = \det(p_{11} z\, t^0) \cdot 1 = p_{11} z = P_A$ (the empty Vandermonde product is 1) and $K = 0$, so $[t^0]H_A = P_A$ holds; the centering (8) is used only for $n \ge 2$. □

The case $n = 2$. Write $\alpha = p_{11}p_{22}z^9$ and $\beta = p_{12}p_{21}z^6$, so that $u_1 = t$, $u_2 = t^2$ and $K = 3$. Then

$$\det \begin{pmatrix} p_{11}z & p_{12}z^2\,t \\ p_{21}z^4 & p_{22}z^8\,t^2 \end{pmatrix} = \alpha\,t^2 - \beta\,t,$$
$$H_A(t,z) = (\alpha t^2 - \beta t)(t - t^2) = -\beta t^2 + (\alpha+\beta)t^3 - \alpha t^4,$$
$$\widehat{H}_A = 1 - \beta t^2 + (\alpha+\beta)t^3 - \alpha t^4 + t^6.$$

The degree is 6, the central index is 3, and $[t^3]\widehat{H}_A = p_{11}p_{22}z^9 + p_{12}p_{21}z^6 = P_A$. All exponents of t in H_A lie in $\{2,3,4\} \subseteq \{1,\dots,5\}$. For $n = 3$ one has $K = 26$, and for $n = 4$, $K = 255$; I checked the identity $[t^K]H_A = P_A$ and the exponent range $[1, 2K-1]$ by direct expansion, with exact rational arithmetic and randomly chosen integer matrices, for $n = 2,3,4,5$. These checks are consistency checks, not part of the proof.

I now count the gates. I use the exponentiation by repeated squaring in the following form.

Lemma 14.6 (Powers by squaring)**.** *For any node v and any integer $m \geq 1$, the power v^m is computed by at most $2\lfloor \log_2 m \rfloor$ gates added to the circuit.*

Proof. Write m in binary with $\ell = \lfloor \log_2 m \rfloor + 1$ digits. Scan the digits from the most significant one: start with v, and for each of the $\ell - 1$ following digits square the current value, and multiply it by v if the digit is 1. This uses $\ell - 1 = \lfloor \log_2 m \rfloor$ squarings and at most $\ell - 1$ multiplications by v. For $m = 1$ no gate is needed. □

Proposition 14.7 (Size of the circuit for $\widehat{H}_A$)**.** *For $n \geq 2$ the polynomials H_A and $\widehat{H}_A$ are computed by scalar circuits over $\mathbb{Q}$ in the variables z, t with at most*

$$2n^4 - n^3 + 7n^2 + 4n\log_2 n \;\leq\; 4n^4$$

gates, whose constants are the p_{ij}, the constants 1, 0, ± 1 and the rationals $-1/k$ $(1 \leq k \leq n-1)$ and $(-1)^{n+1}/n$ of the recurrence of Proposition 7.4. The binary encoding of the circuit has length $O(L + n^4 \log n)$.

Proof. The circuit has the following blocks.

Gates	Number
markers $Z_1, \dots, Z_{n^2-1}$	$n^2 - 1$
entries $b_{ij} = p_{ij} \cdot Z_{\beta_i(j)}$	n^2
$u_1 = t$, then $u_{i+1} = u_i^n$ for $i < n$	$\leq 2(n-1)\log_2 n$
powers u_i^k, $2 \leq k \leq n-1$ $(u_i^k = u_i^{k-1} \times u_i)$	$n(n-2)$
entries $g_{ij} = b_{ij} \cdot u_i^{j-1}$ $(j \geq 2;\ g_{i1} = b_{i1})$	$n(n-1)$
determinant of (g_{ij}) (Section 7)	$\leq 2n^4 - n^3 + 2n^2$
differences $u_i - u_\ell$ $(i < \ell)$ and their product	$n(n-1) - 1$
$H_A = \det \times \prod (u_i - u_\ell)$	1
t^{2K}	$\leq 2 + 2n\log_2 n$
$\widehat{H}_A = (1 + t^{2K}) + H_A$	2

The markers and the b_{ij} are as in Lemma 7.1. The powers $u_{i+1} = u_i^n$ use Lemma 14.6 with $m = n$, at most $2\lfloor \log_2 n \rfloor$ gates each. The determinant is computed over the commutative ring $\mathbb{Q}[t,z]$, in which $1, \dots, n$ are invertible, by Proposition 7.4, whose count does not depend on

the ring. The Vandermonde factor has $\binom{n}{2}$ differences and $\binom{n}{2} - 1$ multiplications to form their product. For t^{2K}, Lemma 14.6 gives at most $2\lfloor \log_2(2K) \rfloor \leq 2\log_2(2n^n) = 2 + 2n\log_2 n$ gates, since $2K < 2n^n$; the powers of t are thus never formed by $2K$ successive multiplications.

Summing, the coefficient of n^2 is $1+1+1+1+2+1 = 7$, the linear terms are $-2n-n-n = -4n$, the constants are $-1-1+1+2+2 = 3$, and the logarithmic terms are at most $2(n-1)\log_2 n + 2n\log_2 n \leq 4n\log_2 n$. Since $-4n+3 < 0$, the total is at most $2n^4 - n^3 + 7n^2 + 4n\log_2 n$. For the bound by $4n^4$: at $n = 2$ the left side is $32 - 8 + 28 + 8 = 60 \leq 64$; for $n \geq 3$, $\log_2 n \leq n - 1$ gives $4n^4 - (2n^4 - n^3 + 7n^2 + 4n\log_2 n) \geq 2n^4 + n^3 - 11n^2 + 4n > 0$.

Encoding: there are at most $4n^4$ gates and at most $n^2 + n + 4$ leaves, so node identifiers have $O(\log n)$ bits and the structure uses $O(n^4 \log n)$ bits; the constants p_{ij} have total length $\leq 2L + O(n^2)$ and the others $O(n \log n)$, as in Proposition 7.6. □

Proof of Theorem 1.7(b). Proposition 14.7. □

The size of the encoding is thus polynomial. What is not polynomial, by the generic method, is the extraction of its central coefficient.

Proposition 14.8 (Cost of coefficient extraction by truncation)**.** *Let $f \in \mathbb{Q}[t, z]$ be computed by a scalar circuit of size s, and let $K \geq 0$. Then the coefficient $[t^K]f \in \mathbb{Q}[z]$ is computed by a circuit in the variable z alone, of size at most $s\,(K+1)^2$.*

Proof. Associate to every node v of the circuit the vector $(v_0, \ldots, v_K)$ of the coefficients of $t^0, \ldots, t^K$ of the polynomial it computes. The leaf t gives $(0, 1, 0, \ldots, 0)$, a constant c gives $(c, 0, \ldots, 0)$, the leaf z gives $(z, 0, \ldots, 0)$, all computed without gates. A $+$ gate is simulated by $K+1$ additions. A $\times$ gate is simulated by the truncated convolution $w_k = \sum_{a=0}^{k} u_a v_{k-a}$ for $0 \leq k \leq K$, which uses $\sum_{k=0}^{K}(k+1)$ multiplications and $\sum_{k=0}^{K} k$ additions, that is $(K+1)^2$ gates. Each gate of the original circuit is replaced by at most $(K+1)^2$ gates, and the component K of the output node is $[t^K]f$. □

For $f = \widehat{H}_A$ one has $s \leq 4n^4$ and $K + 1 = n^n$, so the truncation method gives at most $4n^4\, n^{2n}$ gates, which is not polynomial in n. This is an upper bound for one method, not a lower bound on the complexity of the coefficient.

Proposition 14.9 (Range of separating labels)**.** *Let $r_{ij} \in \mathbb{N}$ satisfy (6). Then $K \geq n! - 1$. In particular no choice of non-negative integer labels with this property brings the central index K down to a polynomial in n, and the index $n^n - 1$ of Lemma 14.4 is within a factor of $n^n/n! \leq e^n$ of this lower bound.*

Proof. By Lemma 14.1 (b), $\rho(\sigma) \in \{0, \ldots, K\}$ for each of the $n!$ permutations, and by (6) these $n!$ values are pairwise distinct. Hence $n! \leq K + 1$. For the comparison, $n^n/n! \leq e^n$ follows from $e^n = \sum_{k \geq 0} n^k/k! \geq n^n/n!$. □

This statement concerns the labels required by the pairing mechanism of Proposition 14.2; it does not bound the size of other circuits for P_A, and another encoding could use cancellations that do not separate the permutations one by one.

Corollary 14.10 (The conjecture is equivalent to the computation of the central coefficient)**.** *Let $\Pi \in \mathbb{N}[X]$. For $n \geq 2$ and $A \in \mathbb{Q}^{n \times n}$, the polynomial P_A is computed by a scalar circuit over $\mathbb{Q}$ in z with at most $\Pi(n)$ gates, the constants being entries p_{ij} or rationals of length at*

most $\Pi(n)$*, if and only if the central coefficient* $[t^K]\widehat{H}_A$ *is. Consequently Conjecture 1.5 holds if and only if the central coefficient* $[t^K]\widehat{H}_A$*,* $K = n^n - 1$*, of the polynomial of Theorem 1.7, which has a circuit of size* $O(n^4)$*, has a circuit of polynomial size (for a possibly different* Π*).*

Proof. By Theorem 1.7 (a), $[t^K]\widehat{H}_A = P_A$ for $n \geq 2$; the two statements are the same statement about the same polynomial. □

Hence the construction of $\widehat{H}_A$ is complete and polynomial, and the only step missing to close Conjecture 1.5 is the computation of the central coefficient $[t^{n^n-1}]\widehat{H}_A$ by a scalar circuit of polynomial size. Corollary 14.10 is an exact reformulation; it does not prove the conjecture.

Remark 14.11 (What would suffice). Two properties would suffice to close the step; neither is established here.

(i) An encoding with a polynomial index. Suppose that $J_A \in \mathbb{Q}[t, z]$ is computed by a scalar circuit of size $s(n) = \mathrm{poly}(n)$, with $[t^{k(n)}]J_A = P_A$ and $k(n) = \mathrm{poly}(n)$. Then Proposition 14.8 gives a circuit for P_A of size $s(n)(k(n)+1)^2 = \mathrm{poly}(n)$. By Proposition 14.9, such an encoding cannot come from the pairing mechanism of Proposition 14.2.

(ii) A recurrence for the coefficients. Suppose that there exist $r = \mathrm{poly}(n)$, vectors $\mathbf{u}, \mathbf{v} \in \mathbb{Q}[z]^r$ and a matrix $M \in \mathbb{Q}[z]^{r\times r}$, given by a circuit with s_0 gates, such that $[t^k]\widehat{H}_A = \mathbf{u}^{\mathsf{T}} M^k \mathbf{v}$ for $k = K$. Computing M^K by repeated squaring takes at most $2\lfloor \log_2 K \rfloor$ products of $r \times r$ matrices, each with $2r^3 - r^2$ gates, and the final product by $\mathbf{u}^{\mathsf{T}}$ and $\mathbf{v}$ takes at most $2r^2$ gates; the total is at most $s_0 + 2\lfloor \log_2 K \rfloor (2r^3 - r^2) + 2r^2$, which is polynomial in n since $\log_2 K < n \log_2 n$. I have not established such a recurrence for $\widehat{H}_A$.

Remark 14.12 (Relation to the literature). The identity of Proposition 14.2 is of Hadamard-product type: two polynomials with small circuits are multiplied, and the coefficients carrying the same label are paired [12, 13]. The question of extracting a coefficient of a polynomial of exponential degree given by a small circuit is the one studied for division and truncation of circuits [14]. I have not verified that the hypotheses of those results hold for the family $\widehat{H}_A$, and I draw no conclusion from them here.

15 The conditional polynomial pipeline

This section proves Theorem 1.6. Let $K \supseteq \mathbb{Q}$ be a field of characteristic 0 containing a primitive N-th root of unity ω (e.g. $K = \mathbb{Q}(\zeta_N)$), extend $\mathcal{C}_n$ K-linearly to $K[z]$ (Definition 12.1), and let $\mathcal{C}_n \otimes \mathrm{id}_\Lambda$ act on $\Lambda = \Lambda_{K[z]}$ by acting on the coefficient of Ω in the top component, i.e. $(\mathcal{C}_n \otimes \mathrm{id})(g\,\Omega) = \mathcal{C}_n(g)\,\Omega$ for $g \in K[z]$.

Proof of Theorem 1.6. (a). By Proposition 12.2, $P_A = \mathcal{C}_n(D_A) = \sum_\sigma \big(\prod_i p_{i,\sigma(i)}\big) z^{E(\sigma)}$, with $n!$ monomials, distinct by Theorem 1.1 (ii), and strictly positive coefficients since $p_{ij} > 0$. The circuit bound $\Pi(n)$ is Conjecture 1.5.

(b). By Theorem 1.2 with $B = B_A \in \mathbb{Q}[z]^{n\times n} \subseteq K[z]^{n\times n}$, $T[k] = \mathbf{1}_{\{k \equiv D^*\}} D_A\,\Omega$ in $\Lambda_{K[z]}$, where $D_A \in \mathbb{Q}[z] \subseteq K[z]$. Since $\mathcal{C}_n$ is K-linear and preserves $\mathbb{Q}[z]$, $T^+[k] = (\mathcal{C}_n \otimes \mathrm{id})(T[k]) = \mathbf{1}_{\{k\equiv D^*\}}\,\mathcal{C}_n(D_A)\,\Omega = \mathbf{1}_{\{k\equiv D^*\}} P_A\,\Omega$. For $k \equiv D^*$ the coordinate is P_A, computed by the circuit of (a), read over $K \supseteq \mathbb{Q}$ (a circuit over $\mathbb{Q}$ is a circuit over K with the same constants); for $k \not\equiv D^*$ it is 0. Over the N slots the total is at most $N\,\Pi(n)$ gates.

(c). The markers and the filter are polynomial by Theorem 1.1 (v),(vi); the spectral projection by Theorem 1.3, with at most $N(2n^4 - n^3 + 3n^2 + 2) + n^2$ gates, or $G(n) < 3n^4$ after

collapsing the N determinants; and the extraction of the coefficient at a prescribed address by Proposition 8.1. None of these uses Conjecture 1.5.

(d). By Proposition 13.11, $\widetilde{T}[k] = \mathbf{1}_{\{k \equiv D^*\}} \gamma_n P_A \widetilde{\Omega}$. For $k \equiv D^*$ the coordinate is $\gamma_n P_A$, obtained from the circuit of (a) by one multiplication by the constant $\gamma_n = \pm 1$, hence with at most $\Pi(n) + 1$ gates; for $k \not\equiv D^*$ it is 0.

Rectangular case. If Conjecture 12.4 holds, the same argument as in (a), with Proposition 12.2, gives a circuit with at most $\Pi(n)$ gates for $\sum_e |[z^e] D_A^{(r)}| z^e$ for all $1 \le r \le n$. □

In terms of the sequence of stages, the pipeline is therefore

$A \longmapsto B_A$	marking: $O(n^2)$ gates,
$B_A \longmapsto D_A$	filtering: $G(n) < 3n^4$ gates,
$D_A \longmapsto T[D^* \bmod N] = D_A\,\Omega$	spectral projection: at most $G(n)$ gates after collapsing,
$D_A \longmapsto P_A$	sign correction: $\Pi(n)$ gates under Conjecture 1.5,

and every stage except the last is unconditionally polynomial.

16 Scope, limits and what is not claimed

(1) *Universality of the rational filter.* For $p_{ij} = 1 + a_{ij}^2$, Theorem 1.1 holds for every finitely encoded $A \in \mathbb{Q}^{n \times n}$, dense or sparse, structured or not. The value $p_{ij} \ge 1$ is used only to exclude vanishing coefficients; Remark 5.6 explains how to keep the amplitudes a_{ij}.

(2) *Circuit versus expansion.* A circuit of $O(n^4)$ gates is not a list of $n!$ monomials; the latter needs at least $n!(n^2 - n + 1)$ bits for the addresses alone (Proposition 6.2). The compactness of Theorem 1.1 (v) is a statement about circuits, not about any expanded form.

(3) *Numerical evaluation.* Specializing z to a number can destroy the separation of the addresses; for instance $z = 1$ sends D_A to $\det(p_{ij})$, and the individuality of the monomials is lost. The variable z must stay formal, or be evaluated in a setting where individual monomials are no longer required.

(4) *Reductions modulo a prime.* Over $\mathbb{F}_Q$ the support of $\bar{D}_A$ may shrink if some $\bar{p}_{ij} = 0$ (Remark 5.6); Theorem 1.2 holds for $\bar{D}_A$ regardless.

(5) *The spectral layer.* Theorem 1.2 is exact in the exterior algebra. Its circuit version (Theorem 1.3) is polynomial in n and N; the naive layer-by-layer evaluation is not (Proposition 11.1). The size bound is polynomial in n only if N is polynomial in n.

(6) *Status of the sign correction.* Conjecture 1.5 is not proved. Nothing stated as a theorem in this paper depends on it except Theorem 1.6, which is explicitly conditional, and no polynomial-size circuit for the corrected polynomial P_A is claimed unconditionally. The double orientation of Section 13 gathers all the signs into the single constant γ_n, but it does not give a compact circuit for the resulting top coefficient: it reformulates the conjecture rather than resolving it. The gate count $4n^2 + n - 2$ of the doubled circuit is a count of gates of the enlarged exterior algebra and must not be confused with the size of a scalar circuit over $\mathbb{Q}[z]$; the scalar compilation of Proposition 13.14 has size $(n-1)2^n + 2n^2$. The

central-coefficient encoding of Section 14 reduces the conjecture, exactly, to the polynomial-size computation of one coefficient of index $n^n - 1$ of a polynomial having a circuit of $O(n^4)$ gates; it is an equivalent reformulation, and the truncation method costs $s(K+1)^2$ gates, which is not polynomial.

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A Summary of the identities

$$E(s) = \sum_i 2^{(i-1)n + s_i - 1}, \qquad E(s) = E(t) \iff s = t,$$

$$D_A(z) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \Big(\prod_i p_{i,\sigma(i)} \Big) z^{E(\sigma)},$$

$$\deg_z D_A = \frac{2^{n(n+1)} - 1}{2^{n+1} - 1},$$

$$G(n) \le 2n^4 - n^3 + 4n^2 - 1 < 3n^4, \qquad \operatorname{bits}(C_A) = O(L + n^4 \log n),$$

$$D_\sigma = D^* \ \ (\sigma \in S_n), \qquad \det\big(B_A \operatorname{diag}(\omega^{f d_j})\big) = \omega^{f D^*} D_A,$$

$$A_i[f] = \sum_j M_{ij} \omega^{f d_j}, \qquad \Psi_n[f] = \prod_i A_i[f] = \omega^{f D^*} D_A\, \Omega,$$

$$T[k] = \frac{1}{N} \sum_f \Psi_n[f] \omega^{-fk} = \sum_{s:\, D_s \equiv k} \prod_i M_{i,s_i} = \mathbf{1}_{\{k \equiv D^*\}} D_A\, \Omega,$$

$$\mathcal{C}_n(z^e) = \chi_n(e) z^e, \qquad \mathcal{C}_n(D_A) = \sum_e \big|[z^e] D_A\big| z^e,$$

$$q_j = x_j y_j, \qquad q_j^2 = 0, \qquad q_i q_j = q_j q_i, \qquad \gamma_n = (-1)^{n(n-1)/2},$$

$$q_{\sigma(1)} \cdots q_{\sigma(n)} = \gamma_n \widetilde{\Omega}, \qquad \mathscr{F}_A = \gamma_n P_A \widetilde{\Omega},$$

$$G_{\mathrm{ext}}(n) = 4n^2 + n - 2, \qquad \mathcal{R}_n(\mathscr{F}_A) = \gamma_n \mathscr{F}_A = P_A \widetilde{\Omega},$$

$$\text{scalar compilation of } \widetilde{C}_A : \le (n-1)2^n + 2n^2,$$

$$H_A = \det\big(p_{ij} z^{w_{ij}} u_i^{j-1}\big) \prod_{i<\ell} (u_i - u_\ell), \qquad u_i = t^{n^{i-1}},$$

$$K = n^n - 1, \qquad \widehat{H}_A = 1 + t^{2K} + H_A, \qquad [t^K] \widehat{H}_A = P_A,$$

$$\text{size of } \widehat{H}_A \le 2n^4 - n^3 + 7n^2 + 4n\log_2 n, \qquad K \ge n! - 1 \text{ for any separating labels.}$$

The circuit size of $\mathcal{C}_n(D_A)$ is at most $\Pi(n)$ if Conjecture 1.5 holds.

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B Postfilter Hadamard coupling and scalarization

Addendum of October 17, 2026.

This appendix connects the corrected output of the oriented collision filter to the spectral computation of the paper. The input to the new block is the actual output wire

$$V_A = \gamma_n F_A = P_A(z)\widetilde{\Omega}.$$

The common-phase identity makes a pointwise product sufficient to construct the frequency table after filtering. I give the circuit, prove each intermediate identity, count its gates, and prove a gate-by-gate scalarization of all subsequent operations.

The complexity statement has two distinct parts. The added spectral block and its scalarization have polynomial overhead. A polynomial scalar circuit producing the initial coordinate of V_A from A is still the hypothesis of Conjecture 1.5. The coefficient map defined below is proved explicitly, together with its cost on an exposed scalar coordinate; it is not inserted as an unimplemented arithmetic gate. Section B.6 recalls the actual scalar compilation already available in Proposition 13.14 and calculates the resulting total bound. Sections B.10 and B.11 add exact hash-memory accounting and two global formulations aimed at a replacement of the scalar-source bound.

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B.1 Input, notation and circuit model

Let $n \geq 1$, let $A = (a_{ij}) \in \mathbb{Q}^{n\times n}$ have total binary encoding length L, and retain the definitions of the paper:

$$p_{ij} = 1 + a_{ij}^2, \qquad w_{ij} = 2^{(i-1)n+j-1}, \qquad B_{ij} = p_{ij} z^{w_{ij}}. \tag{B.1}$$

The variable z is formal unless an evaluation is explicitly specified. The elements

$$q_j = x_j y_j, \qquad \widetilde{\Omega} = x_1 \cdots x_n y_1 \cdots y_n, \qquad \gamma_n = (-1)^{n(n-1)/2}$$

belong to the enlarged exterior algebra. By Theorem 1.4, the q_j commute with one another and with all scalar coefficients, $q_j^2 = 0$, and $q_1 \cdots q_n = \gamma_n \widetilde{\Omega}$. The already constructed filter has output

$$F_A = \prod_{i=1}^{n} \left(\sum_{j=1}^{n} B_{ij} q_j \right), \qquad V_A = \gamma_n F_A = P_A(z) \widetilde{\Omega}. \tag{B.2}$$

Equation (B.2) specifies the value of an existing output wire. In the appended circuit, P_A is not supplied as an extra constant or an oracle.

Let $\mathbb{K}$ be a field containing a primitive Nth root of unity ω, where $N \geq 1$ and N is invertible in $\mathbb{K}$. When $N = 1$, set $\omega = 1$. For characteristic zero, $\mathbb{K}$ contains $\mathbb{Q}$. In finite characteristic, the entries in (B.1) are read modulo an admissible prime as specified in Section B.7. The same identities then concern the reduced polynomial.

For arbitrary integer column labels $d_1, \ldots, d_n$, including negative labels, define

$$D^* = \sum_{j=1}^{n} d_j, \qquad k_* = D^* \bmod N, \quad 0 \leq k_* < N, \qquad g = \omega^{k_*}. \tag{B.3}$$

All frequency and output indices below range over $\{0, \ldots, N-1\}$. In particular, the output slot is the residue k_*.

Definition B.1 (The coefficient module and legal downstream gates)**.** Put $\mathcal{M} = \mathbb{K}[z]\widetilde{\Omega} \subseteq \widetilde{\Lambda}_{\mathbb{K}[z]}$. A downstream circuit uses scalar gates in $\mathbb{K}[z]$ and module gates of the forms $U + V$ and rU, where $U, V \in \mathcal{M}$ and $r \in \mathbb{K}[z]$. Scalars are central. Every arithmetic gate has fan-in two. Multiplication by a constant is counted as a gate; a subtraction is implemented by a multiplication by -1 followed by an addition. Copies and output references are wires and do not count as arithmetic gates. Scalar leaves are z, the original scalar input entries, and the explicitly supplied field constants. Every other scalar coefficient r is computed by a scalar sub-DAG included in the gate count. An arbitrary polynomial in z is not introduced as a free constant.

The representation is a directed acyclic graph (DAG), so a computed value can feed several gates without recomputing its source. The number of gates in an algebra-valued DAG and the number of gates in a scalar DAG are different measures, as already stated in Definition 13.6.

B.2 Common phase after the oriented filter

For compatibility with Definition 13.10, write the original frequency-dependent row value as

$$\widetilde{A}_i[f] = \sum_{j=1}^{n} B_{ij} q_j \omega^{f d_j}.$$

The calculation below verifies why its corrected final output can be obtained by applying a single phase to V_A after the filter. It is an identity proof, not an instruction to enumerate assignments.

Lemma B.2 (A common phase on the surviving output)**.** *For every frequency f,*

$$\gamma_n \prod_{i=1}^{n} \widetilde{A}_i[f] = \omega^{fD^*} V_A = \omega^{fD^*} P_A(z) \widetilde{\Omega}. \tag{B.4}$$

Proof. In a distributed term, let $s(i)$ be the column selected in row i. If s is not injective, some column j is selected at least twice. Because the q_j commute, the marker product contains $q_j^2 = 0$, and this term is zero. These are precisely the terms annihilated by the existing filter.

A surviving selection is a permutation σ. Since each column occurs once, its phase satisfies

$$\begin{aligned} \prod_{i=1}^{n} \omega^{f d_{\sigma(i)}} &= \omega^{f(d_{\sigma(1)} + \cdots + d_{\sigma(n)})} \\ &= \omega^{f(d_1 + \cdots + d_n)} = \omega^{fD^*}. \end{aligned}$$

Its marker product is $q_{\sigma(1)} \cdots q_{\sigma(n)} = q_1 \cdots q_n = \gamma_n \widetilde{\Omega}$. The scalar amplitudes and the powers of z are the same as those in F_A. Thus every nonzero term of the modulated output is its unmodulated counterpart multiplied by ω^{fD^*}. Factoring out this common scalar gives $\prod_i \widetilde{A}_i[f] = \omega^{fD^*} F_A$. Multiplication by γ_n yields (B.4). □

Equivalently, the substitution

$$x_j \longmapsto \omega^{f d_j} x_j, \qquad y_j \longmapsto y_j$$

fixes $\mathbb{K}[z]$ and extends to an automorphism of the enlarged exterior algebra. It preserves all defining exterior relations, and the substitution with the inverse scalar factors is its inverse. It sends q_j to $\omega^{f d_j} q_j$ and hence sends $\widetilde{\Omega}$ to $\omega^{fD^*} \widetilde{\Omega}$. Its restriction to $\mathcal{M}$ is therefore the scalar action $V \mapsto \omega^{fD^*} V$.

Definition B.3 (Pointwise coupling after filtering)**.** Define $H[f] = \omega^{fD^*}$ and $U_A[f] = V_A$. Their pointwise product is

$$\mathbf{Y}_A = \mathbf{H} \odot \mathbf{U}_A, \qquad Y_A[f] = H[f] U_A[f]. \tag{B.5}$$

The frequency index labels the coordinates of this table. The multiplication in each coordinate is scalar multiplication of the already filtered output.

By Lemma B.2, (B.5) is exactly the corrected frequency table of Proposition 13.11. This identity allows the spectral modulation to be placed after the filter without changing any amplitude, address or output slot.

B.3 Explicit shared circuit and uniformity

Proposition B.4 (A circuit for the complete frequency table)**.** *Compute g from* (B.3)*, set $Y_A[0] = V_A$, and append the gates*

$$Y_A[f+1] = g\,Y_A[f] \qquad (0 \le f < N-1). \tag{B.6}$$

Then every output is

$$Y_A[f] = g^f V_A = \omega^{fD^*} P_A(z)\widetilde{\Omega}.$$

Proof. The initial wire satisfies $Y_A[0] = g^0 V_A$. If $Y_A[f] = g^f V_A$, the next gate computes $Y_A[f+1] = g(g^f V_A) = g^{f+1} V_A$. Induction proves the first equality for all output indices. There is an integer r such that $D^* = k_* + rN$; hence

$$\omega^{fD^*} = \omega^{fk_*}(\omega^N)^{fr} = \omega^{fk_*} = g^f.$$

This also holds for negative r, since ω is invertible. Substitute $V_A = P_A(z)\widetilde{\Omega}$ to obtain the stated value. □

Proposition B.5 (Exact displayed count and additive composition)**.** *Define*

$$\ell_{\mathrm{bin}}(0) = 0, \qquad \ell_{\mathrm{bin}}(k) = \lfloor \log_2 k \rfloor + \mathrm{popcount}(k) - 1 \quad (k \ge 1).$$

Binary exponentiation for g, followed by (B.6)*, uses exactly*

$$h(N, k_*) = \ell_{\mathrm{bin}}(k_*) + N - 1 \tag{B.7}$$

multiplication gates before simplification and no addition gates. If the input DAG producing V_A has s_V gates, the combined DAG has at most $s_V + h(N, k_*)$ gates in the same algebraic model.

Proof. For $k_* = 0$, the value of g is the constant 1 and no exponentiation gate is required. For $k_* \ge 1$, write

$$k_* = (1 b_{t-1} \cdots b_0)_2, \qquad t = \lfloor \log_2 k_* \rfloor.$$

Initialize the accumulator to the input ω. For each of the t remaining bits, square the accumulator; if that bit is 1, multiply once more by ω. The exponent is updated from a to $2a + b$ at each step, so the final exponent is k_*. The number of squarings is t, and the number of other multiplications is $\mathrm{popcount}(k_*) - 1$.

There are $N - 1$ recurrence gates, one for each new table value after $Y_A[0]$. The initial value is a reference to the upstream output. The upstream DAG is included once. Appending gates to its output therefore adds their number to s_V and does not multiply s_V by N. □

For $N \ge 2$,

$$h(N, k_*) \le N - 1 + 2\lfloor \log_2(N-1) \rfloor \le N - 1 + 2\lceil \log_2 N \rceil.$$

For $N = 1$ the table is the single wire V_A and the added count is zero. If $k_* = 0$ with $N > 1$, the displayed recurrence multiplies by 1; after constant folding all its outputs can refer directly to V_A.

Corollary B.6 (Composition with the existing oriented circuit)**.** *Let $\varepsilon_n = 1$ when $n \equiv 2, 3 \pmod 4$ and $\varepsilon_n = 0$ otherwise. With the p_{ij} as leaves, the complete algebra-valued frequency-table circuit has size at most*

$$4n^2 + n - 2 + \varepsilon_n + \ell_{\mathrm{bin}}(k_*) + N - 1. \tag{B.8}$$

Starting from the a_{ij} instead adds $2n^2$ gates.

Proof. Proposition 13.8 gives $4n^2+n-2$ gates for F_A. Multiplying by the global sign costs exactly ε_n additional gates in the stated construction. Apply Proposition B.5. Finally, computing $p_{ij} = 1 + a_{ij}^2$ takes one square and one addition per entry. □

Uniformity and description length. Let ℓ_d be the total binary length of the integer labels. A constructor first computes their sum and its residue modulo N. Integer addition and Euclidean division do this in time polynomial in $\ell_d + \log(n+1) + \log(N+1)$. It then reads the bits of k_*, emits the exponentiation gates, emits $N-1$ recurrence gates in increasing frequency order, and writes the N output references.

If the original DAG has encoding length B_V, node identifiers in the combined DAG require $O(\log(s_V + N + \log(N+1) + 2))$ bits. The added graph and output references therefore have length

$$O((N + \log(N+1)) \log(s_V + N + \log(N+1) + 2)) .$$

Together with B_V, the label encoding and the supplied field data, this bounds the construction time by a polynomial in their lengths and in N. This is a bound in N, not just in $\log N$, because the requested table contains N output references.

For the concrete choice

$$N = 2^{\lceil \log_2(n+1) \rceil}, \qquad n < N \le 2n, \tag{B.9}$$

the added frequency-table size is $O(n)$. To check the upper inequality, put $N = 2^m$. Minimality gives $2^{m-1} < n+1$; both sides are integers, so $2^{m-1} \le n$ and $N \le 2n$. No requirement on separating individual surviving permutations is needed for this choice: filtering has already taken place.

B.4 Inverse transform and its exact cost

Proposition B.7 (Exact support after the inverse transform)**.** *With the normalization of the paper, put*

$$Z_A[k] = \frac{1}{N} \sum_{f=0}^{N-1} \omega^{-fk} Y_A[f].$$

Then

$$Z_A[k] = \delta_{k,k_*} V_A = \delta_{k,k_*} P_A(z) \widetilde{\Omega}. \tag{B.10}$$

Proof. Substitution of Proposition B.4 gives

$$\begin{aligned} Z_A[k] &= \frac{1}{N} \sum_{f=0}^{N-1} \omega^{-fk} \omega^{fk_*} V_A \\ &= \frac{1}{N} \left(\sum_{f=0}^{N-1} \omega^{f(k_*-k)} \right) V_A. \end{aligned}$$

If $k = k_*$, all N summands in parentheses are 1. Their sum is N, and multiplication by N^{-1} yields V_A. If $k \neq k_*$, exact order N and the chosen representatives imply $r = \omega^{k_*-k} \neq 1$. Also $r^N = 1$. The identity

$$(r-1)(1 + r + \cdots + r^{N-1}) = r^N - 1 = 0$$

and invertibility of $r - 1$ in the field give $1 + r + \cdots + r^{N-1} = 0$. Therefore $Z_A[k] = 0$. For $N = 1$, the only output is the input and the conclusion is immediate. □

Thus the composition has exactly the slot and amplitude proved in Proposition 13.11, including the already applied sign correction. All intermediate sums and scalar products in this inverse transform remain in $\mathcal{M}$.

Proposition B.8 (Explicit inverse-transform circuit)**.** *Suppose $N = 2^m \geq 2$. When the required powers of ω and the constant N^{-1} are available, the inverse transform of an arbitrary table in $\mathcal{M}^N$ has a circuit with at most*

$$b_{\text{inv}}(N) = 2N \log_2 N + N \tag{B.11}$$

gates of the forms in Definition B.1. All powers of ω can alternatively be generated from $1, \omega$ with at most $N - 2$ additional multiplication gates.

Proof. First compute the inverse transform without its factor $1/N$. For $0 \leq k < N/2$, separate the even and odd input indices:

$$E_k = \sum_{j=0}^{N/2-1} (\omega^2)^{-jk} Y_A[2j], \qquad O_k = \sum_{j=0}^{N/2-1} (\omega^2)^{-jk} Y_A[2j+1].$$

The first output in a pair is

$$E_k + \omega^{-k} O_k.$$

For the output indexed by $k + N/2$, the even-index terms are unchanged. The odd-index terms acquire the factor $\omega^{-N/2} = -1$. Indeed $\omega^{N/2}$ has order two; in a field with N invertible it equals -1. Hence the second output is

$$E_k - \omega^{-k} O_k.$$

The root ω^2 has exact order $N/2$, so each of E and O is an inverse transform of length $N/2$ of the same kind.

For each pair, form $t = \omega^{-k} O_k$, form $-t = (-1)t$, and form $E_k + t$ and $E_k + (-t)$. These are four fan-in-two gates. If $T(N)$ denotes the unnormalized count, then

$$T(1) = 0, \qquad T(N) \le 2T(N/2) + 2N.$$

There are $\log_2 N$ recursion levels. Each level has $N/2$ butterflies and costs $2N$ gates, so $T(N) \le 2N \log_2 N$. Multiplication of each of the N outputs by N^{-1} adds N gates. This proves (B.11).

For the constant pool, take $W_0 = 1$, $W_1 = \omega$ and append $W_{j+1} = \omega W_j$ for $1 \le j < N - 1$. These are $N - 2$ gates and produce $W_j = \omega^j$. Every twiddle at every recursion level is one of these values: negative powers are read using their residues modulo N. The pool is shared by all butterflies. □

Corollary B.9 (One bound including phase preparation)**.** *For $N = 2^m \ge 2$, the complete postfilter frequency-table and inverse-transform block can be built from $1, \omega, -1, N^{-1}$ with at most*

$$B(N) = 2N \log_2 N + 3N - 3 \tag{B.12}$$

additional arithmetic gates.

Proof. Prepare the pool W_j with $N - 2$ gates and select $g = W_{k_*}$ by a wire. Use the $N - 1$ gates of (B.6). Apply the inverse circuit of Proposition B.8, which uses the same pool. The sum is

$$(N - 2) + (N - 1) + (2N \log_2 N + N) = 2N \log_2 N + 3N - 3.$$

The pool replaces binary exponentiation here; its cost is not added to a second charge for computing g. □

B.5 Scalarization, proved gate by gate

Lemma B.10 (The normalized scalar coordinate)**.** *The map*

$$\lambda : \mathcal{M} \longrightarrow \mathbb{K}[z], \qquad \lambda(c(z)\widetilde{\Omega}) = c(z) \tag{B.13}$$

is a well-defined $\mathbb{K}[z]$-linear isomorphism. Its inverse is $\iota(c) = c\widetilde{\Omega}$.

Proof. The ordered exterior monomials form a free $\mathbb{K}[z]$-basis of $\widetilde{\Lambda}_{\mathbb{K}[z]}$, and $\widetilde{\Omega}$ is one of these basis elements. Thus $c\widetilde{\Omega} = d\widetilde{\Omega}$ implies $(c-d)\widetilde{\Omega} = 0$, which implies $c - d = 0$ by linear independence. This proves uniqueness of the coordinate and well-definedness of λ.

For $U = c\widetilde{\Omega}$, $V = d\widetilde{\Omega}$ and $a, b \in \mathbb{K}[z]$,

$$\begin{aligned} \lambda(aU + bV) &= \lambda((ac + bd)\widetilde{\Omega}) \\ &= ac + bd \\ &= a\lambda(U) + b\lambda(V). \end{aligned}$$

Finally $\lambda(\iota(c)) = c$ and $\iota(\lambda(c\widetilde{\Omega})) = c\widetilde{\Omega}$. These identities prove the inverse relations and the isomorphism. □

Theorem B.11 (Polynomial scalarization of a represented downstream DAG)**.** *Let a downstream DAG contain b scalar or module arithmetic gates as in Definition B.1. Suppose its initial module inputs U_j are supplied through scalar circuits computing $c_j = \lambda(U_j)$, with a shared total of s scalar gates. Then every module output can be scalarized by an explicit scalar DAG with at most $s + b$ gates over the same coefficient ring. The compiler is uniform and preserves sharing.*

Proof. Process the downstream DAG in a topological order. Keep all scalar input wires and scalar arithmetic gates. For every module wire U, maintain a scalar wire u satisfying the invariant

$$U = u\widetilde{\Omega}. \tag{B.14}$$

For each initial input U_j, use the output wire computing c_j; the invariant holds by hypothesis. There are two kinds of module gates.

1. If a gate computes $W = U + V$, the predecessor invariants give $U = u\widetilde{\Omega}$ and $V = v\widetilde{\Omega}$. Hence

   $$W = u\widetilde{\Omega} + v\widetilde{\Omega} = (u + v)\widetilde{\Omega}.$$

   Append the single scalar gate $w = u + v$. It restores (B.14) at this output.

2. If a gate computes $W = rU$, use the unchanged scalar wire for r and append the single gate $w = ru$. Since

   $$W = r(u\widetilde{\Omega}) = (ru)\widetilde{\Omega},$$

   the invariant again holds.

Induction proves the invariant at every module gate. At every output, the produced scalar is therefore its unique coordinate, by Lemma B.10. Each original arithmetic gate contributes at most one scalar gate. When a wire is reused, its scalar counterpart is reused; no subcircuit is expanded into a formula. Including the s input gates proves the bound $s + b$.

To implement the compiler, store the index of the scalar counterpart of each original wire and scan the topological gate list once. Each gate emits at most one new record with predecessor indices. The number of records is linear in the input and output graph sizes, and their binary encoding length is polynomial. This proves uniform construction as well as the gate bound. □

Corollary B.12 (Only one scalar coordinate is required)**.** *If an actual scalar circuit of size s_{sc} computes $c_A = \lambda(V_A)$ from the original data, then the complete scalar frequency table has a circuit of size at most*

$$s_{\mathrm{sc}} + \ell_{\mathrm{bin}}(k_*) + N - 1. \tag{B.15}$$

For $N = 2^m \geq 2$, the frequency table and its inverse transform together have a circuit of size at most

$$s_{\mathrm{sc}} + B(N) = s_{\mathrm{sc}} + 2N \log_2 N + 3N - 3. \tag{B.16}$$

Proof. All module gates in Propositions B.4 and B.8 have the legal forms of Definition B.1. Apply Theorem B.11 with the single module input V_A. The coordinate circuit of size s_{sc} is included once. Proposition B.5 gives (B.15); the pooled construction of Corollary B.9 gives (B.16). □

In particular, the scalar gates are explicitly

$$S_A[0] = c_A, \qquad S_A[f+1] = gS_A[f], \qquad S_A[f] = \omega^{fD^*} P_A(z). \tag{B.17}$$

Linearity also proves that scalarization commutes with the inverse transform. For every k,

$$\begin{aligned} \lambda(Z_A[k]) &= \lambda\left(\frac{1}{N} \sum_{f=0}^{N-1} \omega^{-fk} Y_A[f] \right) \\ &= \frac{1}{N} \sum_{f=0}^{N-1} \omega^{-fk} \lambda(Y_A[f]) \\ &= \mathrm{INTT}(\mathbf{S}_A)[k] = \delta_{k,k_*} c_A. \end{aligned}$$

Thus one may read the initial coordinate and then perform the scalar spectral computation, or read the final peak. Both give $P_A(z)$.

Remark B.13 (What the theorem assumes about the input representation)**.** If V_A is already encoded as its scalar coordinate together with the fixed basis label $\widetilde{\Omega}$, λ is a wire selection on this encoding and requires no arithmetic gate. The cost of computing that coordinate is the input cost s_{sc}. If V_A is supplied as the output of a symbolic circuit over $\widetilde{\Lambda}_{\mathbb{K}[z]}$, its coordinate has not thereby been exposed as a scalar wire. Theorem B.11 does not replace that input cost by the algebra-valued gate count.

The distinction is forced by the operations involved. For $n \geq 1$, $\widetilde{\Omega}^2 = 0$ because a second copy repeats every exterior generator. Yet $\lambda(\widetilde{\Omega}) = 1$. Consequently

$$\lambda(\widetilde{\Omega}^2) = 0 \quad \text{while} \quad \lambda(\widetilde{\Omega})\lambda(\widetilde{\Omega}) = 1.$$

The map λ is a module isomorphism, not a multiplicative homomorphism for arbitrary algebra products. The downstream scalarization works because its only products involving module values are products by scalars.

B.6 The scalar input computed from the original data

The preceding scalarization theorem is an explicit polynomial compiler for the downstream block. To obtain a complete construction from A, its initial scalar wire must also be constructed. The following is the scalar compilation already given in Proposition 13.14; it is recalled here to make the composition and its actual cost fully explicit.

Proposition B.14 (Existing scalar compilation of the filtered coordinate)**.** *Set $F_{0,\varnothing} = 1$. For $1 \le i \le n$ and $U \subseteq [n]$ with $|U| = i$, compute*

$$F_{i,U} = \sum_{j \in U} B_{ij}\, F_{i-1,U\setminus\{j\}}. \tag{B.18}$$

Then $F_{n,[n]} = \lambda(V_A) = P_A(z)$. With the p_{ij} as leaves, this displayed scalar circuit has at most

$$s_{\mathrm{DP}}(n) = (n-1)2^n + 2n^2 \tag{B.19}$$

gates. Computing the p_{ij} from the entries of A adds $2n^2$.

Proof. For each i and U, consider the polynomial whose terms choose one entry in each of the first i rows, using the columns of U exactly once. For $i = 0$, the only choice is empty and its product is 1.

For $i \ge 1$, partition these choices according to the column j used in the last row i. Removing this last choice leaves exactly a choice on the first $i-1$ rows using $U \setminus \{j\}$. Conversely, every such smaller choice extends uniquely after j is fixed. Its amplitude and marked power are multiplied by $B_{ij} = p_{ij} z^{w_{ij}}$. Summing over $j \in U$ proves (B.18). Induction on i establishes the intended polynomial at every state. For $i = n$ and $U = [n]$, the choices are exactly the valid surviving permutations, all with positive sign in the corrected output. Hence $F_{n,[n]} = P_A = \lambda(V_A)$.

For the count, first compute the marked powers by

$$Z_0 = z, \qquad Z_{r+1} = Z_r^2 \quad (0 \le r < n^2 - 1).$$

These are $n^2 - 1$ gates. Each of the n^2 values B_{ij} then needs one multiplication by p_{ij}. At a state (i, U), (B.18) uses i multiplications and $i - 1$ additions, a total of $2i - 1$ gates. There are $\binom{n}{i}$ such states at layer i.

The total number of recurrence gates is therefore

$$\begin{aligned}\sum_{i=1}^{n} (2i-1)\binom{n}{i} &= 2\sum_{i=1}^{n} i\binom{n}{i} - \sum_{i=1}^{n}\binom{n}{i} \\ &= 2n2^{n-1} - (2^n - 1) \\ &= (n-1)2^n + 1.\end{aligned}$$

Here $\sum_{i=1}^{n}\binom{n}{i} = 2^n - 1$ counts all nonempty subsets, and $\sum_{i=1}^{n} i\binom{n}{i} = n2^{n-1}$ counts pairs (j, U) with $j \in U$: for fixed j, each of the remaining $n - 1$ elements can independently belong to U. Adding the marking and entry gates gives

$$(n^2 - 1) + n^2 + (n-1)2^n + 1 = (n-1)2^n + 2n^2.$$

The last claim follows from the two gates used to compute each p_{ij} in (B.1). □

Corollary B.15 (Unconditional complete scalar circuit)**.** *For $N = 2^m \geq 2$, the existing scalar compiler followed by the postfilter spectral block gives a complete scalar circuit of size at most*

$$(n-1)2^n + 2n^2 + 2N\log_2 N + 3N - 3 \tag{B.20}$$

with the p_{ij} as leaves, or at most

$$(n-1)2^n + 4n^2 + 2N\log_2 N + 3N - 3 \tag{B.21}$$

from the a_{ij}.

Proof. Use Proposition B.14 for the single scalar input c_A, and then use (B.16). The same scalar input wire is shared by all frequencies. Therefore the two sizes add. □

Why the terminal one-dimensional module does not settle the input cost. At the last product gate of the original filter, the desired coordinate satisfies

$$F_{n,[n]} = \sum_{j=1}^{n} B_{nj}\, F_{n-1,[n]\setminus\{j\}}.$$

The n preceding coordinates require coordinates from the $(n-2)$-row product, and so on. At row i, the above compilation has one state for every i-element column subset. That is the source of the sum counted in Proposition B.14. The fact that the final layer has one coordinate does not make the earlier coordinate computations disappear from this circuit.

For $n \geq 2$, the final row-product gate of the source circuit multiplies algebra values of exterior degrees $2(n-1)$ and 2. It is not a downstream gate of the forms $U + V$ or rU with r scalar. The initial coordinate required in Theorem B.11 therefore cannot be obtained by applying that theorem backwards to this gate.

The bound (B.20) is an upper bound for this specified compilation and is exponential in n. It is not a lower bound for all possible scalar circuits. A compiler satisfying the exact size, constant-length and uniformity requirements of Conjecture 1.5 would establish the input hypothesis of Corollary B.12 and close that conjecture. The appendix does not assume such a compiler in an unconditional gate count. The exact dictionary implementation is given in Section B.10; the global alternatives are established in Section B.11.

B.7 Field representation and numerical evaluation

An arithmetic gate count specifies a compact symbolic computation. For a bit-complexity claim, the coefficient representation and the output format must also be specified. This section gives both an exact characteristic-zero representation of the spectral constants and an exact finite-field evaluation bound.

Proposition B.16 (Polynomial rational simulation of the spectral field)**.** *Let $N = 2^m \geq 2$, choose $\mathbb{K} = \mathbb{Q}(\omega)$ with ω primitive, and set $d = N/2$. Use the basis*

$$1, \omega, \ldots, \omega^{d-1}, \qquad \omega^d = -1.$$

When the rational coordinates of all constants are supplied with controlled binary lengths, a scalar circuit over $\mathbb{K}[z]$ with s gates can be simulated over $\mathbb{Q}[z]$ with at most $3d^2 s$ gates. Its construction and constant descriptions have polynomial overhead in d and in the supplied encoding lengths.

Proof. For a power of two N, the cyclotomic polynomial is $X^d + 1$. Its irreducibility also follows directly from Eisenstein at 2 applied to $(X+1)^d + 1$: all intermediate coefficients are even, its constant coefficient is 2, and its leading coefficient is 1. The parity statement follows from $(X+1)^{2^r} = X^{2^r} + 1$ over the field with two elements. Thus the displayed powers form a rational basis.

Represent a wire by its d coordinate polynomials:

$$u = \sum_{i=0}^{d-1} u_i(z)\omega^i.$$

An addition takes d scalar additions. For a multiplication, form the d^2 products $u_i(z)v_j(z)$. Since $0 \leq i + j \leq 2d - 2$, reduction uses only

$$\omega^{i+j} = \begin{cases} \omega^{i+j}, & i + j < d, \\ -\omega^{i+j-d}, & i + j \geq d. \end{cases}$$

Group the resulting signed products by their final basis index. At most d^2 multiplications, d^2 sign multiplications and d^2 additions suffice. Hence $3d^2$ scalar gates per multiplication are an upper bound, which also covers an addition.

Process the original DAG in topological order and retain its sharing. This proves the count and uniformity. Constants are replaced by their supplied coordinate vectors; no unrelated constant with an uncontrolled encoding is introduced. □

For the concrete choice (B.9), $d \leq n$. The representation of the new spectral constants is particularly simple: a power of ω has a single nonzero coordinate equal to 1 or -1, and N^{-1} has rational encoding length $O(\log N)$. Thus the spectral extension itself does not hide an exponential dimension. This statement concerns the spectral coefficient field, and does not replace the scalar compilation of the enlarged exterior algebra.

Proposition B.17 (Local and global finite-field evaluation)**.** *Assume that Q is prime, $N \mid Q-1$, an element $\omega \in \mathbb{F}_Q$ of exact order N is supplied, and every denominator of A is coprime to Q. Evaluate z at a specified $\alpha \in \mathbb{F}_Q$. Let $b = \lceil \log_2 Q \rceil$. If the source circuit is obtained by reducing a rational circuit, require also that every denominator of its rational constants is coprime to Q. If a complete scalar circuit has S gates, its exact modular-arithmetic evaluation work is $O(Sb^2)$ bit operations with elementary arithmetic, in addition to polynomial input preparation and graph handling. The added block of Corollary B.9 contributes $O(N \log N\, b^2)$ of these operations.*

Proof. Reduce rational input numerators and denominators modulo Q. Each denominator has a modular inverse by the hypothesis. The extended Euclidean algorithm computes these inverses in time polynomial in their encoding lengths and in b. Reduction of the labels and preparation of k_* also take polynomial bit time.

Every scalar wire then holds one residue in $\{0, \dots, Q-1\}$, so its stored length is at most b bits. An addition followed by reduction has cost $O(b)$. Elementary multiplication and modular reduction have cost $O(b^2)$. Evaluate the scalar DAG in topological order, once per gate. Each gate therefore costs at most $O(b^2)$, and summing over all S gates gives $O(Sb^2)$. Using $S = B(N)$ for the added block proves its stated contribution. Stored gate values use $O(Sb)$ bits; the output table uses $O(Nb)$. The explicit gate records require $O(S \log(S+2))$ bits, apart from the supplied constants. Reading and indexing these records adds polynomial graph-handling overhead; it is separate from the displayed modular-arithmetic count. □

The admissible field data are part of the specified spectral input. The proof does not include a new procedure for choosing a prime or a primitive root. It computes the reduction of P_A modulo Q; one modular image does not by itself reconstruct $P_A \in \mathbb{Q}[z]$. The inequality $Q > n!$ alone is neither the root-of-unity condition nor a bound for arbitrary weighted amplitudes.

Formal polynomials and exact rational values. Over $\mathbb{Q}[z]$ or $\mathbb{K}[z]$, the output remains a shared arithmetic circuit. No list of coefficients in z is required by the construction. If an exact rational evaluation is requested instead, its bit cost also depends on intermediate and output lengths. For example, with $A = 0$ and $z = 2$, the identity-permutation term has exponent

$$E(\mathrm{id}) = \sum_{r=0}^{n-1} 2^{r(n+1)} \geq 2^{n^2-1}.$$

Its coefficient is 1, and all coefficients of P_0 are positive. Therefore $P_0(2) \geq 2^{E(\mathrm{id})}$, whose binary expansion has at least $2^{n^2-1} + 1$ bits. A claim about a small formal circuit does not promise a polynomial time bound for writing this full integer. This observation concerns an expanded output format; it is not an arithmetic-circuit lower bound.

B.8 Worked examples after filtering

A symbolic check for $n = 2$. The marked matrix has entries

$$B_{11} = p_{11}z, \quad B_{12} = p_{12}z^2, \quad B_{21} = p_{21}z^4, \quad B_{22} = p_{22}z^8.$$

Using the established filter,

$$\begin{aligned} F_A &= (p_{11}zq_1 + p_{12}z^2q_2)(p_{21}z^4q_1 + p_{22}z^8q_2) \\ &= p_{11}p_{21}z^5q_1^2 + p_{11}p_{22}z^9q_1q_2 + p_{12}p_{21}z^6q_2q_1 + p_{12}p_{22}z^{10}q_2^2 \\ &= \left(p_{11}p_{22}z^9 + p_{12}p_{21}z^6\right) q_1q_2. \end{aligned}$$

Here $q_1^2 = q_2^2 = 0$, $q_2q_1 = q_1q_2 = -\widetilde{\Omega}$, and $\gamma_2 = -1$. Thus the corrected output is

$$V_A = c\widetilde{\Omega}, \qquad c = p_{11}p_{22}z^9 + p_{12}p_{21}z^6 = P_A(z).$$

Choose $d_1 = 0$, $d_2 = 1$, $N = 4$, and $\omega^2 = -1$. Then $D^* = k_* = 1$, $g = \omega$, and the recurrence gives

$$\mathbf{Y}_A = (c\widetilde{\Omega},\ \omega c\widetilde{\Omega},\ -c\widetilde{\Omega},\ -\omega c\widetilde{\Omega}).$$

Gate-by-gate scalarization produces

$$\mathbf{S}_A = (c,\ \omega c,\ -c,\ -\omega c).$$

For $k = 1$, every inverse-transform summand is c, so the result is $4c/4 = c$. For $k = 0, 2, 3$, the geometric sum in Proposition B.7 is zero. Hence

$$\mathrm{INTT}(\mathbf{S}_A) = (0, c, 0, 0).$$

This calculation checks the amplitudes, the sign correction and the output slot on an explicit filtered output.

An exact finite-field check for $n = 3$. Take $A = 0$, $\alpha = 1$, $d = (0, 1, 2)$, $N = 4$, $Q = 13$ and $\omega = 5$. The conditions are explicit: $4 \mid 12$, $5^2 = 12 = -1 \pmod{13}$, and $5^4 = 1$. Thus ω has exact order four. The corrected filtered coordinate is $c_A = P_0(1) = 3! = 6$, $D^* = k_* = 3$, and $g = 5^3 = 8 \pmod{13}$.

f	0	1	2	3
g^f	1	8	12	5
$S_A[f] = 6g^f \pmod{13}$	6	9	7	4

Since $4^{-1} = 10 \pmod{13}$, the inverse transform gives

$$\mathrm{INTT}(6, 9, 7, 4) = (0, 0, 0, 6) \pmod{13}.$$

For the nonzero slot, the unnormalized sum is $6 + 5 \cdot 9 + 12 \cdot 7 + 8 \cdot 4 = 167 \equiv 11 \pmod{13}$; normalization gives $10 \cdot 11 \equiv 6 \pmod{13}$. The sums in the other three slots are respectively 26, 182 and 169, all divisible by 13.

B.9 Composition theorem and precise status

Theorem B.18 (The complete composition rule)**.** *Fix the field representation and the legal constants described above. Suppose an explicitly specified scalar compiler, uniform in* (n, A)*, computes*

$$c_A = \lambda(\gamma_n F_A) = P_A(z)$$

from the original data using at most $\Pi(n, L)$ *scalar gates, with polynomial description length and construction time. Then the same data determine, by an explicit uniform circuit, the scalar table*

$$S_A[f] = \omega^{fD^*} P_A(z)$$

and its inverse transform

$$T_A[k] = \delta_{k,k_*} P_A(z).$$

For $N = 2^m \geq 2$*, the total number of scalar gates is at most*

$$\Pi(n, L) + 2N \log_2 N + 3N - 3. \tag{B.22}$$

For the choice (B.9)*, it is at most*

$$\Pi(n, L) + 4n \log_2(2n) + 6n - 3. \tag{B.23}$$

Consequently a polynomial bound for the specified input compiler implies a polynomial bound for the complete scalar composition. Under the admissible finite-field evaluation model, the full bit cost is polynomial as well when $\log Q$ *is polynomially bounded.*

Proof. Include the scalar input circuit once, retaining its output wire. Append the phase pool, the $N - 1$ recurrence gates, the inverse butterflies and the normalization gates constructed above. Corollary B.12 proves correctness at every output and the additive bound (B.22). The construction algorithms scan finite DAGs and write a polynomial number of gate records, so their uniform construction times compose. Their constants have the stated controlled encodings.

For (B.23), use $N \leq 2n$ and $\log_2 N \leq \log_2(2n)$ in (B.22). This proves the polynomial arithmetic bound if Π is a polynomial. Proposition B.17, applied to the total size, proves the finite-field bit bound. If the characteristic-zero spectral circuit must be compiled over $\mathbb{Q}$, use Proposition B.16; $d \leq n$ adds only polynomial overhead. □

Proposition B.19 (The frequency table and its amplitude have the same polynomial scalar–compilation requirement)**.** *Over the same coefficient field and with the same allowed constant encodings, a uniform polynomial scalar circuit for the frequency table gives one for* P_A *by selecting output* $f = 0$*. Conversely, a scalar circuit for* P_A *gives the table with the additional cost* (B.7)*. For an inverse-transformed table, the corresponding output selection is* $k = k_*$*.*

Proof. The defining table identity at $f = 0$ is

$$S_A[0] = \omega^{0D^*} P_A = P_A.$$

Selecting this output adds no arithmetic gate and preserves uniform construction. If desired, remove gates not feeding the selected output; this never increases the size. The converse is the explicitly proved recurrence (B.17). The inverse table has $T_A[k_*] = P_A$ by (B.10) and the linearity of λ. All three statements concern the same coefficient field: in finite characteristic the selected polynomial is the reduction of P_A, not an unproved rational reconstruction. □

Relation to the preceding paper. The input identity and oriented-circuit count come from Theorem 1.4 and Proposition 13.8. The common phase and target slot agree with Proposition 13.11. The new contribution here is the explicit shared adapter, its uniformity and exact gate bounds, and the gate-by-gate proof that its entire downstream computation scalarizes without additional per-gate cost. The actual input-coordinate compilation and its current upper bound are those of Proposition 13.14.

Theorem B.18 permits a general bound $\Pi(n, L)$. To recover Conjecture 1.5 exactly, require a bound $\Pi_0(n)$ independent of L, with the p_{ij} as original leaves and polynomial-in-n bounds on the lengths of auxiliary rational constants. Its construction time is polynomial in (n, L). Preparing the p_{ij} from A then adds the stated $2n^2$ gates.

Established unconditionally. The postfilter pointwise coupling is exact. Its added frequency-table cost is $\ell_{\mathrm{bin}}(k_*) + N - 1$. Including a radix-two inverse transform and shared phase preparation gives the bound $B(N)$. All these downstream gates have an explicit scalar translation once their source coordinate is represented. Combining them with the existing scalar compiler gives the complete bound (B.21) from A.

The remaining assertion needed for polynomial closure. One must construct the initial scalar coordinate from A with a uniform polynomial bound replacing the term $(n-1)2^n$ in the available compilation. Theorem B.18 proves exactly how that bound would close the full pipeline. The identity $\lambda(c\widetilde{\Omega}) = c$ and the final rank-one output prove a coordinate is well defined; they do not supply this input compiler. Therefore this appendix does not claim an unconditional proof of Conjecture 1.5, and it makes no claim that another polynomial compiler is impossible.

B.10 Hash memory for the scalar source

This section implements the source-coordinate computation of Section B.6 with an exact hash dictionary. Its output is the scalar wire $c_A = \lambda(V_A)$ required by the spectral adapter. The invariant below implements the same oriented collision rule and the same scalar coordinate as the paper. The three proposed operations (row accumulation, row NTT, and pointwise multiplication) are connected to this dictionary in Proposition B.22.

Exact keys and the constructed scalar circuit

An entry has a key $U \subseteq [n]$ and a value equal to the scalar coefficient $F_{i,U}$ after i rows. The layer index is implicit in the current dictionary; every key has cardinality i. Store the key as its exact n-bit mask. A hash selects a bucket, and equality of the full masks decides whether two entries are the same state.

In the following pseudocode, `MUL` and `ADD` either append a scalar gate and return its wire identifier, or perform the corresponding operation in an admissible finite-field evaluation. The preparation of B_{ij} is the one counted in Proposition B.14.

```
H = { empty_set : 1 }
for i = 1, ..., n:
    J = empty exact dictionary
    for each (U, c) in H:
        for each j in [n] outside U:
            V = U union {j}
            v = MUL(c, B[i,j])
            if V is already a key of J:
                J[V] = ADD(J[V], v)
            else:
                J[V] = v
    H = J
c_A = H[full_set]
```

Proposition B.20 (Correctness of the exact dictionary)**.** *After row i, the entry at U equals $F_{i,U}$ as defined in* (B.18)*. Consequently the algorithm constructs $P_A(z)$ from the original data, without taking P_A as an input leaf.*

Proof. Initially the empty choice has amplitude 1. Suppose the invariant holds after row $i-1$. A transition from U chooses $j \notin U$; this is exactly the condition that multiplication by q_j survives the oriented filter. The resulting amplitude is $B_{ij}F_{i-1,U}$ and its new support is $V = U \cup \{j\}$. For a fixed V, the predecessors are exactly $V \setminus \{j\}$ for $j \in V$. Accumulating them gives (B.18). The first contribution is assigned directly; later contributions are added. Induction proves the invariant. At $i = n$, $V = [n]$ and the coefficient is P_A. □

The actual number of states and scalar gates

Proposition B.21 (State count and gate count for this implementation)**.** *For the formal polynomial of the paper, the exact dictionary has*

$$r_i = \binom{n}{i}, \qquad R = \sum_{i=0}^{n} r_i = 2^n$$

distinct states over all layers. The displayed algorithm uses

$$E = \sum_{i=0}^{n-1} (n-i)\binom{n}{i} = n2^{n-1}$$

multiplication gates and

$$E - (R-1) = (n-2)2^{n-1} + 1$$

addition gates. Its source-coordinate gate count, including the preparation from the a_{ij}, is at most

$$s_{\text{hash}}(n) = (n-1)2^n + 4n^2. \tag{B.24}$$

Proof. Every i-element subset can be reached by assigning the first i rows bijectively to its columns. All corresponding B_{ij} are nonzero formal monomials. Since $p_{ij} = 1 + a_{ij}^2 > 0$ over $\mathbb{Q}$, their contributions cannot cancel. Thus every such key is present. The binomial theorem gives $R = 2^n$.

Each state in layer i emits $n - i$ transitions. The identity $(n-i)\binom{n}{i} = n\binom{n-1}{i}$ gives $E = n2^{n-1}$. Every transition creates one multiplication. Each of the $R - 1$ noninitial states receives its first contribution by direct assignment, saving one addition. Therefore the recurrence uses

$$E + \big(E - R + 1\big) = (n-1)2^n + 1$$

scalar gates. Preparing the p_{ij} uses $2n^2$ gates; preparing all $z^{w_{ij}}$ uses $n^2 - 1$ squarings; and forming the B_{ij} uses n^2 products. Adding $4n^2 - 1$ proves (B.24). Hashing implements this same shared recurrence, so the earlier recurrence count is not added a second time. □

Reduction modulo Q does not supply a uniform state reduction. Take $A = 0$ and keep z formal. Then $p_{ij} = 1$ and every $F_{i,U}$ is a sum of monomials with coefficient 1. Their exponents encode distinct choices of row-column cells using disjoint binary addresses. Distinct choices therefore give distinct exponents. No coefficient vanishes after reduction modulo any prime Q. Consequently all $\binom{n}{i}$ states remain nonzero in $\mathbb{F}_Q[z]$. Specializing z to a field element can create additional coincidences or zeros on particular inputs; it does not prove a uniform compression of the formal polynomial circuit.

One memory shared by all spectral frequencies

Proposition B.22 (Frequency sharing and exact connection to the NTT)**.** *For the oriented entries $\widetilde{M}_{ij} = B_{ij}q_j$, define*

$$P_i[t] = \sum_{j:d_j \equiv t \pmod N} B_{ij}q_j, \qquad A_i[f] = \sum_{t=0}^{N-1} P_i[t]\omega^{ft}.$$

Write $d(U) = \sum_{j\in U} d_j$. After the first i pointwise row products, the exact identity is

$$\prod_{\ell=1}^{i} A_\ell[f] = \sum_{|U|=i} \omega^{fd(U)} F_{i,U}(z) q_U. \tag{B.25}$$

Thus the coefficients $F_{i,U}$ are shared across frequencies. One dictionary suffices; a separate copy for each f is unnecessary.

Proof. The defining transform gives $A_i[f] = \sum_j B_{ij}q_j\omega^{fd_j}$, since $\omega^{f(d_j \bmod N)} = \omega^{fd_j}$. Suppose (B.25) holds for $i-1$. Multiplication by the row factor eliminates exactly the choices $j \in U$ through $q_j^2 = 0$. For $j \notin U$, it replaces q_U by $q_{U\cup\{j\}}$ and multiplies the phase by ω^{fd_j}. For $V = U \cup \{j\}$ this gives

$$\omega^{fd(V)} \sum_{j\in V} B_{ij} F_{i-1,V\setminus\{j\}} = \omega^{fd(V)} F_{i,V}.$$

The base case is the empty product. This proves the identity by induction, while retaining the same collision filter at every step. □

At the final layer only $U = [n]$ remains, and $q_{[n]} = \gamma_n\widetilde{\Omega}$. The existing orientation correction therefore gives $\omega^{fD^*}P_A\widetilde{\Omega}$. The dictionary has already constructed its scalar amplitude $c_A = F_{n,[n]}$, so the scalar table is generated by the shared recurrence

$$g = \omega^{k_*}, \qquad Y_0 = c_A, \qquad Y_{f+1} = gY_f \quad (0 \le f < N-1).$$

No table of intermediate phases indexed by both U and f needs to be materialized. This is a sharing improvement over independent scalar simulations of all N frequencies. The spectral adapter of Section B.3 already has this sharing.

Appending the existing inverse transform yields $T_A[k] = \delta_{k,k_*}P_A$. For $N = 2^m \ge 2$, the complete bound from A is consequently

$$(n-1)2^n + 4n^2 + B(N), \qquad B(N) = 2N\log_2 N + 3N - 3. \tag{B.26}$$

The source is counted once, and the existing spectral bound is counted once. This establishes the exact interface with the rest of the paper.

Memory, lookup time, and bit cost

Proposition B.23 (Two successive layers suffice for evaluation)**.** *During a forward numerical evaluation, the algorithm needs only the current layer and the accumulating next layer. The maximum number of state entries in these two dictionaries is*

$$\max_{0\le i<n}\left\{\binom{n}{i}+\binom{n}{i+1}\right\}=\binom{n+1}{\lfloor (n+1)/2\rfloor}=\Theta\!\left(\frac{2^n}{\sqrt{n}}\right). \tag{B.27}$$

This bounds active state values; it does not reduce the number of gates in an explicitly retained circuit DAG.

Proof. Every transition reads only the current dictionary and adds to the next one. After all such transitions, the older dictionary is no longer used and can be released. Pascal's identity gives $\binom{n}{i}+\binom{n}{i+1}=\binom{n+1}{i+1}$. The largest binomial coefficient is central, and the standard central-binomial estimate yields (B.27). If values are gate identifiers, releasing identifiers from the dictionary does not delete earlier gates required in the emitted DAG. Thus the storage statement is separate from the circuit-size statement. □

n	States over all layers	Two-layer peak entries
10	1 024	462
20	1 048 576	352 716
30	1 073 741 824	300 540 195

What an exact fast dictionary guarantees. A hash table locates a bucket and compares complete keys inside its collision-resolution scheme. With the usual hashing assumptions and a bounded load factor, dictionary operations have expected amortized constant word-operation cost. Keeping the number of buckets polynomial while inserting exponentially many distinct keys does not preserve this load-factor assumption. Reference [H1] gives the standard analysis.

A key here has n bits. In a bit model, its storage, hashing and comparison must be charged; constant word cost is not constant bit cost for arbitrarily long keys. With $b=\lceil \log_2 Q\rceil$, finite-field values occupy $O(b)$ bits. A conservative active-state bound is $O((n+b)\max_i(r_i+r_{i+1}))$ bits, besides the prepared input values and spectral workspace. The field-arithmetic cost per gate is the one established in Proposition B.17.

The displayed gate count is deterministic. Expected hash lookup time is an implementation bound, not a worst-case polynomial-time proof. Exact deterministic dictionaries give a polynomial overhead in the key length and in the number of stored entries as well. No equality test for arbitrary computed polynomials is assumed.

Which states may share an entry?

Proposition B.24 (Distinct supports are distinguishable by the filter)**.** *Let $U \neq V$ with $|U| = |V| = i$. A summary that replaces the support state completely and must support every subsequent multiplication allowed by the oriented filter cannot identify q_U and q_V.*

Proof. Let $W = [n] \setminus U$. Then

$$q_U q_W = q_{[n]} \neq 0.$$

Since $|U| = |V|$ and $U \neq V$, there exists $j \in V \setminus U$. This j belongs to both V and W, and hence

$$q_V q_W = 0$$

by $q_j^2 = 0$. Equal summaries would have to give the same response to the same continuation q_W, contrary to these two identities. □

Both prefixes in the proposition are valid filtered partial selections. The distinction comes from the remaining columns they permit. It does not reintroduce any eliminated contribution. For example, $q_1 q_2$ and $q_1 q_3$ are both surviving states; multiplication by q_3 preserves the first and annihilates the second.

Even equality of current scalar values is insufficient. Two entries with the same numerical amplitude but different used columns can have different valid continuations. They therefore need distinct support information unless an additional exact aggregation rule is proved. A coincident hash value is only an address collision, and an exact dictionary resolves it by comparing the full keys.

Continuation in the fixed formal instance. For a support U at layer i, define its suffix polynomial

$$C_{i,U}(z) = \sum_{\substack{\sigma:\{i+1,\dots,n\}\to[n]\setminus U \\ \sigma \text{ bijective}}} \prod_{r=i+1}^{n} B_{r,\sigma(r)}.$$

The contribution of a prefix amplitude c at this state to the terminal scalar is $c\,C_{i,U}$. For $U \neq V$ of the same size, the exponent supports of $C_{i,U}$ and $C_{i,V}$ are disjoint: equal exponents would encode the same selected suffix cells, hence the same used suffix columns, and force $U = V$. Their supports are nonempty and their rational coefficients are positive. Thus these continuation polynomials are different. Replacing two amplitudes by their unweighted sum does not preserve this dependence on U.

These are limits of the specified state-merging operation. They do not establish a lower bound for all scalar arithmetic circuits, all representations, or a different algorithm specialized to the single final aggregate.

A precise sparse-state bound and the remaining requirement

Proposition B.25 (Output-sensitive cost of exact memoization)**.** *Suppose a variant has r_i structurally reachable states at layer i, with $r_0 = 1$, and emits E valid transitions in total. Define $R = \sum_i r_i$. If omitted transitions have been justified from the supplied data, the first-insertion rule gives a recurrence with at most*

$$2E - R + 1 \le 2nR$$

scalar gates, where $E \le \sum_i (n-i) r_i \le nR$. Including input preparation and the spectral layer gives

$$s_{\text{data}} + 2E - R + 1 + B(N). \tag{B.28}$$

Proof. Each emitted transition creates one product. Every reachable state other than the initial state has one first incoming transition and therefore saves one addition. There are $R - 1$ such states, giving $E + (E - R + 1)$. A state at layer i has at most $n - i$ possible extensions, proving the upper bounds. The spectral adapter uses the single terminal coordinate, so its gates add to this count. □

Here "structurally reachable" is essential. It does not assume a free test that an arithmetic circuit computes the zero polynomial. For numerical values modulo Q, a zero can be recognized directly; any pruning based on more elaborate identities must include the cost and correctness of its own procedure. If no terminal state is reachable in a sparse variant, its scalar output is the constant 0.

What would establish a polynomial replacement. A state reduction would need an explicitly computable equivalence or other compressed representation, together with exact transition rules and a proof that the total number and encoded size of the retained states are polynomial in the original input parameters. Its construction and final readout must also have polynomial cost. Under those proved conditions, (B.28) or the corresponding compressed-circuit count would supply the source bound required by Theorem B.18.

For the dictionary constructed here, $R = 2^n$ and $E = n2^{n-1}$. It provides exact memoization, reuse across all frequencies, and the two-layer evaluation memory bound. The complete source-plus-spectral gate count is (B.26). The hash dictionary does not by itself prove a polynomial scalar source circuit, and no such claim is inserted into the paper.

References for the data-structure statements

[H1]
MIT 6.006, Fall 2009, Lecture 5, *Hashing I: Chaining, Hash Functions*, pp. 3–5. Official lecture notes.

[H2]
James Aspnes, Yale University, *Dynamic Programming*, Sections 1–2. Official course notes. Memoization shares repeated subproblems; retaining successive layers can reduce active evaluation storage.

B.11 Global evaluation of all filtered contributions

The objective is to replace the exponential scalar-source bound, not to require the subset representation of Section B.10. The number of coordinates in that representation is not a lower bound for arbitrary scalar circuits. For example, $\prod_{j=1}^{n}(1+t_j)$ has 2^n distinct monomials and a circuit with $2n-1$ binary arithmetic gates.

The existing filter already computes a compact global expression

The factorized expression

$$F_A = \prod_{i=1}^{n}\left(\sum_{j=1}^{n} B_{ij}q_j\right)$$

represents all contributions simultaneously. Its products can be organized as a balanced tree. Terms with a repeated column vanish because $q_j^2 = 0$, and every surviving term has the same terminal marker $q_{[n]}$. Hence

$$F_A = P_A q_{[n]}, \qquad \gamma_n F_A = P_A \widetilde{\Omega}.$$

The circuit over the enlarged algebra has the polynomial gate bound given in Proposition 13.8. In particular, its correctness does not require a preliminary list of all states. The remaining construction problem is to obtain its scalar coordinate using ordinary scalar arithmetic gates from the original input, without treating a coefficient-reading operation as free.

The following global reformulation gives an explicit collision projector with a polynomial number of operator factors. It tests whether the simultaneous action of the filter can yield a new scalar compiler. Its operator count and its scalar compilation cost are established separately.

Commuting auxiliary variables and a collision counter

Introduce new commuting variables $t_1, \ldots, t_n$, distinct from the exterior generators x_j, y_j, and define

$$H_A(\mathbf{t}, z) = \prod_{i=1}^{n}\left(\sum_{j=1}^{n} B_{ij}(z)t_j\right).$$

This has degree n in $\mathbf{t}$ and an explicit scalar circuit of polynomial size. A monomial $\mathbf{t}^{\mathbf{m}}$ records the column multiplicities m_j, where $\sum_j m_j = n$. Its number of pairs of rows choosing the same column is

$$\kappa(\mathbf{m}) = \sum_{j=1}^{n}\binom{m_j}{2}, \qquad 0 \le \kappa(\mathbf{m}) \le h_n := \binom{n}{2}.$$

Thus $\kappa = 0$ holds exactly for $\mathbf{m} = (1, \ldots, 1)$. This is the same survival condition as the oriented filter, expressed by one global statistic.

A global projector with polynomially many factors

Proposition B.26 (Exact global collision projector)**.** *Over characteristic zero, let*

$$\mathcal{L} = \frac{1}{2}\sum_{j=1}^{n} t_j^2 \frac{\partial^2}{\partial t_j^2}, \qquad \pi_n(T) = \prod_{r=1}^{h_n}\left(1 - \frac{T}{r}\right).$$

For $n = 1$, the product defining π_n is empty and equals 1. Then

$$\pi_n(\mathcal{L})H_A(\mathbf{t}, z) = P_A(z)t_1 \cdots t_n, \qquad P_A(z) = \pi_n(\mathcal{L})H_A(\mathbf{t}, z)|_{\mathbf{t}=\mathbf{1}}. \tag{B.29}$$

The projector has $h_n = O(n^2)$ operator factors and rational constants with polynomial total binary description length.

Proof. For a monomial of total $\mathbf{t}$-degree n,

$$t_j^2 \frac{\partial^2 \mathbf{t}^{\mathbf{m}}}{\partial t_j^2} = m_j(m_j - 1)\mathbf{t}^{\mathbf{m}}.$$

Consequently $\mathcal{L}\mathbf{t}^{\mathbf{m}} = \kappa(\mathbf{m})\mathbf{t}^{\mathbf{m}}$. Since $0 \le \kappa \le h_n$, the polynomial π_n satisfies

$$\pi_n(\kappa) = \begin{cases} 1, & \kappa = 0, \\ 0, & 1 \le \kappa \le h_n. \end{cases}$$

The operator therefore preserves exactly the squarefree terms of H_A. Its total degree is n in n variables, so its only possible surviving auxiliary monomial is $t_1 \cdots t_n$. Choosing this monomial means choosing every column exactly once. Its coefficient is $P_A(z)$, with all amplitudes retained and no permutation sign. This proves both identities in (B.29).

The factored representation uses the constants $1/r$ for $1 \le r \le h_n$, whose total encoding length is $O(h_n \log(h_n + 1))$. If the polynomial π_n is expanded, its common denominator divides $h_n!$, and each numerator has polynomial binary length as well. No coefficient of P_A is used as an input in this construction. □

The same proof holds over $\mathbb{F}_Q$ when Q is prime and $Q > \max(2, h_n)$: every denominator is invertible and the integers $0, 1, \ldots, h_n$ remain distinct in the field. The spectral conditions on N and ω are imposed separately as before.

There is also a shorter operator factorization on this degree-n space. With $E_j = t_j\partial/\partial t_j$, one has

$$\pi_n(\mathcal{L})H_A = E_1 \cdots E_n H_A = t_1 \cdots t_n\, \partial_{t_1} \cdots \partial_{t_n} H_A.$$

Indeed, $E_1 \cdots E_n$ multiplies $\mathbf{t}^{\mathbf{m}}$ by $\prod_j m_j$, which is 1 for $\mathbf{m} = \mathbf{1}$ and 0 for every other composition of n into n parts. This alternative uses n differential factors and no denominators; the same mixed derivative identity holds in every characteristic.

This is an exact simultaneous filter. Once an ordinary scalar circuit for the projected polynomial has been constructed, substitution $\mathbf{t} = \mathbf{1}$ gives its scalar coordinate, and Theorem B.18 supplies the rest of the pipeline.

Scalar compilation of the operator: the cost to be proved

The h_n operator factors in Proposition B.26 are not scalar multiplication gates. One application of $\mathcal{L}$ to a scalar circuit of size s does have a polynomial compiler, as the following explicit construction shows.

For a fixed j, substitute $t_j \mapsto t_j + v$ and retain the coefficients of $1, v, v^2$. At every gate, store a triple (f_0, f_1, f_2) modulo v^3. Addition is coordinatewise, and a product of two triples is

$$\begin{aligned} h_0 &= f_0 g_0, \\ h_1 &= f_0 g_1 + f_1 g_0, \\ h_2 &= f_0 g_2 + f_1 g_1 + f_2 g_0. \end{aligned}$$

A multiplication gate therefore requires at most six scalar multiplications and three additions. At the output, $f_2 = \frac{1}{2}\partial_j^2 f$. Repeating this construction for $j = 1, \ldots, n$ and summing $t_j^2 f_{j,2}$ constructs $\mathcal{L}f$ with $O(ns + n)$ gates.

Applying this compiler successively to

$$H^{(0)} = H_A, \qquad H^{(r)} = H^{(r-1)} - \frac{1}{r}\mathcal{L}H^{(r-1)} \quad (1 \le r \le h_n)$$

gives only a bound of the form

$$\operatorname{size}(H^{(h_n)}) \le (cn)^{h_n}\big(\operatorname{size}(H_A) + n + 1\big)$$

for a fixed constant $c > 1$. This is an upper bound for the displayed repeated compiler; it is not a lower bound for an optimized global construction. A polynomial number of applications of a compiler whose size increase is multiplicative does not prove a polynomial final circuit size. The shorter Euler factorization has n first-derivative steps. A straightforward repeated first-derivative compiler gives a bound $c_1^n(s + n + 1)$ for a fixed $c_1 > 1$, which again does not establish polynomial size.

Why the collision statistic alone does not give a closed scalar update. At a product gate,

$$\mathcal{L}(fg) = (\mathcal{L}f)g + f(\mathcal{L}g) + \sum_j t_j^2(\partial_j f)(\partial_j g).$$

The cross terms depend on which columns the two factors use. For example, $f = t_1$ and $f = t_2$ have identical values $f(\mathbf{1}) = 1$ and $(\mathcal{L}^r f)(\mathbf{1}) = 0$ for every $r \ge 1$. Multiplying each by $g = t_1$ nevertheless gives

$$(\mathcal{L}(t_1^2))(\mathbf{1}) = 1, \qquad (\mathcal{L}(t_1 t_2))(\mathbf{1}) = 0.$$

Thus these scalar summaries of the inputs do not determine the summary of their product. A different exact update rule would need additional information whose production and size are counted.

A second global candidate: two scalar determinants

Proposition B.27 (Global determinant contraction)**.** *Set* $E_n(z) = \det(z^{w_{ij}})_{i,j\in[n]}$. *If* $\odot$ *denotes coefficientwise Hadamard multiplication in the variable* z*, then*

$$P_A(z) = D_A(z) \odot E_n(z). \tag{B.30}$$

Both determinants on the right have scalar circuits of polynomial size, including the shared preparation of their marked powers.

Proof. For $e_\sigma = \sum_i w_{i,\sigma(i)}$, the distinct binary cell addresses give $e_\sigma \neq e_\tau$ whenever $\sigma \neq \tau$. The coefficients at e_σ in the two determinants are respectively

$$\operatorname{sgn}(\sigma) \prod_i p_{i,\sigma(i)}, \qquad \operatorname{sgn}(\sigma).$$

Their coefficientwise product is $\prod_i p_{i,\sigma(i)}$, because the signs square to 1. All other coefficients are zero. This proves (B.30). The shared squaring circuit prepares the powers $z^{w_{ij}}$, after which the division-free determinant construction of the paper has polynomial size. □

This identity acts on all surviving contributions simultaneously. It is a scalar description of the same final coordinate as the doubled oriented filter. The operation to compile is now a single coefficientwise contraction. The existence of small input circuits does not itself give a small scalar circuit for that contraction.

More explicitly, introduce a Laurent variable v. Then

$$P_A(z) = [v^0]\, D_A(v) E_n(z/v).$$

For a primitive root η of order T, with T invertible, averaging gives

$$\frac{1}{T} \sum_{\ell=0}^{T-1} D_A(\eta^\ell) E_n(z\eta^{-\ell}).$$

This equals the desired constant term when $T > \max(\deg D_A, \deg E_n)$, since then the root-of-unity average eliminates all nonzero exponent differences. The generic bound on T is exponential: $\deg E_n \geq w_{nn} = 2^{n^2-1}$. A smaller T can also retain pairs of distinct exponents congruent modulo T; their absence must be proved before using a short transform for this contraction.

Here T is a proposed source-contraction parameter, distinct from the already polynomial postfilter transform length N. Coefficientwise Hadamard multiplication is also different from pointwise multiplication of the values in an NTT. The constructive Hadamard theorems in [G1] have specific noncommutative branching program or circuit hypotheses; those hypotheses are not supplied merely by the two determinant circuits in (B.30).

Cases where the global contraction really is polynomial

The preceding analysis does not rule out simultaneous scalar evaluation. The following two cases give actual positive examples.

Proportional row factors. If legal supplied scalars satisfy $B_{ij} = u_i v_j$, then

$$F_A = \left(\prod_i u_i\right)\left(\sum_j v_j q_j\right)^n = n!\left(\prod_i u_i\right)\left(\prod_j v_j\right) q_{[n]}.$$

The last equality follows because a surviving choice uses each column once and there are $n!$ orders of these n commuting factors. Its scalar coordinate is computed with $O(n)$ products, including the computation of $n!$. For the original input family, $A = 0$ and the evaluation $z = 1$ give the concrete example $P_0(1) = n!$.

This condition is not present for the full marked formal matrix: for every rational A, its determinant $D_A(z)$ is nonzero, because its permutation terms have distinct exponents and nonzero coefficients. Thus $B_A(z)$ has rank n over $\mathbb{Q}(z)$. The proportional-row example proves that a global contraction can avoid the subset table; it does not impose that structure on the general formal instance.

Reduction in characteristic two. When the coefficients admit reduction modulo 2,

$$\overline{P_A}(z) = \det\big(\overline{B_A}(z)\big),$$

because every permutation sign equals 1 in characteristic two. The shared power preparation and a division-free determinant circuit therefore give an actual polynomial scalar source construction for this modular image. No expanded polynomial or coefficient oracle is needed.

The postfilter phase identities continue to hold over a field containing an appropriate root of unity. For example, let $m = \lceil \log_2(n+2) \rceil$ and $N = 2^m - 1$, so $n < N \le 2n + 1$. In $\mathbb{F}_{2^m}$, a primitive Nth root exists and N is invertible. The same shared phase recurrence followed by a direct inverse transform uses $O(N^2)$ field operations. This is a complete polynomial pipeline for the characteristic-two image. It uses odd N, rather than the radix-two choice in characteristic zero or odd prime characteristic. Since $2^m = O(n)$, the defining irreducible polynomial and a primitive element can be found by exhaustive trials on polynomially many $O(\log n)$-bit candidates, with polynomial-cost irreducibility and order tests. A single modular image does not reconstruct the original rational polynomial, and rational denominators must be invertible for the stated reduction.

Status of the general construction. Global evaluation is mathematically legitimate. The exact identities (B.29) and (B.30) provide concrete ways to seek a scalar source circuit without an explicit subset table. Neither displayed general compiler currently has a proved polynomial scalar cost. The polynomial overhead of the downstream spectral block remains the one in Theorem B.18; the missing bound is the scalar realization of the global source operation itself.

Reference

[G1]
V. Arvind, P. S. Joglekar and S. Srinivasan, *Arithmetic Circuits and the Hadamard Product of Polynomials*, FSTTCS 2009, pp. 25–36, Theorems 4 and 10. Publisher's article.